Drishti दृष्टि

Drishti: see, aggregate and analyse any data, live, in one platform

An entity terminal, a BI and analytics platform, and risk computed on demand, on the bank's own data

See what your data hasn't told you yet.

ANY DATA. ANY DOMAIN. ONE GRAMMAR.

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Java 21 or later · console on any device

IN THIS DECK

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TL;DR

Executive summary

What is Drishti?	A data, aggregation, BI and analytics platform: every record live, any aggregate of it, analysed and explained, on the bank's own lakes.
What does it replace?	The screen coded per product, the data mart built per question, the BI tool on copies and the risk engine with its own store: one engine and one grammar over the data instead.
How does it work?	Sutras describe screens; scopes and hierarchies aggregate any set of trades from its members; models serve BI; VaR, ES and exposure are computed when asked, with their working shown.
What is there today?	11 packs, 219 Sutras, 106 commands with data contracts, 19 source plugins; four lenses; VaR by any hierarchy, sharded, on Delta or Iceberg; exposure; backtesting; signed reports; playgrounds.
Where is it going?	One theme at a time: memory, scale and mobile; oversight; Java 25; regulatory reports; counterparty engines; capital; native speed; XVA; executive lenses.

Why one platform

01

Today a new question means a new screen, a new mart and a new report

1 A new screen
a developer codes a page for the product or the desk; it is right on the day it ships, then rots

→

2 A new data mart
the rows are copied out of the systems of record, reshaped and loaded overnight for that question

→

3 A new report
a BI tool reads the copy; its totals, filters and masks are written again, in its own language

→

4 A new risk store
the risk engine keeps its own finished numbers; nobody can re-derive one of them later

→

5 The numbers disagree
the screen, the report and the risk figure each read a different copy, at a different time

What it costs

Time
each question waits for a screen, a mart and a report to be built in turn, by three teams

Trust
a figure in the morning meeting has three versions; reconciling them is a job of its own

Security
every copy needs its own entitlements and masks, and each is a new place for data to leak

Change
a new product or desk means new code in every layer; a hierarchy change means a rebuild

Key insight One platform reads the bank's own data once, and every screen, aggregate and analysis is computed from it.

THE PROBLEM

Every entity kind gets a bespoke screen — and the screen rots

A screen per product, in the documents' own words

- **Hundreds of entity kinds**
  “Front-office, risk and operations staff look at hundreds of entity kinds.”
- **Written once, rotting for ever**
  “Each one traditionally gets a bespoke screen, written once and rotting forever.”
- **Data outruns screens**
  “The data changes shape faster than screens can be rebuilt.”

And what people do in the meantime

- **Tab between systems**
  “A trade lives in one system, its netting set in another, its curve in a third … the desk tabs between them.”
- **Wait for a screen**
  “Every new product type waits for a screen.”
- **Read raw JSON**
  “A new feed or product has no screen at all, so on day one people read raw JSON or wait.”

Key insight **Nobody here is short of data — they are short of a screen for it that is dense, live and linked.**

WHAT DRISHTI REPLACES

Four things a bank builds again and again, and what takes their place

What a bank builds today	What Drishti does instead	Where it shows
The bespoke screen per product	A Sutra: a short YAML file per product family; inference lays out what it leaves out. One pipeline draws every entity.	219 Sutras in 11 packs
The data mart per question	Scopes, hierarchies and groupings computed from the lake's own rows when asked; the Data view downloads any dataset.	Delta or Iceberg, read in place
The BI tool on copies	Rūpaka BI on the same data: datasets rooted in packs, one model per report, row rules and masks before any sum.	the same entitlements as a view
The risk engine with its own store	VaR, ES, exposure and the attributes computed for any scope from trade-level inputs; official records kept, never edited.	any scope, bank to one trade

Key insight Not four products glued together: one engine, one grammar, one set of rules over the bank's own lakes.

THE SHAPE OF THE FIX

From a coded screen per product — to one grammar and inference

Your systems

Trades, curves, tickets, variants

→

A screen per kind

Coded by a developer, per product

→

A backlog

New products wait for a screen

→

Drift

The data changes; the screen does not

Your systems

Read where they are, through plugins

→

DRISHTI

Sutra ⊕ inference

A short YAML Sutra describes the view; scored, explainable rules read the data's shape and fill the rest

→

A live view

Dense, linked, explained — day one

Key insight A brand-new product with no Sutra still renders a usable view on day one; a Sutra makes it beautiful.

Six goals the design is held to

1

One grammar

Every view is a Sutra in Rachana, or inferred; no screen code.

2

Any source

Sources plug in through a small, Spring-free interface.

3

Linked

Every id on a screen is navigable, across systems.

4

Live

Changes reach the open view; p99 is measured and shown in the top bar.

5

Provenance

Every view shows how it was built and where its data came from.

6

Keyboard first

<MNEMONIC> <ID> <GO>, F-keys for panels, F9 for raw JSON.

Key insight Six goals, each one testable — the p99 is not a claim on a slide, it is in the top bar of every view.

Each family does part of the job, and is good at it

Family	Examples	Good at	What it leaves you
Market data terminals	Bloomberg Terminal, LSEG Workspace, FactSet	Dense, keyboard-driven, live screens	The vendor's screens, for the vendor's data
Trading and risk platforms	Murex, Calypso, in-house systems	Deep screens for every product they book	A coded screen per product type, for that platform's data
Low-code internal tools	Retool, Appsmith, Power Apps	Internal screens assembled quickly	A person still assembles each screen
BI and dashboards	Tableau, Power BI, Grafana	Aggregates and charts	Weaker at one entity, live, with its links

Key insight **The screens that are dense and live are coded per product; the screens that are quick to make are not terminals.**

BENEFITS

What a grammar of screens buys

Day one, every product	Inference shows a new feed at once; nobody reads raw JSON.
Screens as reviewed data	A Sutra is versioned, tested, diffed and approved like code — without being code.
One place to look	A trade, its curve and its netting set from three systems on one linked screen.
Trust in the number	Provenance and freshness on every view; masks applied for every reader.
Anywhere it is needed	The console on any device, the API, the CLI, and the same view inside other apps.

Key insight **The cost of a new screen falls from a project to a paragraph — and to nothing at all on day one.**

The platform in one picture

02

THE PLATFORM IN ONE PICTURE

Four layers: data in, one engine, many experiences, and trust around them all

Experiences what people use	**Terminal** any record, live, linked	**BI Studio** reports, models, boards	**Playgrounds** every figure as formulas	**Lenses** Risk, Trading, Market, World	**Phone** the same views, tiered	**Elements** views inside other apps
Engine one grammar	**Rachana** Sutras and inference	**Aggregation** scopes, hierarchies, totals	**Computed measures** VaR, ES, exposure, groupings	**Live** one subscription per entity	**Drishti*net** several servers as one	
Data the bank's own	**Lakes** Delta and Iceberg, local or S3	**JSON Lines** the one data standard	**Contracts** one per command	**Connectors** 19 source plugins	**Public packs** SEC, GLEIF, World Bank	

Trust, around every layer sign-in and roles · masks before arithmetic · audit of who saw what · data contracts · official records that never change · the evidence: tests, drills, benchmarks

Key insight **Every experience asks the same engine, which reads the same data under the same rules.**

ONE ENGINE, THREE KINDS OF ANSWER

See a record, aggregate any scope, analyse a model: the same request path

See

```
TRD MX-20000001
```

- a kind and an id
- the Sutra matched, the layout from cache
- links fetched in parallel
- a ViewModel in JSON, live by patches

Aggregate

```
VAR BANK - DESK:CREDIT
```

- a scope written as text
- resolved to ordinals on the date
- parts summed by position runs
- VaR, ES and contributions that add up

Analyse

```
BI · a report on a model
```

- one model per report
- row rules and masks injected first
- one query plan, live or as of a date
- the browser only draws

Shared by all three, so they always agree

- the same connectors and business dates; the same entitlements, masks and audit; the same data versions and caches keyed by them
- the same answer on every member of a cluster (Drishti*net), and the same answer over the REST API, the CLI and in the console

Key insight Seeing, aggregating and analysing are three questions to one engine, not three systems.

Eight primitives carry the whole model

Primitive	What it is	Where you meet it
Pack	A folder for one domain: kinds, mnemonics, links, connectors, Sutras, samples, help text; packs inherit	config/packs/ — 11 shipped, e.g. trading, market-risk, funds
Sutra	A short, versioned YAML description of one view, in the Rachana grammar	219 Sutras in the shipped packs; fragments from the UI kit
Match	kind + where (a closed expression) + priority; the highest match wins	match: { kind: trade, where: …, priority: 10 }
Strip	Up to eight headline figures across the top of a view	Source, side, notional, status …
Panel	One of 22 kinds on a 12-column grid, each with an F-key	table, ladder, pivot, surface, map …
Inference	Explainable rules that read the data's shape and lay out what no Sutra covers	infer: true; 'completed by inference'
ViewModel	Plain JSON the server returns and the browser draws	GET /api/v1/views/{kind}/{id}
Frame	A patch of what changed, sent over one live channel per browser	Server-sent events

ARCHITECTURE

From your systems to a view, in one direction

Stores and lakes

Delta, JDBC, Aerospike, Cassandra, MongoDB, Redis, Iceberg, DuckDB

Streams and queues

Kafka, ActiveMQ, RabbitMQ

Files and services

JSON Lines, CSV (deprecated), S3, REST

Public market data

SEC EDGAR, GLEIF, N-PORT, NY Fed, ECB, FRED

DRISHTI SERVER · JAVA 21 · ONE OF A CLUSTER

Fetch, match, infer, link, bind

Virtual threads fetch in parallel; the engine matches a Sutra, merges inference, links ids and binds values; BI answers on DuckDB; VaR and exposure engines compute any scope — 19 source plugins

SAME SERVER

Identity, packs, audit, Drishti*net

Users, roles and masks; audit; a signed pack registry; commands switched off for everyone; peers by gossip on their own port, HTTP/2 between servers

Console

Web terminal, any device

REST API

/api/v1, OpenAPI at /api/docs

CLI

tools/drishti.py, 22 groups

Other apps

<drishti-view>, <drishti-report>

SEVEN STAGES

One command, seven stages, one view

1

Command

TRD MX-20000001 <GO>, or plain words

2

Fetch

Sources in parallel on virtual threads

3

Classify

Which kind is this record?

4

Fingerprint

The data's shape, hashed

5

Layout

Sutra ⊕ inference, cached by shape

6

Bind

Values into panels, in parallel

7

Link

Every id navigable, time-boxed

0.6 ms
Server work for a warm view

3.0 ms
Warm view over HTTP, p99 (gate: 50 ms)

7.2 ms
Structured search over 750 trades

Key insight **The expensive part — deciding the layout — happens once per shape, not once per request.**

A swap's whole screen, in one short file

murex/irs-fixfloat.v1.sutra.yaml (trimmed)

```
rachana: 1
sutra: irs-fixfloat
doc: { summary: "Exchanges fixed for floating …" }
match: { kind: trade, priority: 10, where:
  "$.SOURCE_SYSTEM == 'MUREX' &&
   $.PRODUCT_TYPE == 'IRS_FIXFLOAT'" }
title: { pill: "Rates · Interest rate swap",
         id: $.TRADE_ID }
strip:
  - { label: Source, bind: $.SOURCE_SYSTEM }
  - { label: Maturity, bind: $.MATURITY_DATE }
panels:
  - { use: trading/deal-ticket@1 }
  - { use: trading/legs@1, with: { rate-fmt: pct4 } }
  - { use: data-provenance@1 }
  - { use: risk-links@1 }
```

- **When it applies**
  A MUREX trade whose PRODUCT_TYPE is IRS_FIXFLOAT; the highest priority match wins.
- **What it shows**
  A linked header, eight headline figures and fragments: the pack's deal ticket and legs, the UI kit's provenance and risk links; 132 pack files reuse the deal ticket.
- **Documented in itself**
  Every Sutra and fragment carries a doc: block (summary, params, examples) that the designer, the help and AI read.
- **Shipped through review**
  Rachana-EL has no side effects; a bad edit keeps the last good version live; in production a second person approves.

LIVE, END TO END

One subscription per entity, one channel per browser

Source
subscribe()

→

TopicHub
One topic per entity; latest wins

→

ViewStream
One per open view; cached layout

→

PatchDiffer
Old vs new ViewModel

→

Browser
Patches over one SSE channel

2.4 ms
Tick to frame, p50

11.2 ms
Tick to frame, p99 (gate: 40 ms)

50 ms
A topic sends at once if it has not sent within this window

1
Live channel per browser, however many views

Key insight A view does not poll and does not redraw — only the figures that changed travel, and only to the views that show them.

Several servers, one Drishti

Console
Given one server; learns the rest; fails over

→

Any node
Signed calls; your roles checked

→

Capability table
Who renders each kind, who holds its data

→

Forward or fetch
Newest pack served; masked on both nodes

→

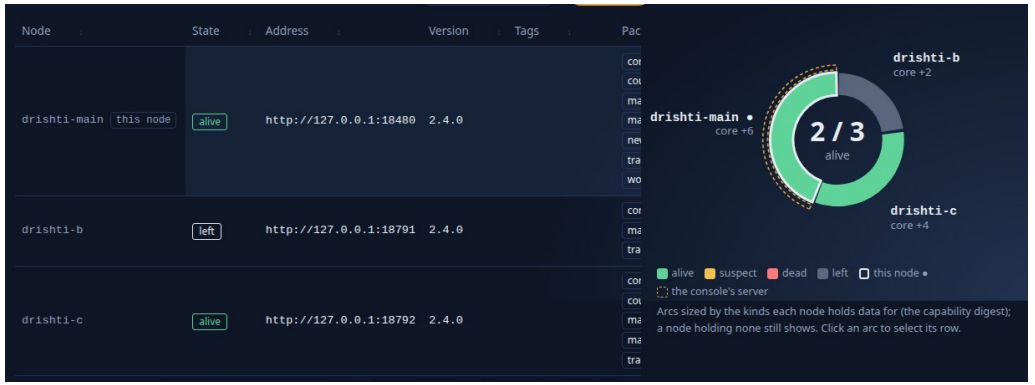
Scatter-gather
A search asks every shard, once

52 ms
Three servers converge after the third starts

0 wrong
of 146,498 answers checked in a 2-hour soak with kills

117 ms
p99 between servers on HTTP/2 (370 ms on HTTP/1.1)

13,542
Signed calls through a key rotation, none refused



Key insight Each node keeps its own packs, connectors and data — the user cannot tell which one answered; an administrator can.

See: every record, live

03

JAVA 21 OR LATER · ANY DEVICE

Drishti — overview

11
Packs: core, trading, market data and risk, counterparty risk, funds, US equities, GLEIF, economy, news

219
Sutras shipped in those packs, built from the UI kit

106
Commands, each with a machine-readable data contract

19
Source plugins behind one interface

- **A terminal on any device**

Command line, history ranges, FUNC, DATAMAP, Knowledge and WHAT over production-shaped trades from ENDUR, CALYPSO and MUREX and public data from SEC, GLEIF and N-PORT; phones.

- **Risk computed, never stored**

VaR and exposure for any scope, baskets and pre-deal what-ifs, official end-of-day records, in BI; VaR from scenario PVs in any reporting currency.

- **Build and analyse without code**

The four-mode Sutra designer and the UI kit; BI Studio with runs, statements and charts of any table; Python data prep; forms that write back under four eyes.

- **Run it, on one server or several**

Pack bundles with rollback, data-load signals, JSON Lines in and out, Delta and Iceberg side by side, a CLI for every step, Drishti*net clusters.

What is a live entity terminal?

- **One record, one dense screen**
  A swap, an FX trade, a netting set, a ticket or a gene variant, with its headline figures in a strip and its detail in up to twenty-two kinds of panel.

- **Live, not refreshed**
  When the source changes, the open screen changes: only the changed figures are sent, measured at p99 11 ms from tick to frame.

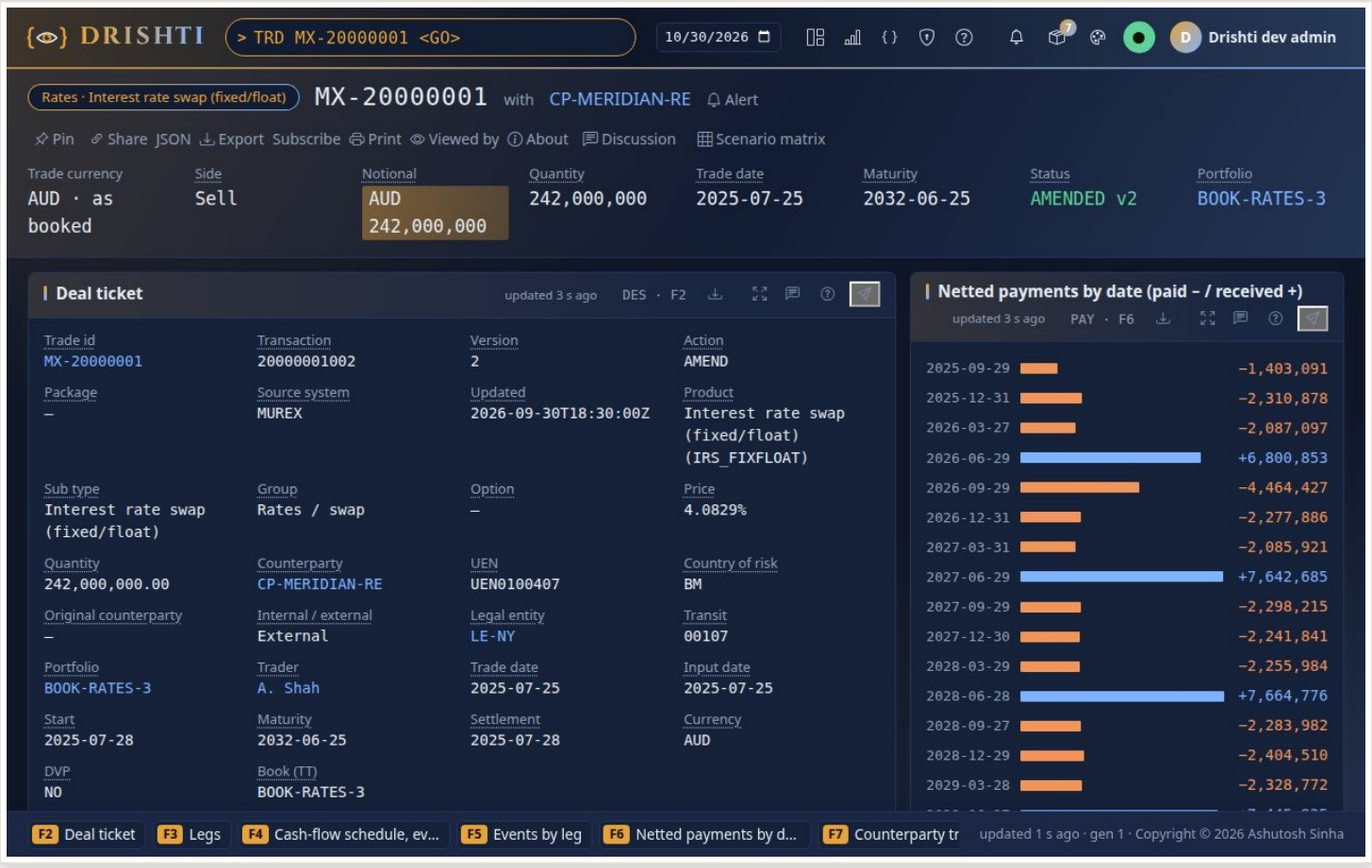
- **Linked**
  Every id on the screen is a link: a trade's counterparty, its netting set, its curve — one keystroke away, across systems.

- **Explained**
  Every view says where its data came from, how fresh it is, and why it is laid out the way it is.

Key insight **The terminal is the standard for dense, live screens — Drishti makes one for any record you have, not only the vendor's.**

THE VIEW

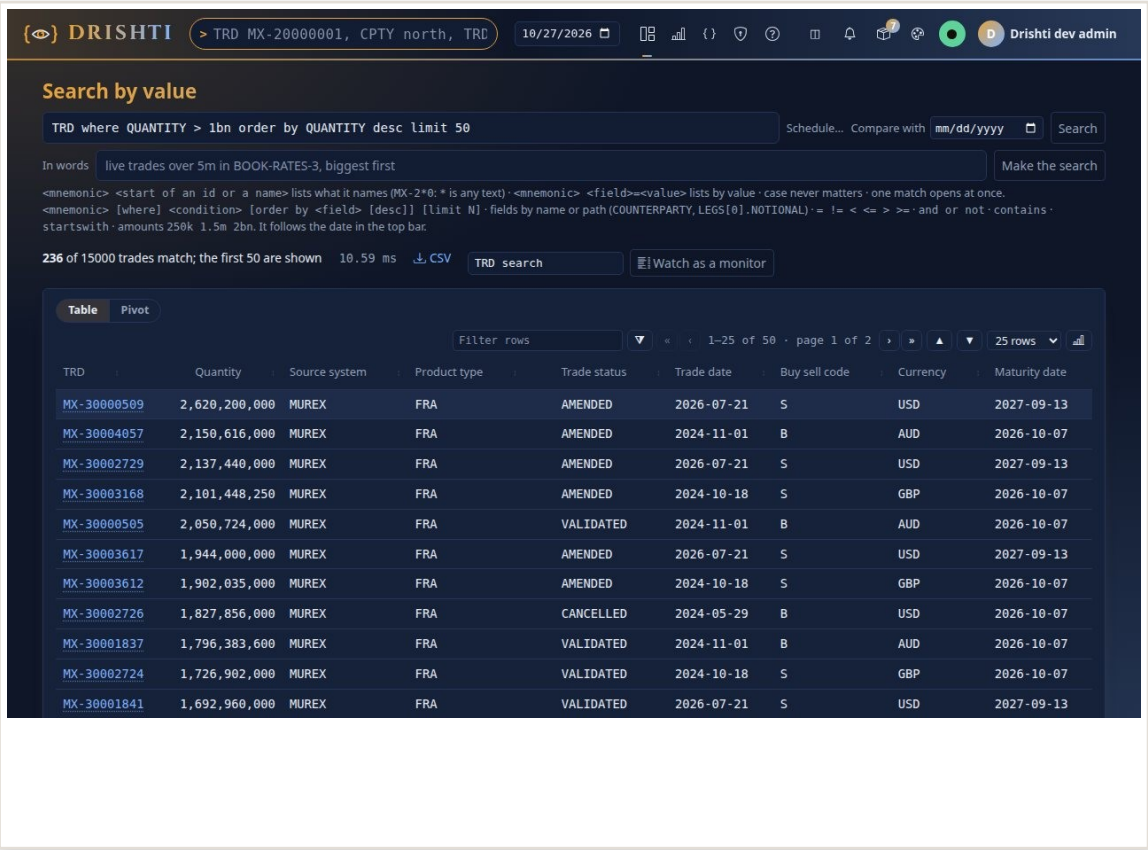
One trade, linked and explained



A MUREX interest rate swap, UPPER_SNAKE as production data is; captured on the demo lakes (tutorial 1).

- **Strip**
  Source, side, notional, quantity, dates, status and portfolio.
- **Panels by key**
  F2 deal ticket, F3 legs, F4 cash flows, F6 netted payments; Alt+L to rearrange.
- **Linked**
  Counterparty, portfolio, legal entity and trader open in one click.
- **Any day**
  The business date in the top bar; Compare shows what changed between two days.

Ask in a grammar or in plain words; pivot what comes back




```
TRD where QUANTITY > 1bn order by QUANTITY desc
amended trades over 5m in BOOK-RATES-3
FORM trading <GO>
```

134 ms

Search all 1M trades a day, 2,017 matches

29 ms

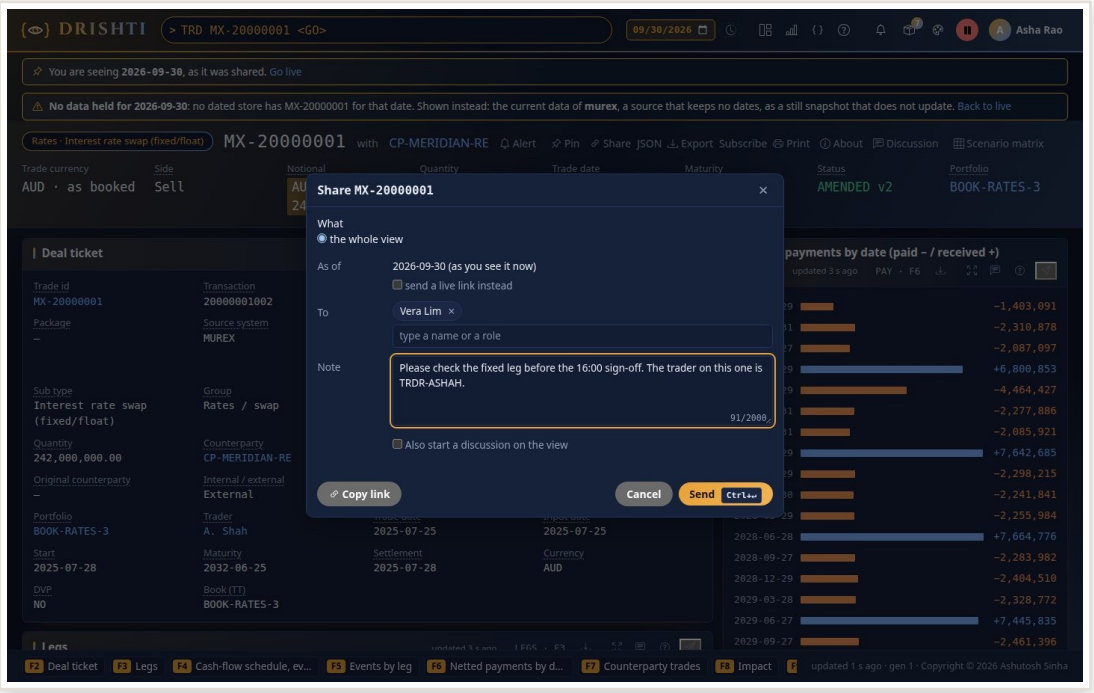
Type-ahead at 1M trades a day

Measured on a 24-core workstation, 1,000,000 trades a day for three days (5.2 GB).

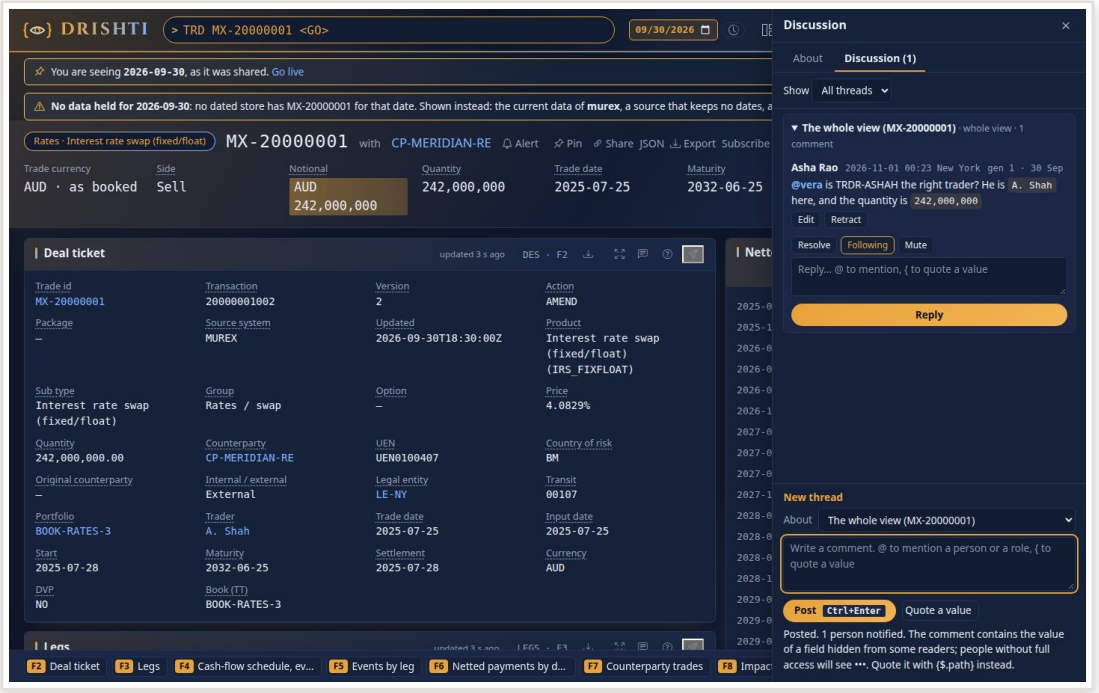
Key insight **Search is a question in the terminal's own grammar — and its answer is a view you can pivot, compare and share.**

COLLABORATION

Share what you see, discuss it, and prove what was said



Share with a note, to people or roles.



A discussion thread with @mentions and quoted fields.

Key insight **Every comment is hash-chained and sealable; retract and hide never delete — evidence, not just conversation.**

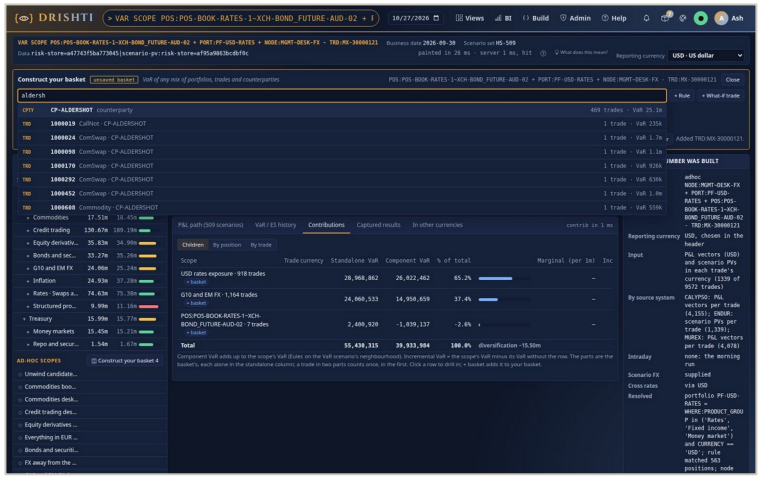
Explain the data, then let anyone build the portfolio they mean

Knowledge pages

- Market risk
- Counterparty credit risk
- Trades and products
- Entities and ownership
- Market data
- Data and contracts
- Economies
- Companies, funds and ETFs

Each with the plain idea, how the records connect, a worked example on the demo data and the questions to ask.

`WHAT <term>` 162 terms; 'What does this mean?' beside the VaR and exposure explorers, USEQ, FUND, HELD and TRACK.



Construct your basket

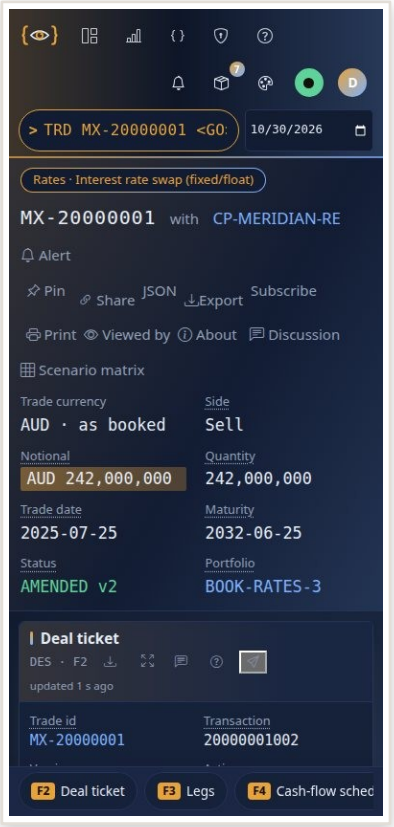
Pick portfolios, desks, books, legal entities, counterparties, positions, trades or a rule as you type; see VaR, ES and diversification live; save it. Alt+B.

The pre-deal what-if

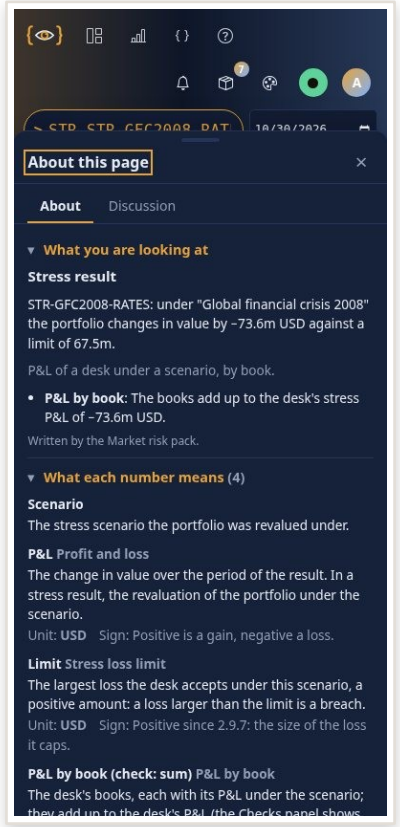
A hypothetical trade — a booked one copied and scaled, −1 its unwind — shows VaR before and after in the basket; the exposure explorer has it on the counterparty and the netting set. Exact methods run on the compute workers; trade pages link to the trade's P&L vector, VaR and exposure, or say why not.

Key insight An id that matches nothing, or a trade that is not live, is said in words — never shown as a VaR of 0.

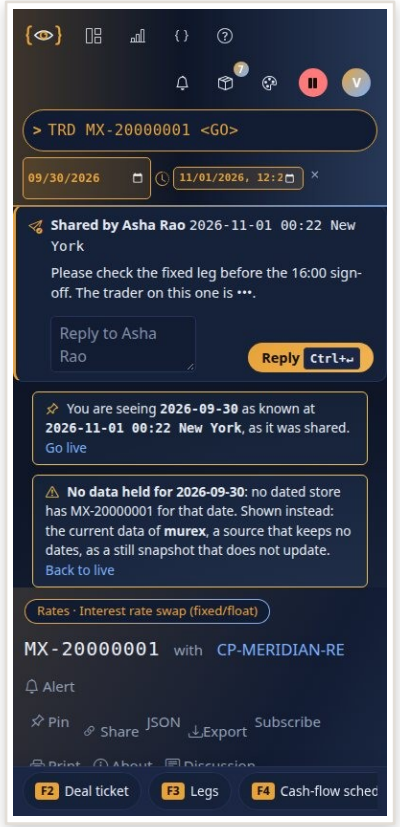
The same terminal in a pocket



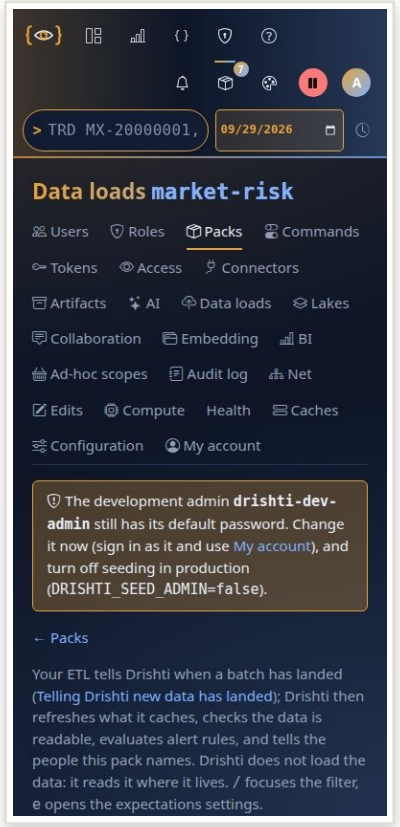
A live trade



About this page



A shared view



Data loads

- **44 px**
  Every control, on touch screens only
- **390 px**
  Every page tested at phone width
- **3 engines**
  Embedded views in Chromium, Firefox, WebKit

Must, glance, more and desk: a phone loads what matters first

must	**shown in full, loaded with the page, live** the header strip, key figures, status and breaches
glance	**a compact form: a few rows, a sparkline** ranked bars; a long table as its top 5 rows
more	**listed under More on this page; loaded on tap** contributions, cash-flow ladders, provenance
desk	**not on a phone; Open on desktop says it exists** designers, wide pivots, editing forms

Set in five layers, the most specific wins

- the panel kind's default
- the fragment's own (a UI kit piece)
- the Sutra, per panel
- the site, the pack or the edition (it can cap a heavy panel)
- the person: pin to my phone

Loading less, not hiding more

- the server binds only must and glance panels for a phone
- About this page says which layer set each tier
- the lens menus fold into one menu named after the lens

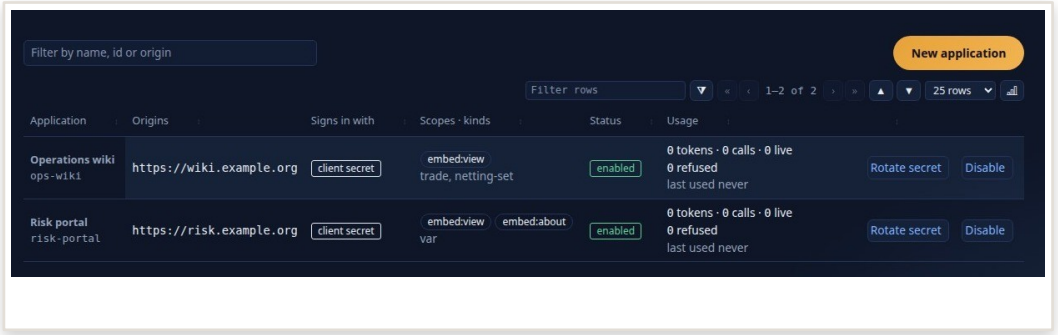
DESIGNED

Key insight **Configured, never coded: masks and roles are unchanged, and desktop is untouched.**

The same live view, inside other web applications

```
<script type="module" src=
  "/embed/v1/elements/1.4.3/drishti-elements.js">
</script>

<drishti-view server="https://drishti.example"
  kind="trade" entity="MX-20000001"></drishti-view>
```



Admin → Embedding: host apps, origins, scopes and usage.

HTML

Over the wire: the console's own templates, sanitised, in a Shadow DOM

1 stream

Shared by every element on the host page

@container

Responsive to its own box: a sidebar gets the narrow layout

Masked

Always, whatever the user's roles; every call audited

Key insight Other teams get the view, not a copy of it — still live, still masked, still audited.

Aggregate: any scope, any hierarchy

04

From a scope written as text to totals that add up, on any date

Scope as text
BANK - DESK:CREDIT, a node, a portfolio, a rule, a basket, a list of trades

→

Resolve
on the date, to a set of trade ordinals; set algebra, so a trade counts once

→

Entitle
intersected with what the viewer may see; the rest counted as hidden

→

Runs
a position is one run of ordinals; a node holds a few runs, not every trade

→

Sum
each scenario's P&L summed in float64 over the runs; parts summed once

→

Answer
VaR, ES, contributions that add up, provenance; cached by data version

What any scope can be

Word	Example	Resolves to
node	DESK:RATES, LE:NY, LL:LL-0003	the node's trades
portfolio	PORT:PF-USD-RATES	its stored definition
rule	WHERE:CURRENCY == 'EUR'	whole positions that match
list	TRD:..., POS:..., CPTY:...	those trades or positions
algebra	A + B, A - B, A & B	union, difference, both

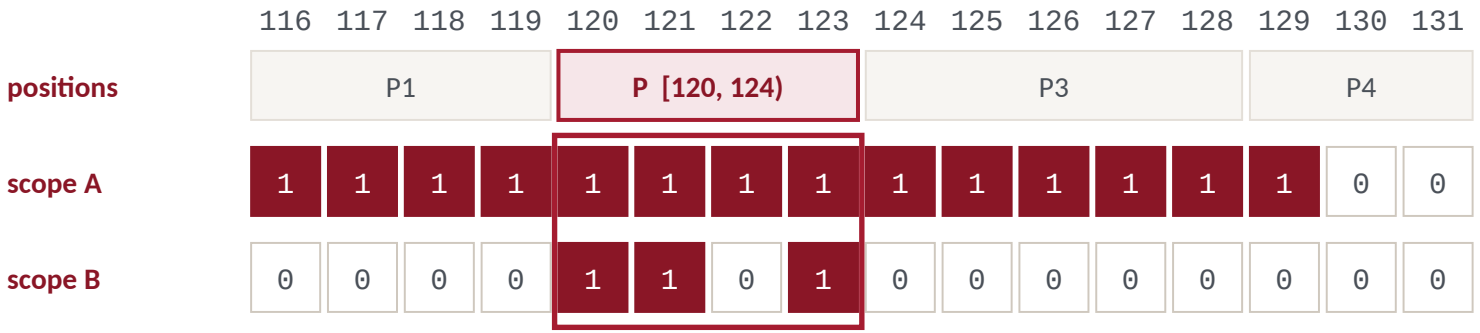
Why it scales

- **Membership is ordinals and runs:** a scope is a set of positions in the date's order, each one contiguous run of trades
- **Parts are summed once:** a desk's parts and their total come from the same pass, so the components add up exactly
- **The display limit is only for display:** a table shows the largest parts; the total is over every part
- **Shards add up:** in a cluster each member sums its shards and the sums are added in shard order

Key insight **Any set of trades is a sentence; the engine turns it into a number in milliseconds, never a stored copy.**

A scope is a set of ordinals; a position is one run

A business date's trades, numbered 0 ... N − 1, sorted by hierarchy cell and position



Is P whole in the scope? `nextClearBit(from) ≥ to`

A: the first ordinal from 120 not in A is 130; 130 ≥ 124, so A holds P whole.

B: the first ordinal from 120 not in B is 122 < 124: B holds P in part, its trades the runs [120, 122) and [123, 124).

A scan of the position's own bits, (to − from) ÷ 64 words, with nothing allocated: the cost of a position is its own trades, whatever the size of the date.

Ordinals

a date's trades numbered in hierarchy and position order: a position's trades are one contiguous run [from, to); an intraday batch's trades follow, listed with their position

A scope is bits[1]

one bit a trade, N ÷ 8 bytes: 25 KB at 200,000 trades, 290 KB at 2.3M; union, difference and intersection word by word; a node is one run

Whole or in part

whole: the position's cached standalone figures; in part: only its runs inside the scope

Cost[2]

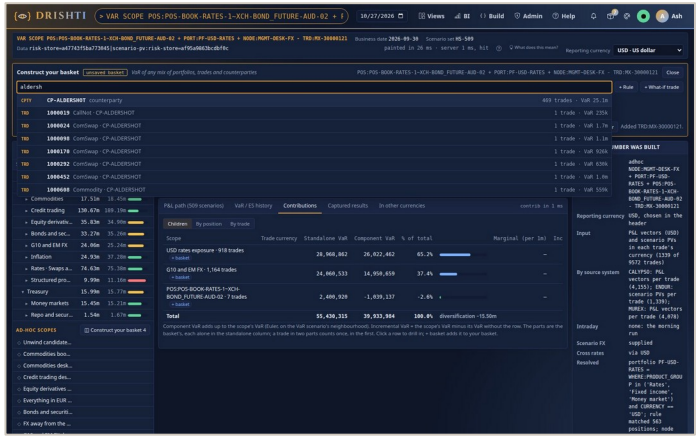
per position: its trades ÷ 64 words; per scope: its positions. Nothing grows with the date

[1–2] How these figures were derived: A-1, in the appendix (How it scales: the algorithms)

Any portfolio a person means, written as a sentence, on 2026-09-30

Scope (as typed)	Kind	VaR 99%, USD	Trades
BANK	node	112,041,646	9,572
BANK - DESK:CREDIT	ad hoc	76,605,857	8,629
DESK:RATES	node	74,633,680	—
DESK:RATES - TRD:MX-30000121	ad hoc	74,720,890	—
WHERE:CURRENCY == 'EUR' and DIRECTION == 'SHORT'	rule	134,918,298	700
NODE:MGMT~DESK-RATES + PORT:PF-USD-RATES - NODE:MGMT~BOOK-RATES-1	basket	live in the tray	—

- **One swap hedges a desk:** the rates desk without MX-30000121 has a VaR 87,210 higher; its incremental VaR is −87,210
- **A rule over positions:** 596 positions, standalone sum 231,439,042, diversification 96,520,744
- **A name that matches nothing** turns red in the tray and says why, never a VaR of 0



Construct your basket: parts added and taken away

- Save it as BASKET:owner/name; it follows its owner across a cluster
- Share it read-only; each reader sees it through their own entitlements
- Nodes and rules are dynamic, lists of ids static; every save audited

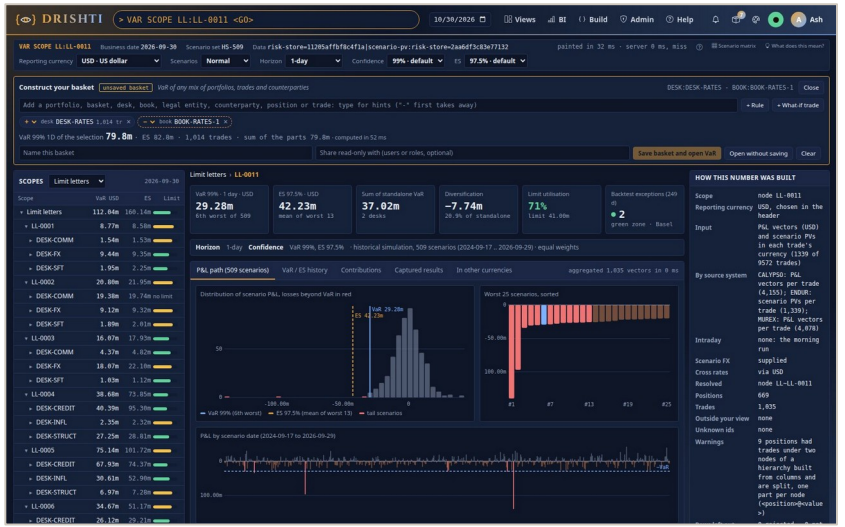
Key insight **A desk without a desk, a rule, a hand-picked basket: each is just another scope, computed when asked.**

Any hierarchy: from the delivered rows, or from the trades' columns

the market-risk pack: hierarchies

```
hierarchies:
  MGMT:  { source: risk-hierarchy,
           limits: var-limit }
  LEGAL: { source: risk-hierarchy }
  LL:    { title: Limit letters,
           levels: "LIMIT_LETTER > DESK",
           word: LL, limits: var-limit }
```

- **Rows or columns:** MGMT and LEGAL from the delivered rows; LL built from every P&L and scenario-PV row's columns
- **A word per hierarchy:** LL:LL-0003 is a scope like DESK:RATES, with its VaR limit from var-limit
- **Proved equal:** MGMT rebuilt from DIVISION > DESK > TT_BOOK gives the delivered tree exactly, to the cent at every node



The limit letters as a tree beside MGMT and LEGAL

Members are rows, not books

a letter may hold part of a book; a position under two nodes is split, and the provenance says how many

A missing column skips one tree

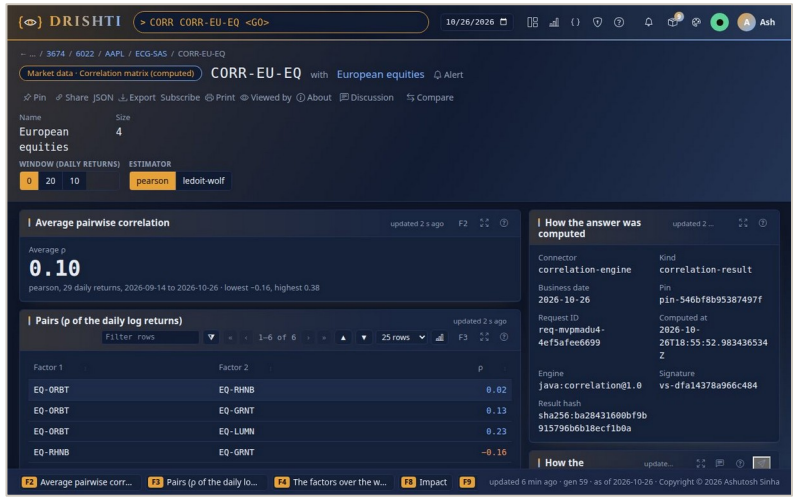
the date still loads; a warning names the column, the kind and the fix

Key insight A new way to cut the bank is configuration: a line of YAML, not a new mart or a new load.

Groupings and analytics computed from their members, not stored

Command	Computed from	Engine
LEIG · LEIC	GLEIF relationships: ownership groups (no member cap), entities per country	gleif-groupings
SIC	the sector's companies: count, revenue, net income	us-equities-groupings
ECG	member economies: GDP summed, growth GDP-weighted, inflation the median	macro-groupings
CORR	price history: Pearson or Ledoit-Wolf over a chosen window	CorrelationEngine
EQB	an equity's beta against its index, from both histories	BetaEngine
USEQ · IDX	ratios, yearly change of a fact, index totals: worked out by the pages	the pages

Every id opens the computed page; a control that feeds a computed panel (CORR's estimator and window) recomputes it at once.



CORR CORR-EU-EQ: estimator and window on the page (tutorial 27).

Every figure can be re-derived

Correlations come from the price history over the chosen window: about 0.1 on the demo, whose price series are independent. The guide reconciles each figure.

Key insight **Retire the stored copy, keep the id: users follow the same links and get a figure they can re-derive.**

Totals over every position; the row limit is for display only

A scope of P positions: one pass, in ordinal order

Each whole position
standalone VaR read from the cache: computed once per date and lens, the first time a scope holds the position, then kept

Each position held in part
the VaR of its trades inside the scope, summed on the fly

Missing from the cache
256 or more: computed in parallel on the engine's sum threads

→

Totals, over every part

`Σ standalone`
the pass on the left

`Σ component = VaR`
no pass at all: the parts partition the scope, so their components add up to its own

`diversification`
`= Σ standalone − VaR`

The row limit: max-parts, 5,000

5,000 of 18,567 positions shown (the largest by gross size); totals are over all 18,567

Position	Standalone	Component
POS-…-0001	4.21m	2.97m
POS-…-0002	3.88m	2.41m
… 4,997 more rows		
Totals, all 18,567	Σ 231.4m	Σ = VaR

- The rows: the largest by gross size, chosen with a heap of max-parts entries, O(P log max-parts), kept in ordinal order
- No total depends on the limit; the answer says truncated, total and shown

Worked: a basket of three positions

Standalone VaRs 100, 80 and 50; the basket's VaR 170.

Σ standalone 230 · Σ component 170 · diversification 60.

With a limit of 2 rows the table lists the two largest by gross size and says 2 of 3 shown; the three totals stay 230, 170 and 60.

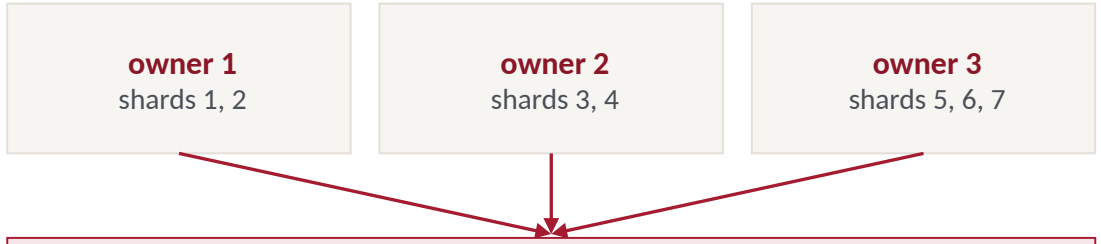
Memory and the k worst[1,2]

8 bytes a position and lens: 1.6 MB at 200,000 positions, 18 MB at 2.3M, in the one cache budget, released with its date. A standalone VaR keeps only the k worst values (k = 6 at 99% of 509) while reading the vector once.

[1–2] How these figures were derived: A-1, in the appendix (How it scales: the algorithms) Figures in the table: illustrative

Sums added shard by shard; a date's files decoded on several threads

In a sharded cluster: partial sums, added in shard order



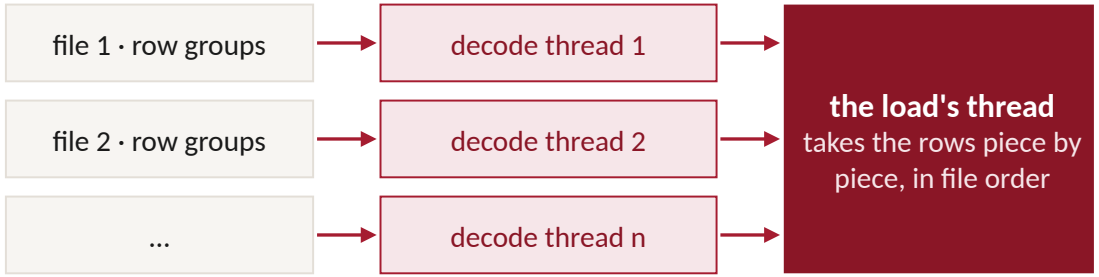
The member answering
adds each shard's partial vector in shard order, then takes VaR and ES once: the same bits as one server

- Exactly one answer a shard; no live owner of a needed shard: refused in words, never a partial VaR
- Every node's aggregate is summed bottom-up as on one server: the bank's VaR is the same bits on every member
- The totals over the parts: computed by the member answering from each position's vector, cached as on one server

Per-shard Σ standalone DESIGNED

each owner returns its shard's partial Σ standalone over the scope's whole positions from its own cache; the sums are additive across shards and, added in shard order, the same on every member

A date's first open: decoded on several threads, in file order

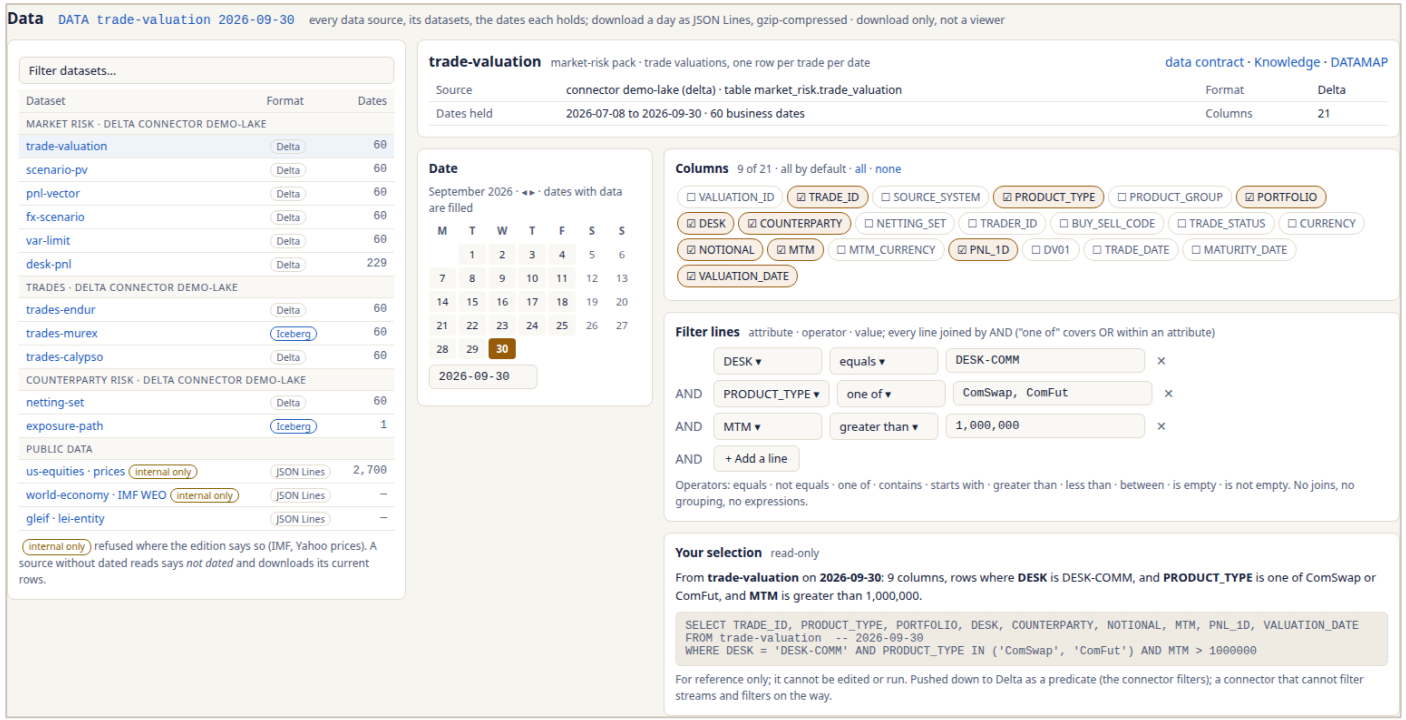


- Only the columns the load uses are read: the id, the vector, the trade's text columns
- The intake, sort, sums and hot file see exactly the rows one thread would give, in the same order: the same book to the bit
- Bounded: drishti.var.read.threads (half the cores, at most 8) and read.buffer-mb (64 MB, at most 1/32 of the heap)
- A piece that cannot be read fails the load with its error, as one thread would

Measured[1]	First open	The read
50,000 positions, one server	1.8 s	0.8 s
200,000 positions, one server	4.5 s	70–73% of CPU
1,000,000 positions, one server	26.3 s	65–69% of CPU
200,000 positions, cluster of 3	10.8 s	by the owners

[1] Measured on one workstation, 24 hardware threads: 50,000 positions at -Xmx4g (VAR_GUIDE 27a.6); 200,000 and 1,000,000: BENCHMARK.md section 4

Any dataset, any date, a few filter lines, streamed down as JSON Lines



The approved design: the datasets by source, a date, columns, filter lines

- **Every stored kind a connector serves:** its source, format (Delta, Iceberg, JSON Lines, a database), the dates held, the columns from its data contract
- **A selection, not SQL:** columns and up to 50 filter lines joined by AND: equals, one of, contains, between, is empty
- **Read back twice:** in plain words and as a SQL-like line, for reference
- **Pushed down and streamed:** comparisons go to the lake's column read; rows stream to the file, never held whole
- **Entitled and audited:** masked columns left out, internal-only data refused where the edition says so, each download one audit record
- **Three doors:** DATA <GO>, GET /api/v1/data/{kind}/rows, drishti.py data get

Key insight Taking data out is a governed, streamed download, not a mart built for one request.

An aggregate kept as a copy drifts; one computed from its members cannot

Overlapping portfolios cannot be added

28 stored portfolios overlap on purpose; a union is computed as a set first, so a position in both counts once. A table of overlapping portfolios never shows a total row.

VaR is not additive

the bank's VaR is not the sum of its desks'. A stored desk figure cannot give the bank minus a desk; the members can.

Stored copies drifted

stored ownership groups were capped at 50 members, stored countries listed 400 entities, stored correlations did not match the prices in the same lake. All are now computed.

Any date, any hierarchy

a stored aggregate exists only for the cuts someone loaded; computed, it exists for every scope on every date held.

Security stays exact

an aggregate is computed over what the viewer may see; a stored total would leak what a mask hides.

Data is not copied

the only stored data is disposable: caches and rollups. Deleting them changes no answer, only the time to it.

Key insight Fundamental data is kept as it is; whatever can be derived from it is computed when asked.

Analyse: BI and analytics

05

Business intelligence on the same data, rules and masks

Models rooted in packs	Datasets and data models built from the packs' kinds, with the relationships the packs suggest.
Three languages, one plan	A Python-like expression language, read-only SQL and Python meet in one plan, your row rules and masks applied.
Servers compute	The Java server and the console's sandbox answer every number; the browser draws; standing queries push live.
Governed	Versions and diffs, four-eyes review, certification, lineage past the connector, bundles with rollback.
Everywhere it is needed	Subscriptions and alerts per reader, PDF, PNG, Excel and CSV, and <drishti-report> in other apps.

Key insight **A report reader sees exactly the rows and masks they see on the terminal — BI cannot leak around them.**

Eleven principles hold the BI design; six matter most here

Four languages only
YAML for structure, the Drishti expression language for measures and rules, read-only Drishti SQL over models, and Python; all compile to one logical plan

The servers compute, the browser draws
every device shows the same answer; no totals, sorts or pivots in the browser

Security before arithmetic
row rules and masks are injected into the plan before optimisation; no language construct removes them

Correct by construction
each measure at its own grain; joins that repeat rows refused; currencies never summed without a conversion

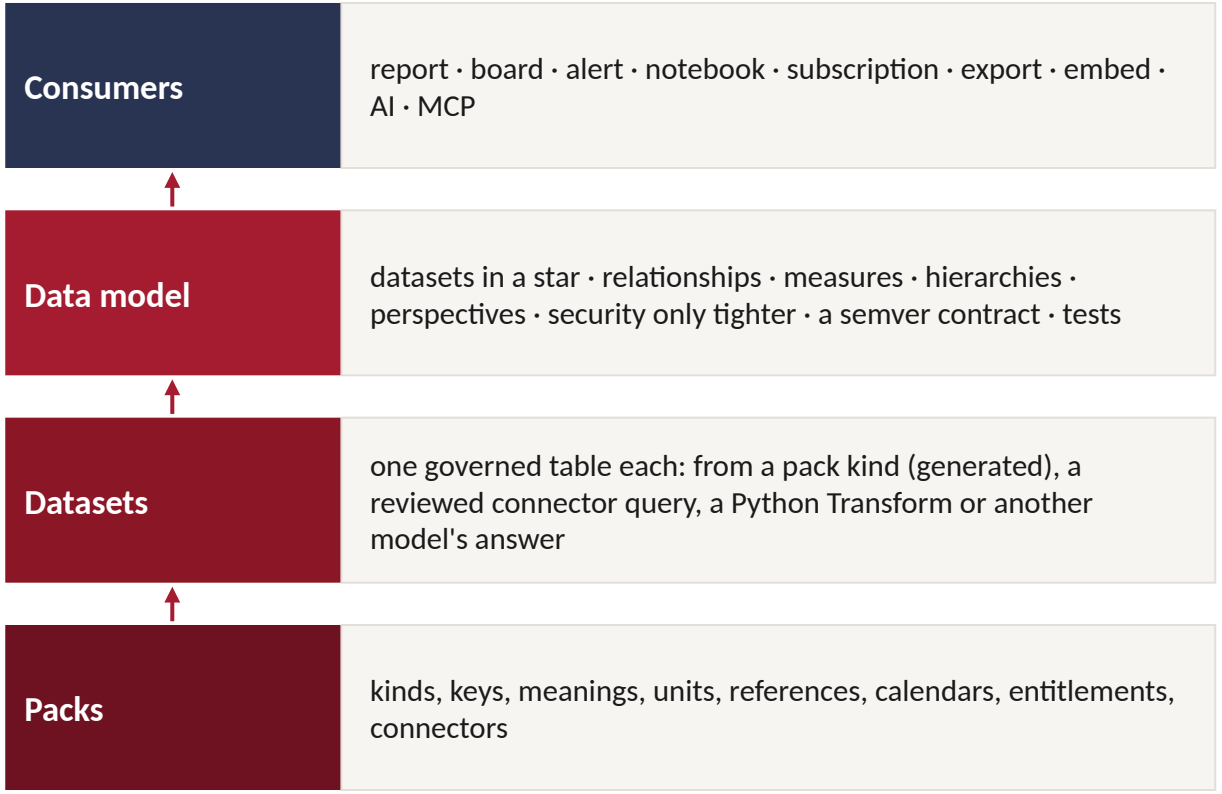
Streaming by nature
time is a parameter of every plan: now, as of a business date, the last hour; any query can be subscribed

Governed like code
versions and diffs, four-eyes review, certification, lineage, bundles between environments

Deliberately not built
A Power Query clone (Python Transforms instead) · DAX and MDX (DEL with a closed filter algebra, a published DAX mapping) · computation in the browser · raw SQL to sources · publish to the web (embedding with tokens) · a second storage engine (DuckDB over Parquet is enough)

Key insight Where Power BI and Drishti differ, it is on correctness, security, live data and governance.

Packs, datasets, one model, many consumers: a mini database

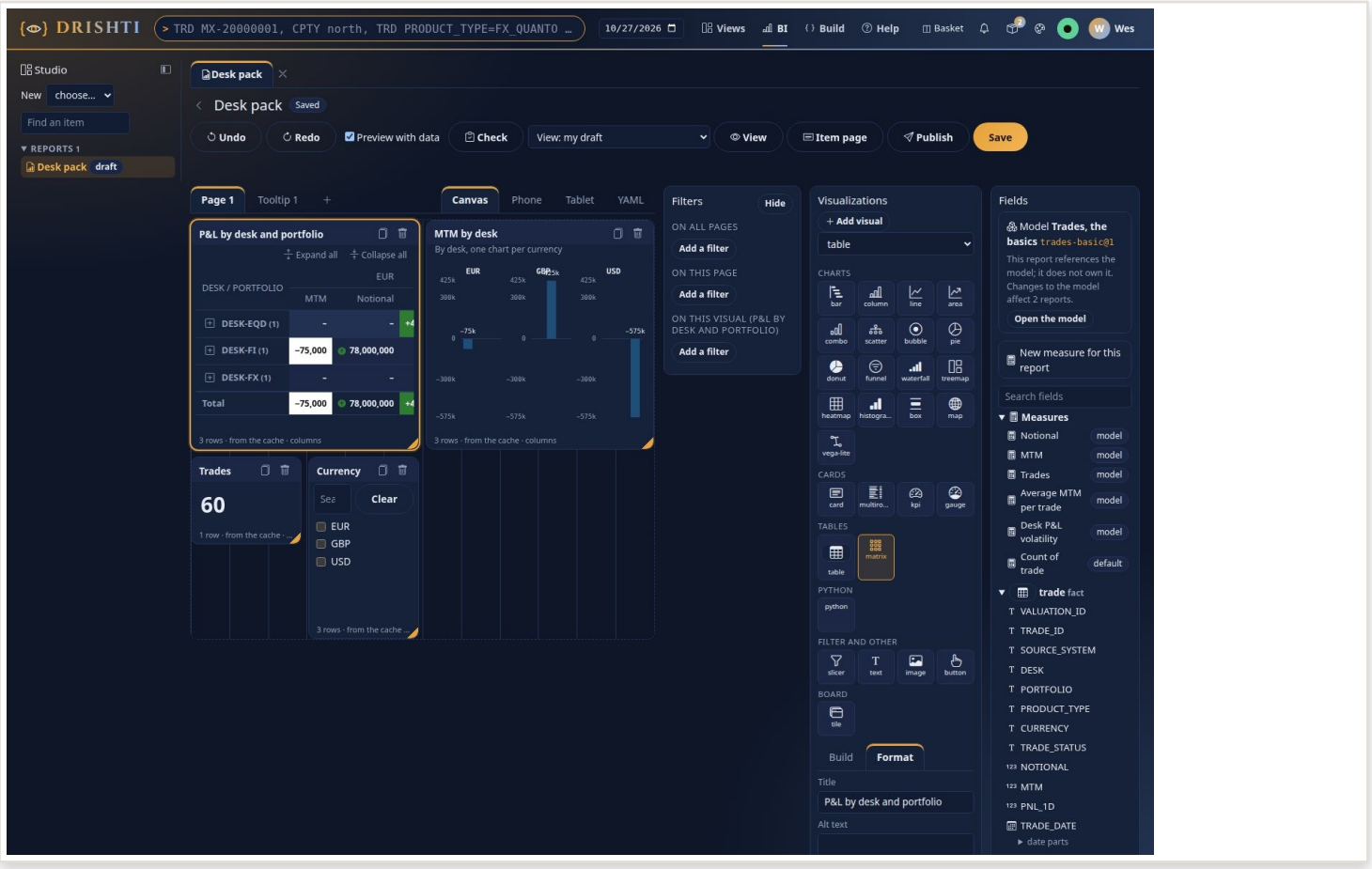


Arrows: each layer reads the one below by reference, never by copy

	Decision
D1	a dataset is one governed table; a data model is datasets, relationships and measures
D2	one model per report, board, notebook, alert, subscription and feed
D3	every pack publishes a dataset for each kind, governed with the pack
D4	the model architecture first, before calculation and authoring depth
D5	one model query language for the UI, the API, AI and MCP
D6	every dataset has a time mode: live, snapshot or history

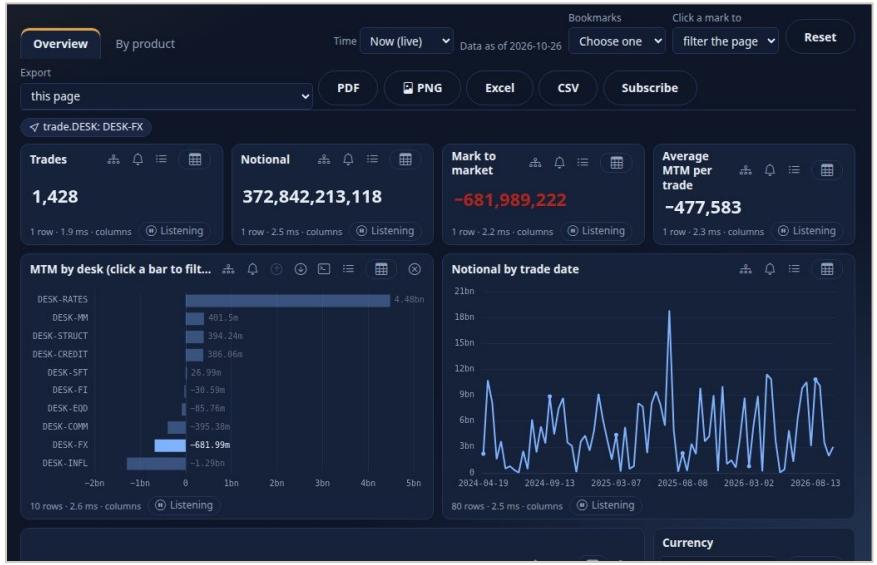
Key insight **A model is the unit of query, security, contract, cache and audit: reports reference it, never own it.**

Reports, models, boards and themes in one studio

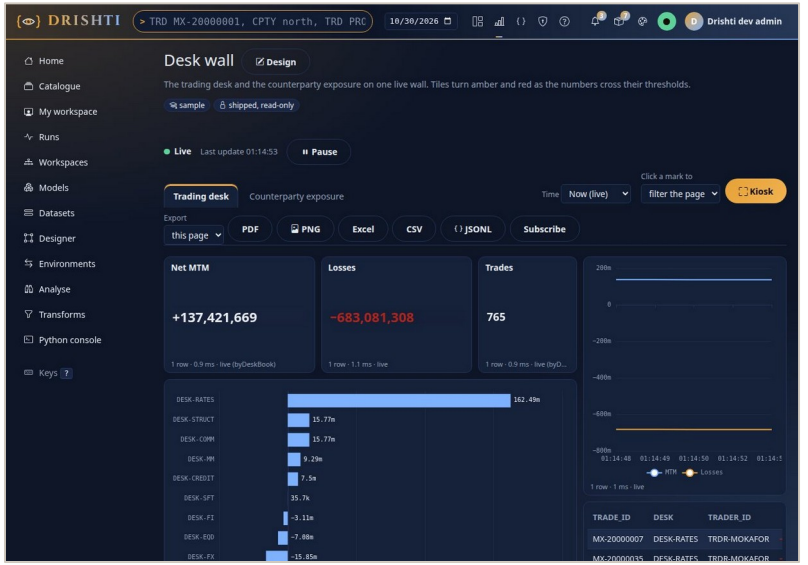


- **Explorer and tabs**
  Every item you may edit, by kind, with New; open items as tabs.
- **Problems and Query**
  The server checks the draft a second after each change; each visual's query and SQL.
- **A generated Format pane**
  Typed formats, conditional formatting, themes, tooltip pages, small multiples, the matrix.
- **Boards are reports**
  Board pages of live tiles beside the report's own visuals, with kiosk mode and alerts.

A report in the designer, cross-filtered, then a live wall for the desk



A click on the map cross-filters the page



A board of live tiles, for a wall in kiosk mode

Model
World economics: seven measures the model owns; GDP per capita is a ratio of sums, not an average of averages

Report
cards, a map drilled from the world to a region, a bar with a top-10 filter, a slicer; publish to a workspace

Live
reports follow the ticks; live tables keep their history; boards of tiles run in kiosk mode

Distribute
subscriptions as PDF, PNG, Excel or CSV, each reader on their own masks; alerts on a tile

Key insight Analysts get the BI tool they know, on the platform's own data and rules.

Every heavy question is sized before it runs, then becomes a visible run

A question
visual, MQ, SQL or statement; Prefer: respond-async

The planner
sizes it from metadata only: Delta row counts, Iceberg manifests, built layers

- **Cheap** answered at once, as before
- **Heavy** a run: Runs tab, bell when it ends, share, export, schedule
- **Very heavy** asks to confirm first
- **Impossible** refused before anything runs (DRS-9029)



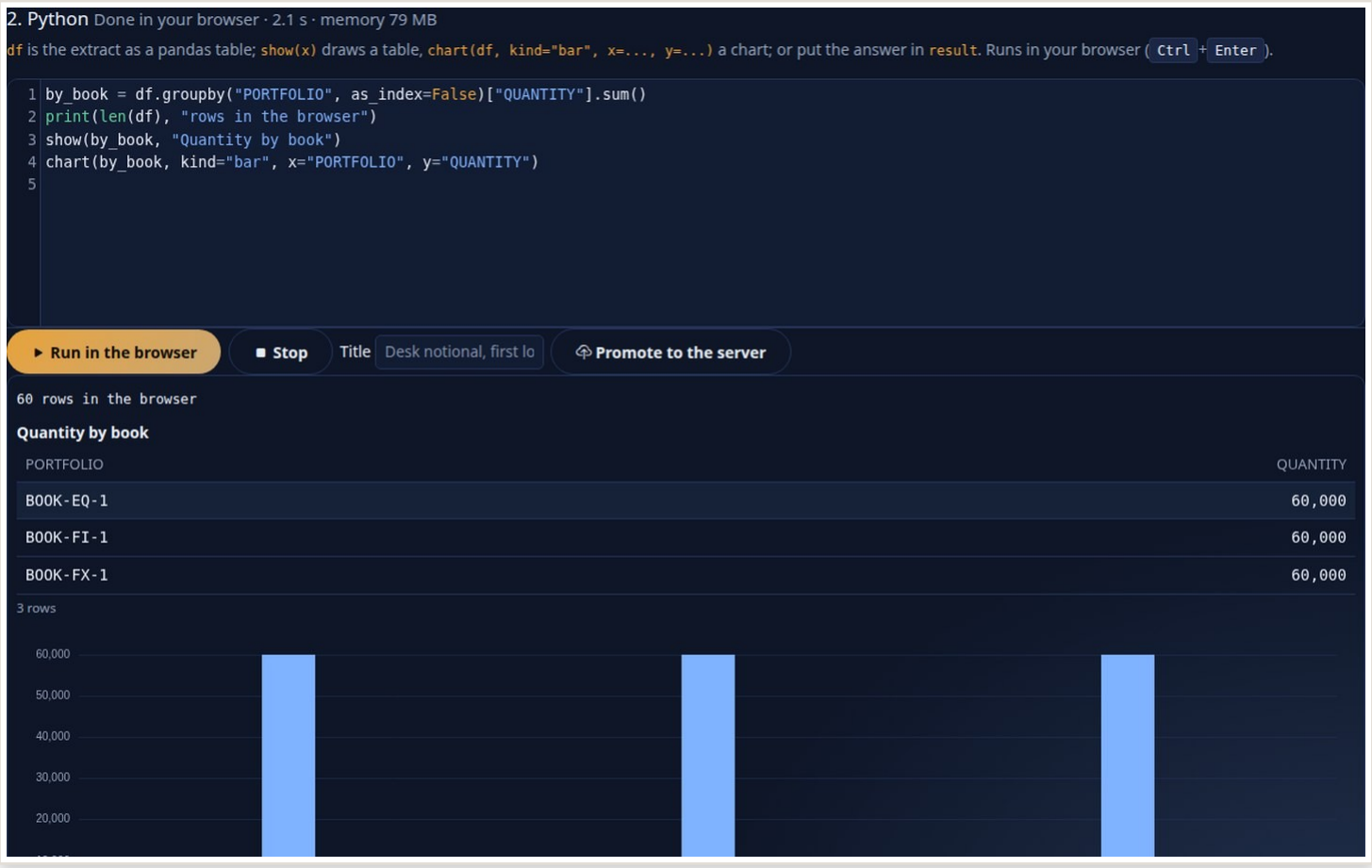
Chart any table
K on view tables, searches, pick lists, admin lists, history and BI tables: six types, Top N, stacking, two axes; PNG, SVG, CSV, Excel. Masked columns are never plotted.

0 connector reads to size a question (RunPlannerTest)

- **Fewer duplicate runs**
  An identical question joins the run already going; a finished answer is reused while the data is unchanged.
- **Masks hold**
  A run computes as its runner; a shared run is the same, re-masked or recomputed as the recipient, decided at every read.

Key insight **Refusing a year of raw trades costs a log read, not a year of builds — and an interactive slot is never held.**

Python where it belongs: on the server, or on your own rows



- **Transforms, step by step**
  Inputs read through a pack's connector as you; 21 tested snippets; a profile after each step.
- **Sandboxed, warm**
  Measures, visuals and Transforms run in a fork-server child with limits.
- **Your own Python**
  BI → Python console: pandas on a masked extract, in your browser.
- **Promote to the server**
  The same code run again on the backend, compared, kept as a notebook draft.

BI, VaR and exposure read the same data under the same entitlements

BI and analytics

models, reports, boards, Analyse with SQL and Python on a masked extract

VaR and ES

any scope, any hierarchy, any currency; the scenario matrix; the backtest

One read of the data

the same connectors, business date, data version, roles, masks and hierarchies

- BI's market-risk-var-official dataset reads the official VaR records, LIMIT_USE per node and date
- a person masked from COUNTERPARTY is masked in BI, in VaR's CPTY: scopes and in the Data view

Exposure

EE, PFE, EPE and EEPE from the paths, netting and collateral path by path

The terminal

the trade, the desk, the netting set: every number linked to its source

Key insight Analyses never disagree about who may see what, because there is only one place that decides it.

Arrow everywhere: DuckDB in the server, Polars in Python, Flight between

Layer	Choice	Why
In-memory and wire format	Arrow everywhere	Arrow IPC to the browser (as today), Arrow C Data between engines in the JVM, Arrow Flight to the Python console for large answers
Storage	Parquet in Delta or Iceberg	unchanged: every engine reads it
Ad-hoc analytics and BI	DuckDB, kept as the one embedded engine	Arrow export removes the row-by-row JDBC read; direct Delta and Iceberg scans remove a copy
Python tools, loaders, compute packs	Polars, with deltalake and pyarrow	multi-threaded JSON Lines and Parquet reads; lazy plans push filters into the scan
VaR and exposure	the Java engines, in the same role	bit-level guarantees, sharding and hot files no general engine has
Anything new and native	a sidecar over Arrow Flight	a crash or a leak costs the sidecar, not the server

Watched, not chosen: DataFusion (its Java binding is young), Velox (no embedding story), chDB (overlaps DuckDB at these sizes), Lance (a candidate for AI embeddings only).

Key insight **The aim: an ad-hoc question over millions of rows answers in tens of milliseconds.**

Compute: risk measures on demand

06

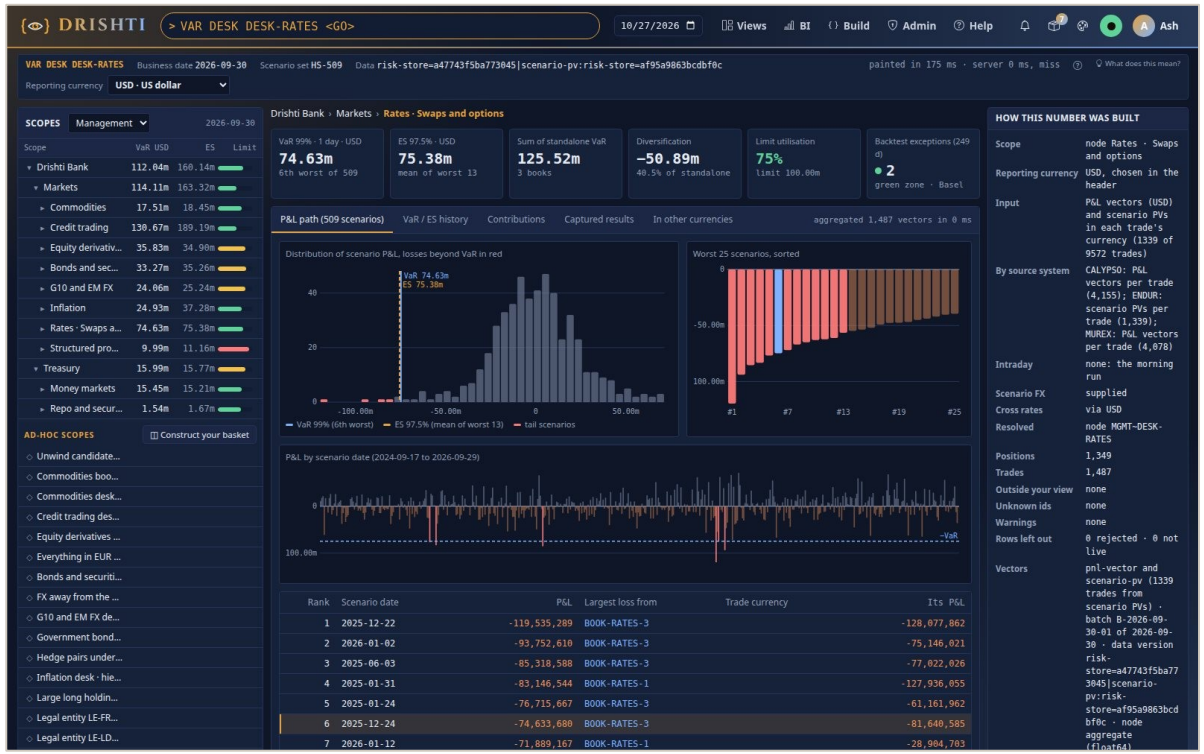
Risk measures are analytics too: computed from inputs, when asked

Measure	Computed from	For	Explained by
VaR and ES, 1-day and 10-day	P&L vectors or scenario PVs per trade or position	any scope, hierarchy, currency	the P&L path, SMAT, the Playground
Stressed VaR	today's positions under a stress period's moves	any scope	the stress window selection (SWIN)
Backtest and period statistics	the official records and the desks' actual and hypothetical P&L	any node, any period	the zone, the exceptions, the lineage
Exposure: EE, PFE, EPE, EEPE	the simulator's paths, netting and collateral path by path	the bank to one trade	the path fan, the collateral tab
Computed attributes	trade-grain inputs on the netting set and counterparty pages	every page	checks on the page, reconciled
Groupings and analytics	GLEIF entities, companies, economies, prices	groups, countries, sectors	LEIG, LEIC, SIC, ECG, CORR, PXH

What they share: inputs kept as delivered; nothing finished is stored; any scope on any date; the viewer's entitlements; an official record when a figure must not change; and, metric by metric, a playground that shows the working.

Key insight A risk measure is one more computed analytic on the bank's data, not a separate system with its own store.

VaR is a computation over the trades, never a stored number



VAR DESK: the scope tree with limits, six tiles, the P&L path and its tabs (VAR_GUIDE §3).

112.04m

bank VaR 99%, 1 day, USD, 2026-09-30

160.14m

ES 97.5%: the mean of the 13 worst

509

historical scenario days, equal weights

9,572

live trades in 7,984 derived positions

131 · 28

nodes in 3 trees · portfolios

< 1 ms

a node's VaR from its aggregate

Any scope: bank, legal entity, desk, book, limit letter, position, trade, a portfolio, a rule or a basket, joined with + − &. Stored VAR retired; official records of every node of every tree, kept 7 years.

See also **In depth: the VaR engine deck — method, data flow, inputs, scale and governance.**

10-day VaR two ways, and any offered confidence, from the same vectors



VAR BANK 10D SUM: the Horizon, Confidence and ES pickers beside the reporting currency; the tiles and the method strip follow the lens.

Method	How	Scenarios
√10, the default	every 1-day P&L × √10: VaR and ES × 3.1623	the same 509
10-day sum	each window the sum of 10 consecutive days	509 − 9 = 500

Demo bank, 99%	VaR	Rank
1-day	112.0m	6th of 509
10-day √10	354.3m	× √10
10-day sum	274.5m	5th of 500

The confidence chosen

`k = ⌈N × (1 − α)⌉`: 99% of 509 is the 6th worst, 95% the 26th, 99.5% the 3rd: another rank of the same sorted vector, nothing stored

- VaR from 95%, 97.5%, 99%, 99.5%; ES from 95%, 97.5%, 99% (configurable)
- Limit use only like for like: a 1-day 99% limit is not compared with a 10-day figure
- Backtesting stays 1-day; the official records stay 1-day at the configured confidence
- Bit-equal whatever the shards and threads that summed the 1-day vector

Figures: the demo bank on 2026-09-30, USD, the committed synthetic test data (VAR_GUIDE 18.6, VarTestDataPrecisionTest)

Stressed VaR: today's positions under a stress period's moves

The same historical simulation, on a second scenario set

Today's positions
today's trades, sizes and sensitivities

→

A stress period's moves
a fixed 12 months, e.g. 2008-07 to 2009-06: about 250 days

→

Stressed P&L vectors
the pricing system revalues today's trades under each stress day

- Not the positions as they were then: a bond bought last month has a stressed P&L for 2008-10-10
- A second set of the business date, through the same feed and contracts: SET_TYPE STRESS (or the HS-STRESS- prefix)
- Its own book and hot file, loaded on the first stressed question; never written to the lake
- VaR at any offered confidence, ES, components, contributions; 1-day and 10-day (√10 or the 241 windows)

On every official record: SVAR, SES and the stress set's id, beside VaR and ES

The 12-week average of a node's official stressed VaR, for the capital rule's max(latest, multiplier × average); official dates without one are named, never counted as zero

The demo's stress set is synthetic. 2008-like moves made by the test-data generator, not market history: the set says so and so does every stressed answer (the method strip reads **synthetic**).

Normal or Stressed, beside the other pickers

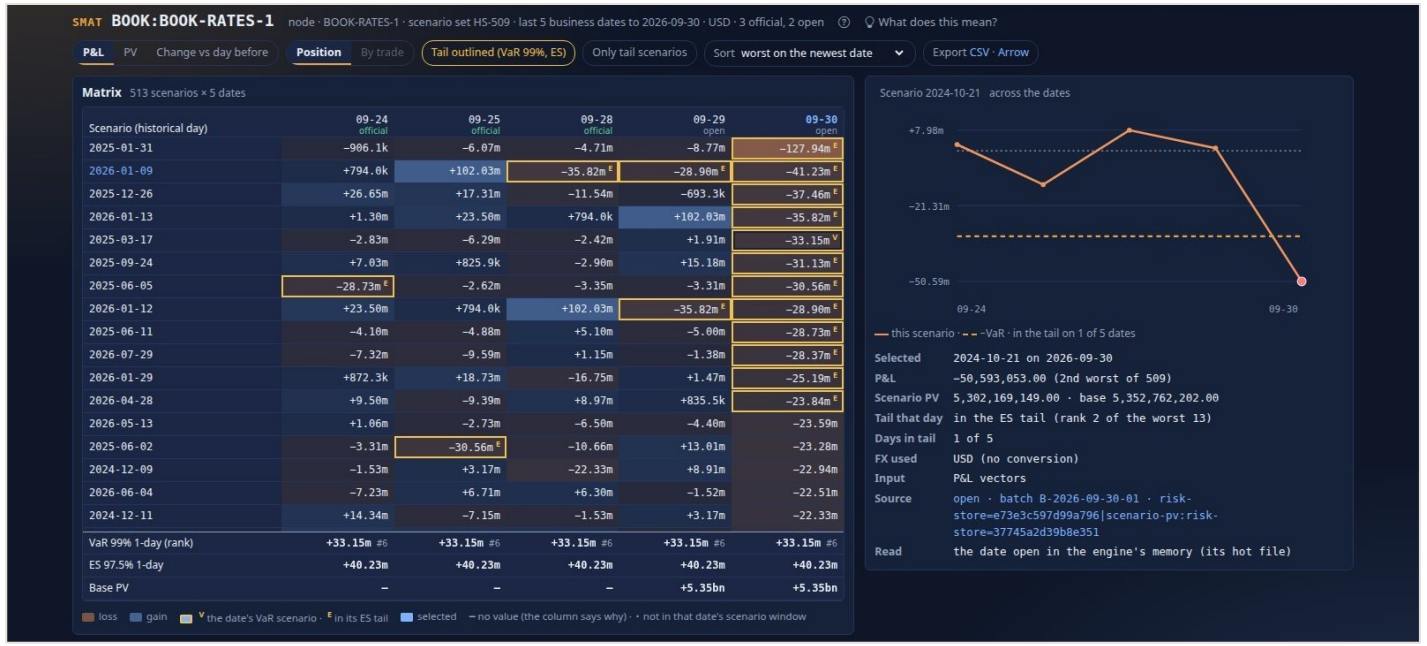


Bank, 99%	Normal	Stressed
1-day	112.0m	213.6m
10-day √10	354.3m	675.6m
10-day sum	274.5m	293.3m

- Reported in USD, the stress set's own currency
- No limit use: VaR limits are set against normal VaR
- Not backtested; intraday batches update the normal set
- Commands: VAR BANK SVAR 10D SUM; the API's set=stressed; the CLI's --stressed

Figures: the demo bank on 2026-09-30, USD; stressed on the synthetic stress set HS-STRESS-2008, 250 days (VAR_GUIDE 18.8, VarComputeFirstTest)

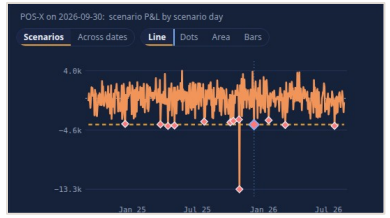
SMAT: one trade, position or node, scenario by scenario, across dates



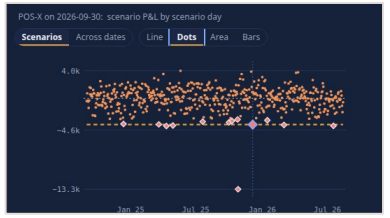
SMAT BOOK:BOOK-RATES-1 5: scenario days down, dates across; V the date's VaR scenario, E its ES tail

- Answers: is this loss scenario new, or in the tail all month? Did VaR move because the book changed or the window rolled?
- P&L, PV or the change against the day before; only the tail scenarios; By trade for a position; CSV or Arrow export
- Each date's numbers name their source: open in memory, read from the lake by id, the official record, or the engine

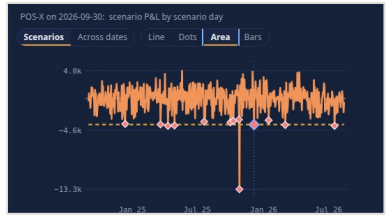
The chart: Scenarios or Across dates



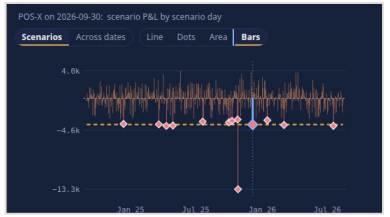
Line



Dots



Area



Bars

- One click switches the type, for both views; remembered in the browser
- The reference line: −VaR for P&L, PV at VaR for PV; the ES tail as diamonds
- The legend names exactly what is drawn

Scenario PVs or P&L, by trade or by position, from files or Kafka

Each system sends	By trade	By position
Scenario PVs in the trade's own currency	yes	yes
P&L vectors any currency	yes	yes

Mix and switch: systems mix on one date and may switch day to day; one form and grain per system per date (DRS-5210); a wrong record is refused and named (DRS-5208), never guessed; old files: drishti.py convert --set.

Intraday: only the changed trades, applied as new − old up the tree

Morning run
FULL: every live trade; the official records

→

Changed trades
files, or Kafka windows of 5 min (500 early)

→

Lake append
one intraday batch, latest COMPUTED_AT wins

→

Apply
book, every node above, every currency

One JSON Lines record, four forms

```
{"VALUE_TYPE":"PV","GRAIN":"TRADE",
 "TRADE_ID":"1000279","CURRENCY":"EUR",
 "BASE_PV":…,"PVS":[…509…],
 "COMPUTED_AT":"…","SOURCE_SYSTEM":"…"}
```

The guarantee
Incremental equals the same data loaded as one run, to the cent, on every node; replaying the Kafka topic after a restart writes nothing twice.

Cost of a batch
∝ trades changed × nodes above × currencies built — never the size of the book. Official numbers come only from full runs.

Key insight Source systems keep their own formats; Drishti turns every input into one internal form, a P&L vector per row.

The same trades, a different VaR per currency — exact to the cent

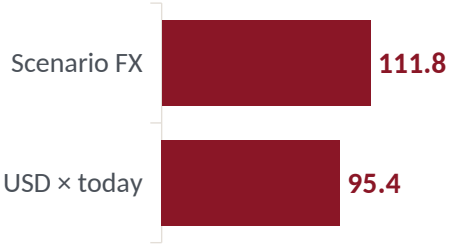


VAR DESK DESK-RATES IN EUR: every panel in EUR, the FX named.

P&L(s) = PV(s)·FX(s) − PV(base)·FX(base), every cross via USD; each node keeps its PVs summed per trade currency.

Ccy	Bank VaR 99%	Bank ES 97.5%
USD	112,041,646	160,143,809
EUR	111,837,045	137,631,546
CAD	189,801,404	230,680,413
JPY	20,883,226,318	23,578,418,999
GBP	104,139,554	125,595,958

Bank VaR, EUR m



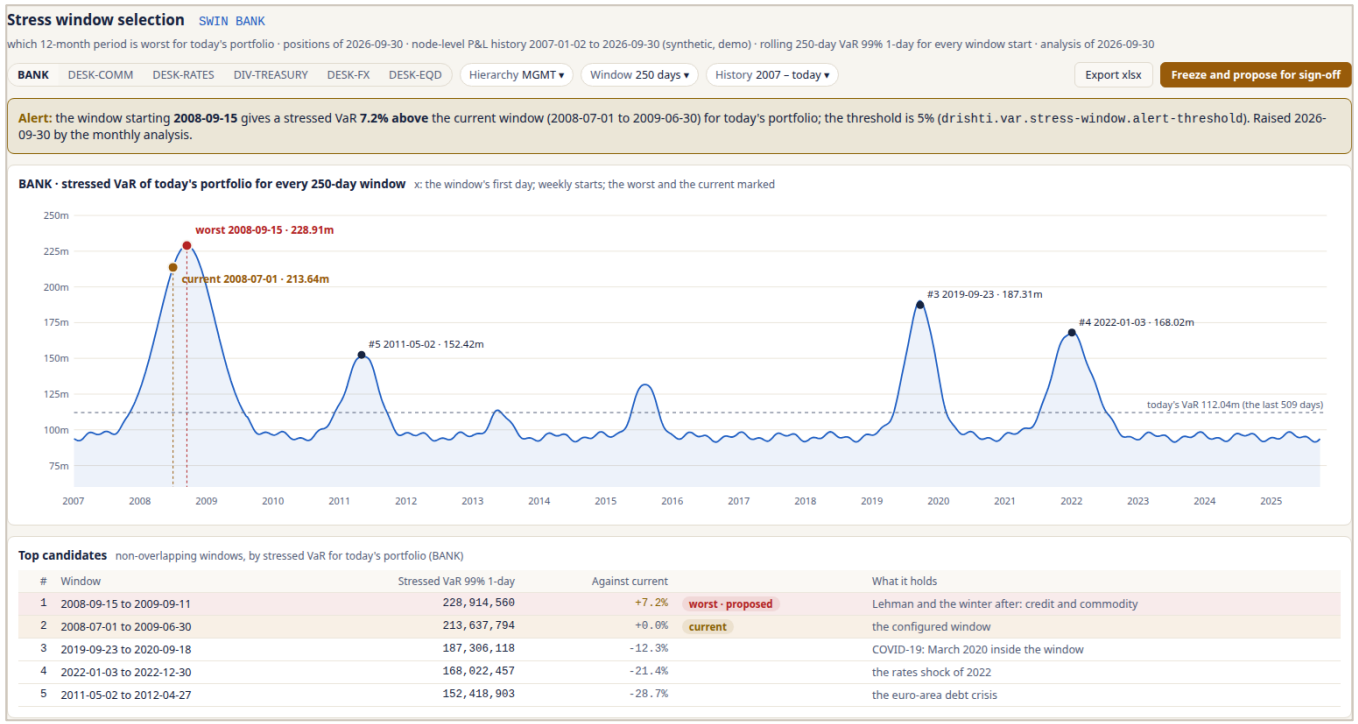
A net 3.65bn USD[1] of value moves against the euro on the scenario days.

Setting	What it does
`drishti.var.scenario-window`	all (509) or the newest N; window=250 per request, labelled, never official
`DRISHTI_VAR_CONFIDENCE`	VaR 99% by default, 95% by setting; ES 97.5%; every label follows
`drishti.var.currencies`	USD, EUR, CAD, JPY, GBP built ahead; any other currency on first ask

Key insight A euro-based owner carries a risk a dollar-based one does not: EUR VaR is not USD VaR times today's rate.

[1] the demo bank's net value on 2026-09-30, from its positions (VAR_GUIDE.md section 15); it is why EUR VaR (111.84m) differs from USD VaR at today's rate (95.4m EUR)

Every 12-month window since 2007, valued for today's portfolio



The approved design: the alert, the history, the worst and current windows

Window (demo, synthetic)	SVaR, bank	vs current
2007-11-30 to 2008-11-28	249,444,106	+16.8%
2008-07-01 to 2009-06-30 (current)	213,637,794	0.0%
2021-10-08 to 2022-10-06	187,951,560	−12.0%

- **Rolling:** each window of 250 days valued as a stressed VaR, the 3rd worst at 99%
- **Apart:** candidates at least 250 days from each other, so each is a different period
- **Alert:** when the worst beats the current by more than 5%
- **By node:** each node's own worst window; one is chosen for the bank
- **Frozen and approved:** the choice is frozen with its inputs and approved by a second person

Key insight **The stress period is chosen by computation and approved by people, not carried forward by habit.**

250 days of VaR against actual and hypothetical P&L, by node



The approved design: 250 days, the zone card and the multiplier

3 and 3

exceptions at 99% against actual and against hypothetical P&L, 250 days to 2026-09-30

green · 3.00

the bank's zone and multiplier (the US table)

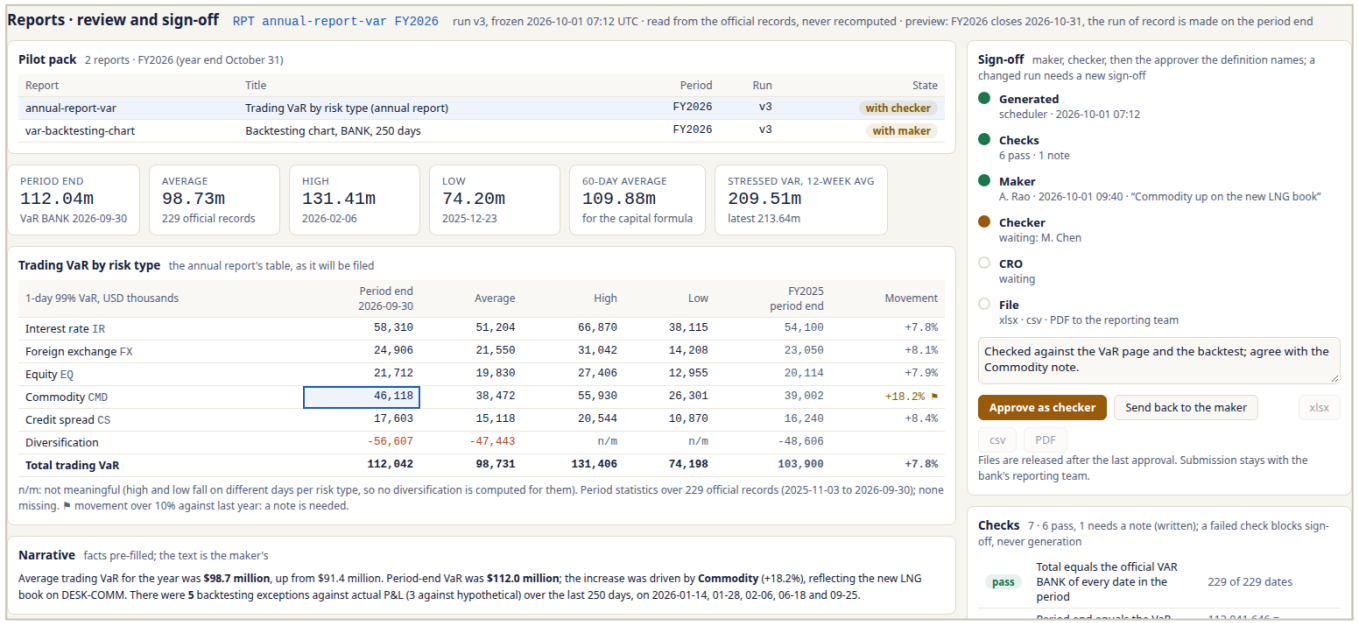
amber · 3.50

the inflation and money-market desks, 6 exceptions each at 99%

- **Three P&Ls:** actual and hypothetical from the bank's desk P&L feed; risk-theoretical from the official record
- **The rule as configured:** green 0–4, amber 5–9, red 10+; the multiplier table, the count read (US: actual; OSFI: the higher); Kupiec and Christoffersen on the actual P&L; the desks that drove each exception

Key insight VBT BANK <GO>: the backtest is computed from the records when opened, for any node.

Period statistics, a report filled cell by cell, checked, frozen and signed



The approved design: the report, its checks, the lineage of a cell, the sign-off

FY2026, BANK, from the official records

Statistic	VaR 99%
period end, 2026-09-30	112,041,646
average, 228 records	151,812,526
high, 2026-01-13	167,372,692
low, 2025-12-10	87,874,268
60-day average	166,450,508

On the demo's synthetic history; every seeded record says so, and is never published.

Definition one YAML file a report → **Run** every cell filled, with lineage → **Checks** a failure blocks sign-off → **Sign-off** maker, checker, approver → **Files** xlsx, csv, PDF

Key insight Reports are produced automatically and signed by people; a restatement makes a new run, never an edit.

Exposure computed for any scope, down to a trade, collateral path by path

Paths
every trade's MtM on every path and date (exposure-path)

→

Net
per netting set, path by path

→

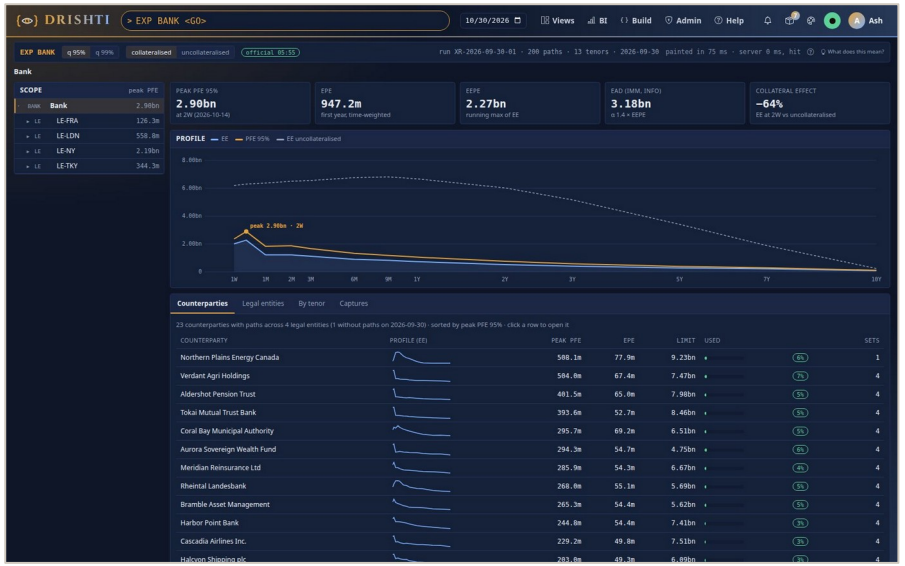
Collateral
each CSA: threshold, MTA, IA, rounding, margin period of risk

→

Measures
EE, ENE, effective EE, PFE 95/99, EPE, EEPE, peak PFE, EAD

→

Any scope
NSET, CPTY, GROUP, LE, DESK, BOOK, TRD, BANK; + − &



74
netting sets on the demo, 8,022 live trades

85.1m
peak PFE, NS-ALDERSHOT-LDN

61 ms
the bank of 9,901 sets (measured)

0.06 ms
one netting set (measured)

Explorer, pre-deal what-if (exact on the run's own paths, or a marked proxy), captures, official results, tiered retention: trade paths 20 days and month-ends; statistics 7 years.

Key insight The stored peaks summed to 92.6bn; computed with the CSAs on the same sets, 7.6bn — the stored profile is retired.

Derived on demand from trade-grain inputs, never stored

Trade valuations
live status, a netting set, maturity not before the date

→

Derived kinds
built once per date and data version, shared

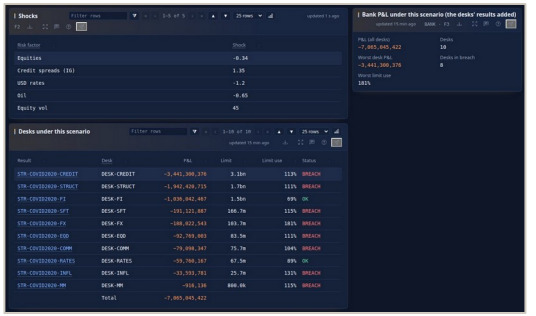
→

Beside the stored
computed, stored and their difference on the page

Command	One page per	What it computes
NSMTM	netting set	net MtM, trades, gross ±, netting benefit, net-to-gross, uncollateralised CE, by product group
CPMTM	counterparty	net MtM, its netting sets, Σ max(set net, 0) as uncollateralised CE
SSPNL	stress scenario	bank P&L (Σ desks), worst desk, breaches, limit use per desk

Live, one rule everywhere: a live-status trade past its maturity is in no total; it is counted and listed as a data-quality warning. Every total says its sum currency (USD).





Key insight **A figure on the page is a computation over the inputs, shown beside the stored copy it will replace.**

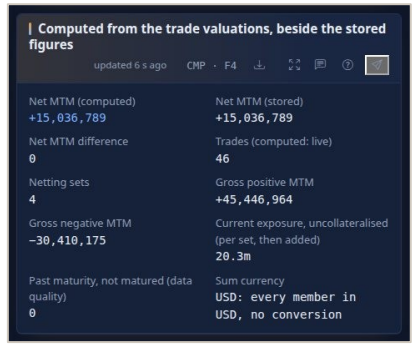
Checks on the pages; exposure on its real sources



Collateral balance: held, posted and net after haircut, from the positions



Credit limit: headroom, utilisation = used ÷ limit, and its difference



Counterparty: computed from the trade valuations, beside the stored

The exposure engine reads	From
A netting set's members and MtM	trade-valuation, live status
Collateral held today	the positions' value after haircut
Holdings of the 76 balances	6,229.6m after haircut (6,304.4m market value)

BANK, 2026-09-30, USD	Computed
Peak PFE 95%	2,897.6m
EPE	947.2m
CE	1,586.8m

Key insight Exposure reads its fundamental sources: live trades' valuations and collateral after its haircut.

Explain and trust

07

Every figure as real formulas that equal the engine to the cent

1 Open
PLAY VAR <scope> or Playground on the VaR page: the same date, currency, horizon, confidence

2 Grid
positions across, scenarios down; the formula bar shows each cell's formula

change a PV in the sheet and every figure moves; the check cells say whether it still equals the engine

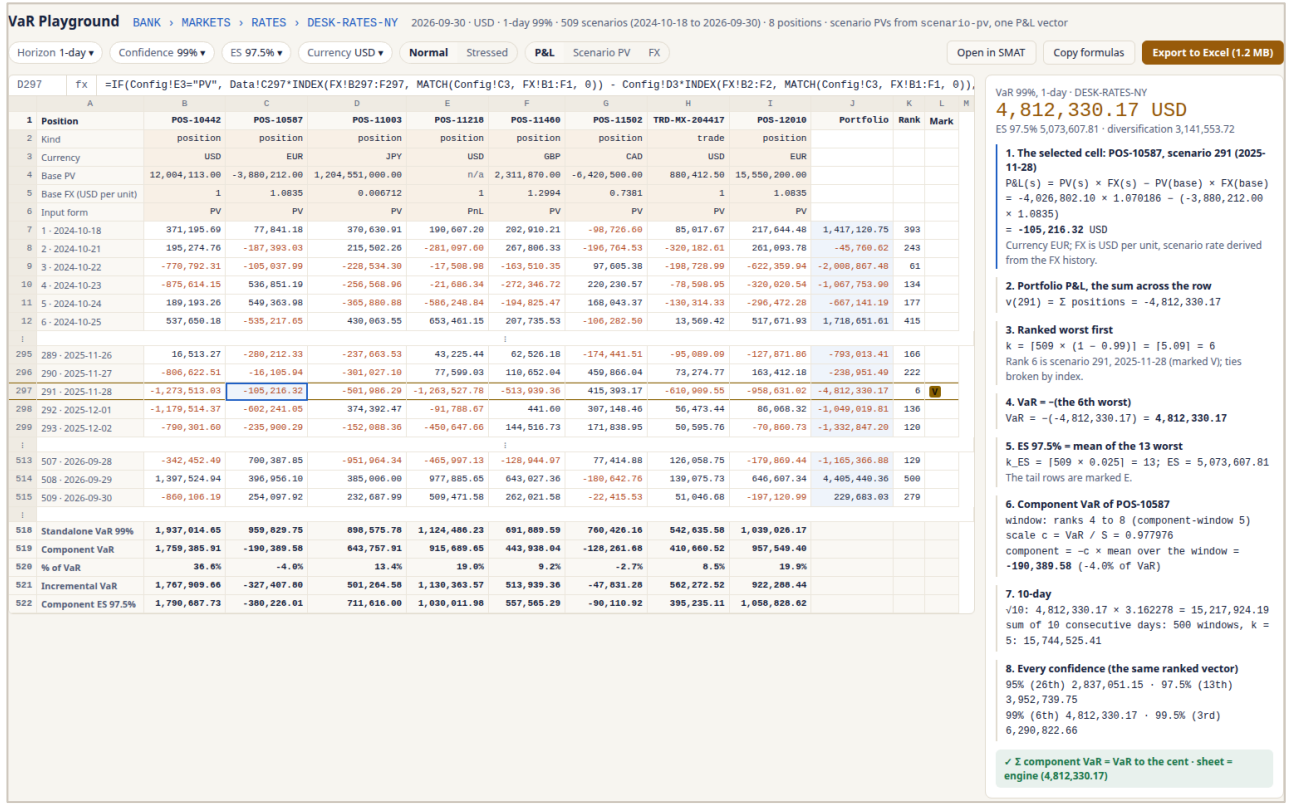
The engine's figure
VaR 4,812,330.17 USD = sheet

3 Explain
the steps in words with the real numbers: PV × FX, the sum, the rank, k, VaR, ES

4 Workbook
an .xlsx of real formulas: Main, Config, Data, FX, PnL, Explain, Provenance; check cells

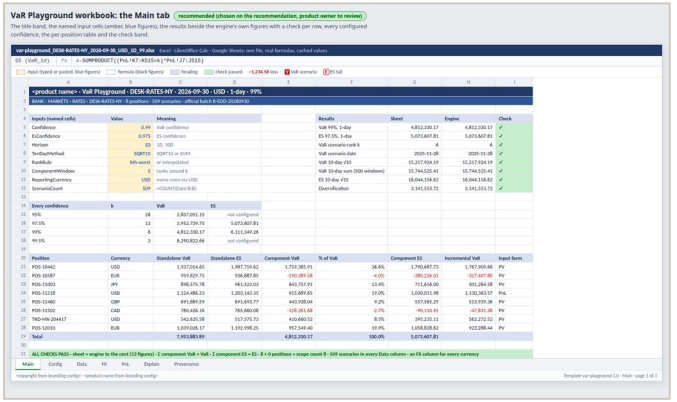
Key insight Trust us becomes check it yourself: offline, in Excel, LibreOffice or Google Sheets.

The grid with its explanation on screen; the same calculation as a workbook



	Position	POS-10442	POS-10587	POS-11003	POS-11218	POS-11460	POS-11502	TRD-MX-204417	POS-12010	Portfolio	Rank	Mark
2	Kind	position	position	position	position	position	position	trade	position			
3	Currency	USD	EUR	JPY	USD	GBP	CAD	USD	EUR			
4	Base PV	12,004,113.00	-3,880,212.00	1,204,551,000.00	n/a	2,311,870.00	-6,420,500.00	880,412.50	15,550,200.00			
5	Base FX (USD per unit)	1	1.0835	0.006712	1	1.2994	0.7381	1	1.0835			
6	Input form	PV	PV	PV	PnL	PV	PV	PV	PV			
7	1 · 2024-10-18	371,195.69	77,841.18	370,630.91	190,607.20	202,910.21	-98,726.60	85,017.67	217,644.48	1,417,120.75	393	
8	2 · 2024-10-21	195,274.76	-187,393.03	215,502.26	-281,097.60	267,806.33	-196,764.53	-320,182.61	261,093.78	-45,760.62	243	
9	3 · 2024-10-22	-770,792.31	-105,037.99	-228,534.30	-17,508.98	-163,510.35	97,605.38	-198,728.99	-622,359.94	-2,008,867.48	61	
10	4 · 2024-10-23	-875,614.15	536,851.19	-256,568.96	-21,686.34	-272,346.72	220,230.57	-78,598.95	-320,020.54	-1,067,753.90	134	
11	5 · 2024-10-24	189,193.26	549,363.98	-365,880.88	-586,248.84	-194,825.47	168,043.37	-130,314.33	-296,472.28	-667,141.19	177	
12	6 · 2024-10-25	537,650.18	-535,217.65	430,063.55	653,461.15	207,735.53	-106,282.50	13,569.42	517,671.93	1,718,651.61	415	
295	289 · 2025-11-26	16,513.27	-280,212.33	-237,663.53	43,225.44	62,526.18	-174,441.51	-95,089.09	-127,871.86	-793,013.41	166	
296	290 · 2025-11-27	-806,622.51	-16,105.94	-301,027.10	77,599.03	110,652.04	459,866.04	73,274.77	163,412.18	-238,951.49	222	
297	291 · 2025-11-28	-1,273,513.03	-105,216.32	-501,986.29	-1,263,527.78	-513,939.36	415,393.17	-610,909.55	-958,631.02	-4,812,330.17	6	V
298	292 · 2025-12-01	-1,179,514.37	-602,241.05	374,392.47	-91,788.67	441.60	307,148.46	56,473.44	86,068.32	-1,049,019.81	136	
299	293 · 2025-12-02	-790,301.60	-235,900.29	-152,088.36	-450,647.66	144,516.73	171,838.95	50,595.76	-70,860.73	-1,332,847.20	120	
513	507 · 2026-09-28	-342,452.49	700,387.85	-951,964.34	-465,997.13	-128,944.97	77,414.88	126,058.75	-179,869.44	-1,165,366.88	129	
514	508 · 2026-09-29	1,397,524.94	396,956.10	385,006.00	977,885.65	643,027.36	-180,642.76	139,075.73	646,607.34	4,405,440.36	500	
515	509 · 2026-09-30	-860,106.19	254,097.92	232,687.99	509,471.58	262,021.58	-22,415.53	51,046.68	-197,120.99	229,683.03	279	
518	Standalone VaR 99%	1,937,014.65	959,829.75	898,575.78	1,124,486.23	691,889.59	760,426.16	542,635.58	1,039,026.17			
519	Component VaR	1,759,385.91	-190,389.58	643,757.91	915,689.65	443,938.04	-128,261.68	410,660.52	957,549.40			
520	% of VaR	36.6%	-4.0%	13.4%	19.0%	9.2%	-2.7%	8.5%	19.9%			
521	Incremental VaR	1,767,909.66	-327,407.80	501,264.58	1,130,363.57	513,939.36	-47,831.28	562,272.52	922,288.44			
522	Component ES 97.5%	1,790,687.73	-380,226.01	711,616.00	1,030,011.98	557,565.29	-90,110.92	395,235.11	1,058,828.62			

The approved design: grid, formula bar and the explanation beside it



The workbook's Main tab: results beside the engine's, a check each

- **Up to 500 positions** on screen; the rest of the scope in one column, so the sheet's VaR is the page's
- **Export 100, 500, 1,000, 10,000 or All,** sized before the download; positions as rows beyond 16,384 columns
- **Every confidence and ES,** components, incremental VaR, both 10-day methods

Key insight **The workbook is built from the engine's own vectors, read as the person, and proved equal in every drill.**

Every view explains itself, in five layers



- **What you are looking at**
  From the pack's own help text and glossary.
- **Where the data came from**
  Source, health, generation and freshness.
- **Why it looks like this**
  Which Sutra matched, which lost, what inference added.
- **Masked for you**
  Says which fields your role cannot see.

One contract per command, and everything else is generated from it

config/packs/<pack>/contracts/<kind>.contract.yaml

```
kind · mnemonic · grain · key
columns: name, type, unit,
  meaning, source, example
delivery · validation
entitlements · retention
kind of data: fundamental,
  engine input, own, computed
```

drishti.py contract list, show, check, docs, bootstrap; kept current by tests

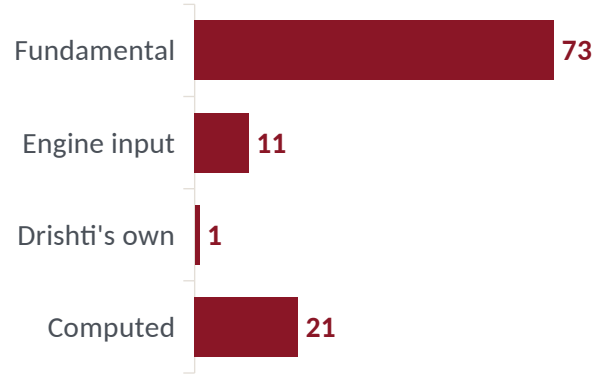
Generated pages docs/contracts and the COMMAND_CATALOG: every command, what it does, an example

Every pack's guide ends with its contracts and where each kind's data comes from

DATAMAP joins between kinds drawn from the contracts and checked against the data

FUNC Views → Functions: every command you may use, by pack; FUNC <words> filters

The 106 commands, by kind of data



A computed figure is never stored in a fundamental or input table: a guard test fails the build.

Admin → Commands
per command

Switch off
any command, for everyone, with a reason

→

Stored and gossiped
the Drishti database, every server

→

'Not active' everywhere
page, API, CLI, search, exports, BI, AI: 403 DRS-1016

Key insight **A command's meaning lives in one file; the catalog, the guides, the map and the switches read it.**

Official results other systems can read — and edits that never overwrite

Official end-of-day results: VaR and exposure

Morning full run
every node of every hierarchy, as the server

→

Captured, append-only
a restatement adds a result version; kept 7 years

→

risk-results
var-official, exposure-official, -cpty: Delta or Iceberg, a folder or S3; JSONL beside

→

BI reads them
read-only kinds VOFF and XOFC; the Market risk VaR model and report

Write-back: forms save through an edit log, with four eyes (off by default)

Submit…
the draft and a review sheet: what you saw, what is there now, your change

→

Approve
a change request; the approver is never you

→

Edit log
appended to its record's lane, acknowledged after commit; 64 lanes, epoch-fenced

→

Apply
overlay, owned value or the source through an outbox; rights re-checked

→

Recompute
reactive recomputes and live updates after each change

Conflicts are named, never overwritten: a change whose fields moved since you saw them is stopped with the three-way difference in words; DRS-9301 to 9306; Review → Changes; tutorial 24.

Key insight Engines never read the published results (a guard test): the record is for reporting, the lake stays the input.

RECONCILED AND GUARDED

55 of 76 netting sets agree at once; the other 21 are explained to the dollar

55 / 76
netting sets: derived net MtM equals the stored, at once

827
live-status trades past maturity in the other 21 (MtM 16.7m)

76 / 76
agree to the dollar with those trades added back

112,041,646
VAR BANK, USD, 2026-09-30: unchanged

Retired field	Why	Where it lives
`netting-set.UTILISATION`	a stored ratio the page can compute	dropped; BI's card shows trades, largest net MtM
`risk-hierarchy.LIMIT_VAR`	VaR limits kept in two places	only in var-limit (same values)
`portfolio.LIMIT_VAR`	the same	var-limit, hierarchy PORTFOLIO

The guard

`tools/contracts.policy.yaml` removed: fails the build if a retired field returns.

No VaR figure moved

Trades, valuations, vectors, scenario PVs and paths are untouched by the computed attributes.

PLANNED Retiring the stored netting-set and counterparty figures (NET_MTM, TRADES[], COLLATERAL, LIMIT) with the netting-set engine.

Key insight Each difference has a named cause; a retired field cannot come back without failing the build.

A faster path must give today's answers, and prove it at start-up

Path	Rule	Why this rule
Today's code (the baseline)	bit-exact, run to run	the reference every other path is measured against
SIMD sums (the Vector API)	bit-exact against a plain-Java twin that adds in the same tile and lane order	same order, same bits; the baseline moves to that order once, measured
BLAS (OpenBLAS)	each scenario total within a relative 1e-12; the same scenario selected as the k-th worst	on a total of 100 million, a ten-thousandth of a cent; VaR's date and every component unchanged
Any breach	the path switches off, back to today's code, with the reason in Admin → Health → Native	found by the start-up canary or the release drill's parity suite

Why sums can differ at all

floating-point addition is not associative; any change of order can move the last bits

Tiles are more accurate

pairwise or tiled sums err about log2(n) × 1.1e-16; one by one, up to n × 1.1e-16

The selected scenario matters most

a near tie is broken by the lower scenario index on both paths

Key insight Speed is never bought with a different number: a path that cannot prove parity does not run.

Shape it

08

One setting turns Drishti into a focused product

config/editions/risk.yaml (abridged)

```
edition: risk
title: Drishti Risk
packs: { enabled: [core, market-data, trading,
          counterparty-risk, market-risk],
         allow-only: true }
commands:
  allow: [VRES, VWIF, VOFF, EXP, NSET, …]
  words: { BI: on, FUNC: on, RUPAKA: off }
features: { var: on, exposure: on, designer: off }
bi: { datasets: [market-risk-var-latest, …] }
menus: { views: [terminal, search, var, …] }
home: { landing: desk,
        landings: [desk, command, map] }
branding: { product: Drishti Risk }
```

- **One file sets it all**
  Packs (allow-only), commands and words, menus and features, BI datasets and models, Home and its landings, help, Admin pages and branding.
- **DRISHTI_EDITION, on every server**
  drishti.edition: risk names the file; blank is the full Drishti, exactly as without editions. The console learns it from the server.
- **The site still wins, key by key**
  The edition is a layer under the site's configuration: adjust one item without forking it. Switching is configuration and a restart, never code.
- **The engines do not change**
  VaR, exposure, intraday, official records and the what-if run the same; their whole suites run again under the edition.

Key insight A focused product is one more file in config/editions/, not a fork of the code.

The whole instance, or one person's view of it

	Edition	Lens
Set by	configuration and a restart: drishti.edition (DRISHTI_EDITION)	a link, per person: Open Drishti Risk / World / Market
Scope	the whole instance, every member of a cluster	one person's session, kept in their settings
Packs	only the edition's (allow-only)	all stay loaded
Offered by	the administrator	the site's drishti.lenses (DRISHTI_LENSES): risk,world,market
Secures?	refuses other packs; hides commands and pages	no: it hides, it does not secure

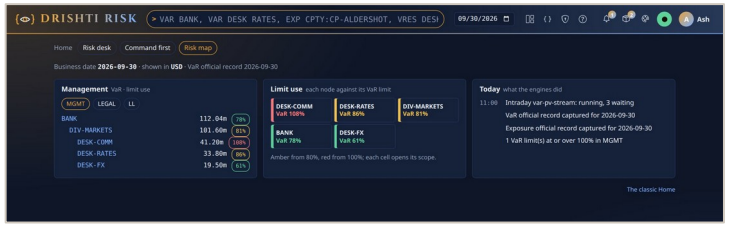
Views → Open …
Drishti Risk, World or Market, with each one's description

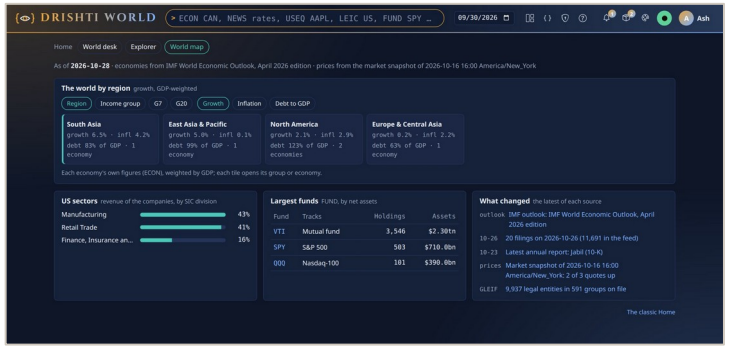
→

Leave … · Switch to …
inside one lens: leave it, or switch in one step

→

Every call
X-Drishti-Lens: commands, FUNC, type-ahead and BI follow

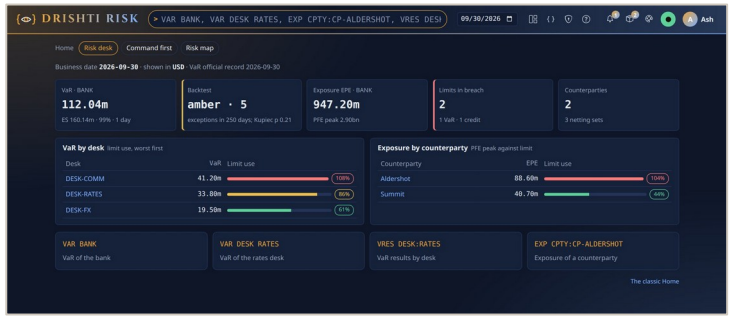


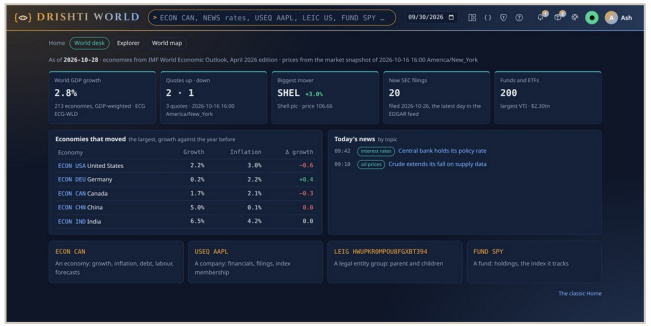


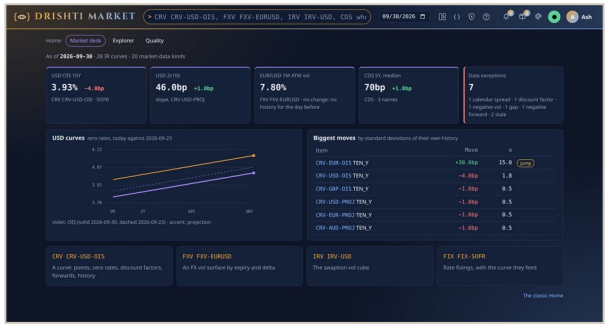
The map landing, in Risk and in World: it follows the person from one lens to the other.

Key insight **A lens hides; roles, entitlements and masks still decide what anyone may read.**

Drishti Risk, World and Market, each with three landings of its own







Drishti Risk

VaR and exposure first

5 packs: core, market-data, trading, counterparty-risk, market-risk; 36 commands

Landings **desk · command · map**

Drishti World

Economies, markets, companies and funds

5 public-data packs: world-economy, news-and-markets, us-equities, gleif, funds; 32 commands

Landings **desk · explorer · map**

Drishti Market

The bank's curves, surfaces and fixings

market-data and core; 22 commands; no trades, no VaR; quality checks

Landings **desk · explorer · quality**

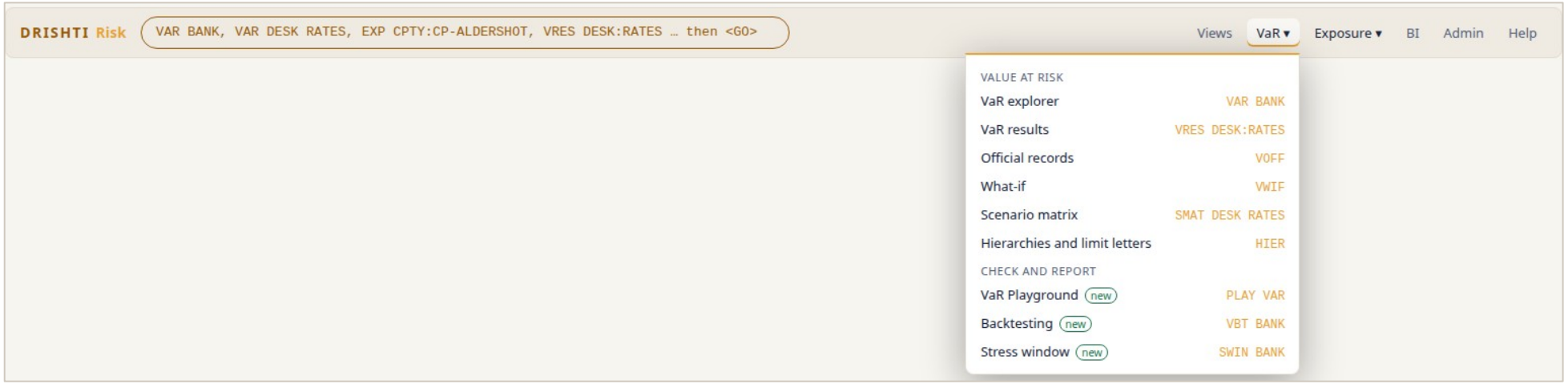
Role hooks

`home.landing-by-role` is read and wins over a person's choice, e.g. risk-manager: map, trader: command. `lens.roles` (who sees a lens) is documented, not wired.

PLANNED

Key insight **Three products from one engine: each new edition was one file and its landings, no engine change.**

Four lenses, each with its own subject menus in the bar



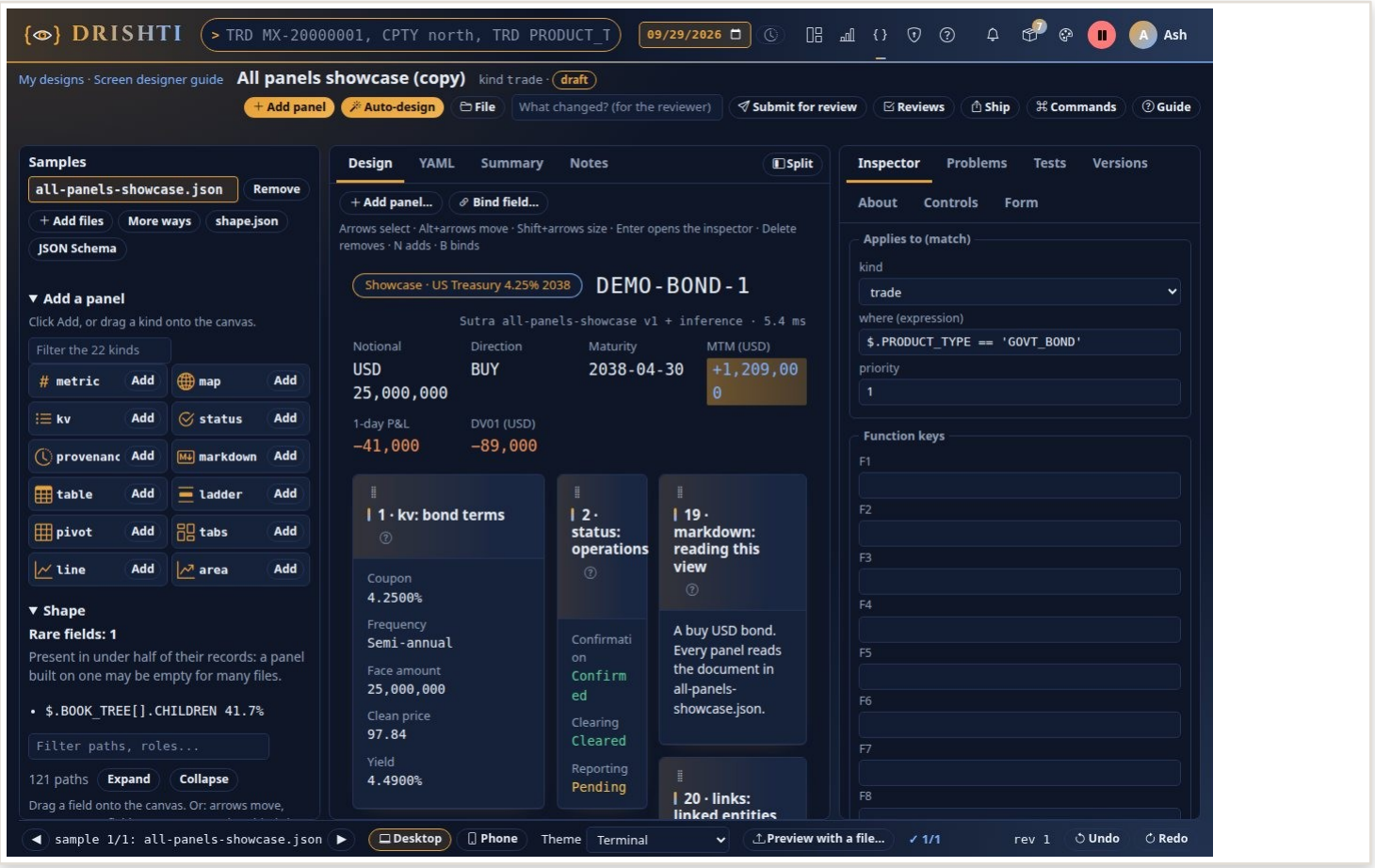
Drishti Risk: the VaR menu open, every item also a command

Lens	For	Subject menus in the bar
Drishti Risk	risk managers near the desks	VaR · Exposure
Drishti Trading	traders and desk heads	Blotter · Positions · Risk · Counterparties
Drishti Market	the market-data team	Curves · Volatility · FX · Credit
Drishti World	macro, research and context	Economies · Markets · Companies

- menus.top in the edition file orders them; configured, not coded
- an item appears only when its command is on for the lens
- on a phone they fold into one menu named after the lens

Key insight A lens hides; roles, entitlements and masks still decide what anyone may read.

Design a screen without writing code — and ship it through review



- **From data you have**
  Start from JSON files, a schema or stored entities.
- **Drag, or type**
  Palette and inspector, or the YAML tab with completion.
- **Tested before shipped**
  Sutra tests run on sample data in the tests tab.
- **Four eyes**
  An approver reviews the evidence and cannot approve their own.

Screens that respond, built from one UI kit

Events and children

A click on a row opens linked records under it, beside, below or in tabs, from any enabled pack (emits, @panel.event, @parent); masked ids stay hidden.

Every pack screen is built from fragments, each proven to draw the same as the screen it replaced.

Refresh and freshness

refresh: per panel or screen; pause, resume and interval per person; every panel says 'updated N s ago' and names a source that is not answering.

The Drishti UI kit

config/ui-kit: 17 reusable fragments (row-picker@1, field-list@1, cashflow-ladder@1 …), pack fragments, theme tokens and brand; every fragment and Sutra has a doc: docstring

Layout

Named areas on a 12-column grid, splits, tabs, sections, a sticky header, phone order; user-can-arrange per reader.

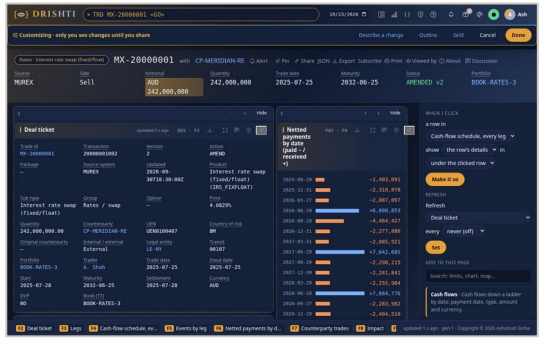
On a cluster, one server per screen; when a server goes away the page says 'your selections were reset' in words.

Forms

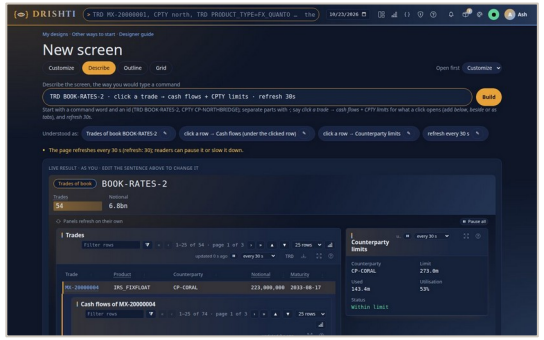
Edit beside a table, in its cells, or New in a guided drawer; linked-record fields, rules and masks; saved through write-back.

Key insight A screen is a paragraph of documented, reusable fragments that respond to the user.

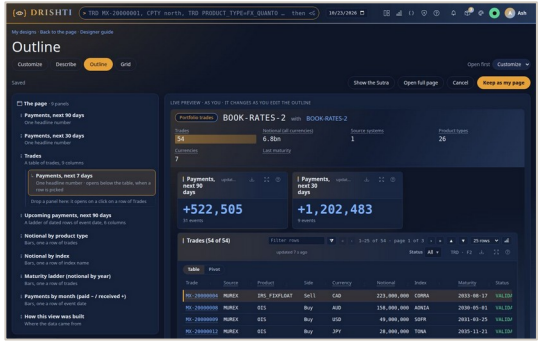
Four modes over one Sutra: customize, describe, outline, grid



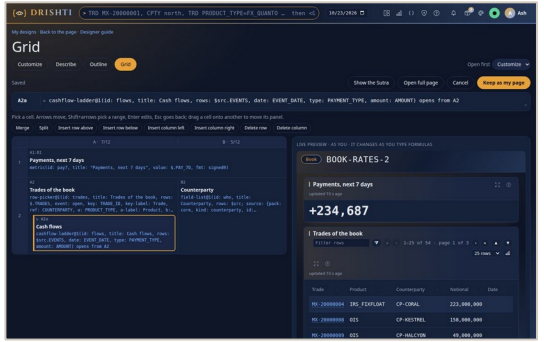
1 Customize the live page



2 Describe a screen in one line



3 Outline: the page as a tree



4 Grid, with formulas per cell

A cell is a fragment call

```
B2 = field-list@1(...;
     source: {... id: @A2.open.ref})
```

- **One Sutra, four views**
  Every mode edits the same file; switch at any time.
- **My version of this page**
  Each person keeps a customised copy; the shared page is untouched.
- **Two levels of click-opened cells; off by default**
  ui.designer; tutorial 22 builds a screen with it.

Key insight Designing a screen in plain words still produces a reviewed, versioned Sutra — not a one-off.

CONFIGURATION, NOT CODE

The product's name and the bank's names come from configuration

Branding

`ui.product · drishti.branding.product`

`ui.owner · ui.owner_email`

No template, script or server message writes the product's name; features follow it: <product> SQL, <product> Elements, <product>*net. The landing headline and tagline too.

Guard `test_branding_guard.py · BrandingLiteralTest`

The bank's names

`config/demo/names.yaml`

`lake demo --names FILE --dry-run`

The bank, its short name, the legal entities LE-NY, LE-LDN, LE-FRA, LE-TKY and branches, renamed as the demo loads, in every kind. Defaults change nothing; LEIs and BICs stay; committed test data is never rewritten.

Guard `tools/lake/test_names.py, in the drill`

The VaR tree and method

`DRISHTI_VAR_ROOT_NAME`

`scenario-window · confidence · currencies`

The top of every VaR tree named by a setting (per hierarchy too); the window, confidence and reporting currencies are settings, and every answer carries its labels.

Guard `VarRootNameTest`

Key insight A bank sees its own names in a demo by editing one file — and no code names the product or the bank.

Case studies: Drishti at work

09

Ten people, ten jobs, one platform

	Case	Who	What they open
1	Market risk: VaR from the bank to a trade	a market risk manager at month-end	VAR, SMAT, PLAY VAR
2	Regulatory VaR reporting	the reporting team and the CRO	VBT, VAR STATS, RPT
3	The Risk lens	risk managers near the desks	the Risk desk landing
4	The Trading lens	traders and desk heads	the Trader desk, VWIF, XWIF
5	The Market lens	the market-data team	CRV, FXV, the Quality landing
6	The World lens	economists and credit analysts	ECON, LEI, FUND, USEQ
7	Counterparty exposure	a counterparty credit officer	EXP, NSET, XWIF
8	BI and analytics	a business analyst	BI Studio: model, report, board
9	Data and operations	a data engineer	lake build, data landed, DATA, DATAMAP
10	Oversight (designed)	the second line, for the CRO	why VaR moved; breaches

Key insight Every figure in these cases is the demo data's, quoted from the guides and tutorials.

Month-end: is the bank's VaR right, and who makes it?

Who they are

Business-aligned market risk manager
answers for the VaR of the businesses they cover: right, explained, within limits

Model validator and auditor
must check how a figure was worked out, independently of the engine

What they need

- the bank's VaR and ES on the official date, with limit use
- the desk that moves it, down to the trade
- what-if cuts: the bank without a desk, a desk without a hedge
- proof of a figure a validator can recompute

Their day

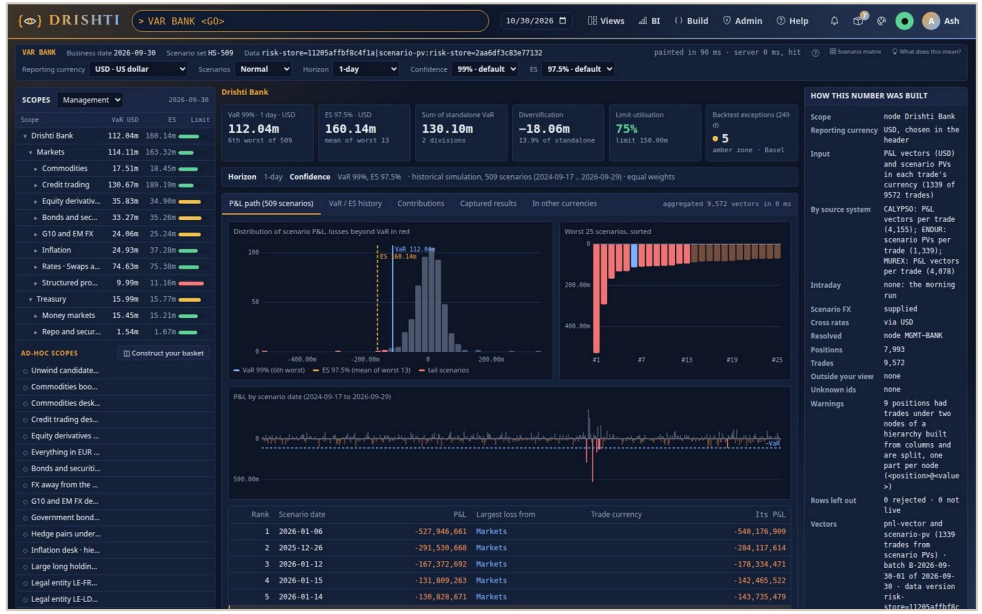
07:00	the official run: VaR by desk, limit use, breaches
09:00	the morning meeting: why it moved
Intraday	a scope someone asks about; a hedge to test
Month-end	backtest, limit review, stressed VaR
On request	the validator's workbook

What was painful before

- finished VaR figures stored by a batch, which nobody could re-derive
- a new cut of the bank meant a new hierarchy load, or a spreadsheet
- explaining a figure took a day of extracts and formulas

Key insight The manager asks in sentences; the engine answers from the trades, and shows its working.

From VAR BANK to the trade, on 2026-09-30



VAR BANK: the explorer on the bank, its hierarchy and its tabs

The screens and commands, in order

`VAR BANK`
VaR 99% 112,041,646 USD, ES 97.5% 160,143,809; limit 150m, use 74.7%; 9,572 trades

`drill: LE-NY › Credit trading`
144,262,258 › 130,670,864, each node with its limit use

`VAR BANK - DESK:CREDIT`
76,605,857: the bank without a desk, as one scope

`VAR DESK RATES · parts by book`
components 30,612,282 + 25,280,825 + 18,740,573 = 74,633,680

`SMAT DESK RATES · PLAY VAR DESK RATES`
scenario by scenario, then the workbook

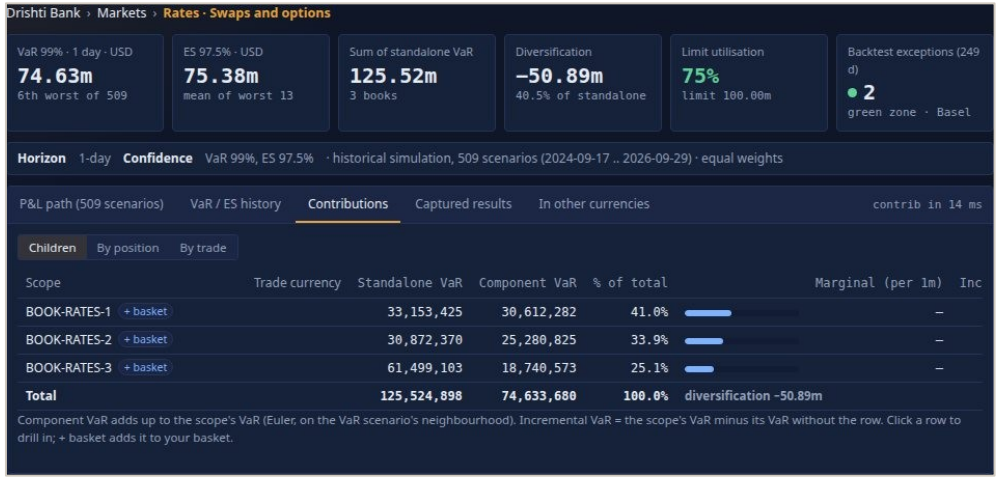
Computed and explained

the P&L path: VaR at 99% is the 6th worst of 509 days (509 × 1% = 5.09, rounded up); ES at 97.5% the mean of the 13 worst.

The result

- diversification across divisions 18,060,250
- the 6th worst of 509 scenario days, to the dollar
- every component sums to the desk's VaR exactly

Who makes the desk's VaR, scenario by scenario, and in a workbook



Contributions: the desk by book, adding up



SMAT: the desk scenario by scenario

DESK:RATES by book	Standalone VaR	Component VaR	% of VaR	Incremental VaR
BOOK-RATES-1	33,153,425	30,612,282	41.0	−5,131,638
BOOK-RATES-2	30,872,370	25,280,825	33.9	21,654,002
BOOK-RATES-3	61,499,103	18,740,573	25.1	26,690,191

- the components add up to 74,633,680 to the dollar
- BOOK-RATES-1 lowers the desk's VaR: a hedge
- PLAY VAR DESK RATES exports the same calculation as formulas

Key insight A validator recomputes the desk's VaR on their own machine and gets the engine's figure to the cent.

The fiscal year closes: the VaR table, the backtest, three signatures

Who they are

Regulatory reporting team
prepares the annual report's VaR table and the backtesting chart for the period

The checker and the CRO
review, send back or approve; one person a step, four eyes

What they need

- period end, average, high and low of VaR for every row
- the backtest's exceptions, zone and multiplier
- every cell traceable to the record it came from
- a signed, frozen run; a restatement as a new run

Their day

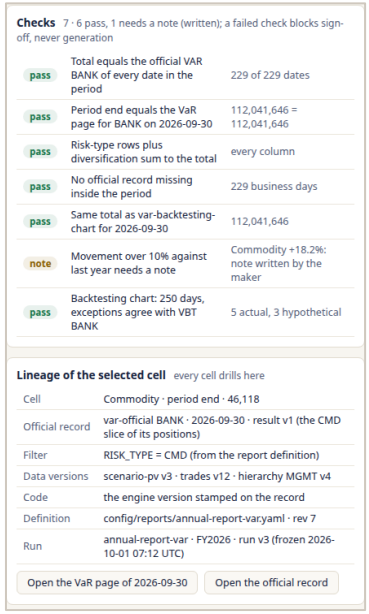
Daily	official records captured at the close
Period end	the run generated from the records
Checks	totals, rows, missing days, the backtest
Sign-off	maker, checker, then the CRO
Filing	xlsx, csv and PDF to the reporting team

What was painful before

- figures re-keyed from risk systems into spreadsheets each quarter
- no lineage from a filed cell back to the day and the data version
- a restated input silently changed an earlier figure

Key insight Drishti fills, checks and freezes; people write the narrative, approve and file.

From the official records to a signed run, FY2026



Checks 7 · 6 pass, 1 needs a note (written); a failed check blocks sign-off, never generation

pass	Total equals the official VAR BANK of every date in the period	229 of 229 dates
pass	Period end equals the VaR page for BANK on 2026-09-30	112,041,646 = 112,041,646
pass	Risk-type rows plus diversification sum to the total	every column
pass	No official record missing inside the period	229 business days
pass	Same total as var-backtesting-chart for 2026-09-30	112,041,646
note	Movement over 10% against last year needs a note	Commodity +18.2%: note written by the maker
pass	Backtesting chart: 250 days, exceptions agree with VBT BANK	5 actual, 3 hypothetical

Lineage of the selected cell every cell drills here

Cell	Commodity · period end · 46,118
Official record	var-official BANK · 2026-09-30 · result v1 (the CMD slice of its positions)
Filter	RISK_TYPE = CMD (from the report definition)
Data versions	scenario-pv v3 · trades v12 · hierarchy MGMT v4
Code	the engine version stamped on the record
Definition	config/reports/annual-report-var.yaml · rev 7
Run	annual-report-var · FY2026 · run v3 (frozen 2026-10-01 07:12 UTC)

Open the VaR page of 2026-09-30 | Open the official record

Checks with their evidence, and a cell's lineage

The screens and commands, in order

`VBT BANK`
250 days to 2026-09-30: 3 exceptions against actual P&L, 3 against hypothetical at 99%: green, 3.00

`VAR BANK STATS FY2026`
228 records; period end 112,041,646; average 151,812,526; high 167,372,692; low 87,874,268

`drishti.py var report run annual-report-var FY2026`
every cell filled from the records, checks run, frozen

`VAR REPORT annual-report-var FY2026`
the run on screen; sign maker, checker, approver; files released

Computed and explained

the backtest from the records of the day before and the desks' P&L; statistics from the records, never stored beyond them.

The result

- seven checks: a failure blocks sign-off, never generation
- each cell keeps its node, statistic, record, data versions
- the definition is one YAML file per report

Risk managers near the desks: VaR and exposure first

Who they are

Market risk manager
the VaR of the businesses they cover: right, explained, within limits

Counterparty credit risk officer
exposure against limits, pre-deal approval, collateral and netting

Risk analyst
risk reporting: the figures and the evidence behind them

What they need

- the bank's VaR and exposure and the limits in breach, first thing
- a pre-deal answer while the trader waits
- the official close, kept for years
- the stress window reviewed once a year

Their day

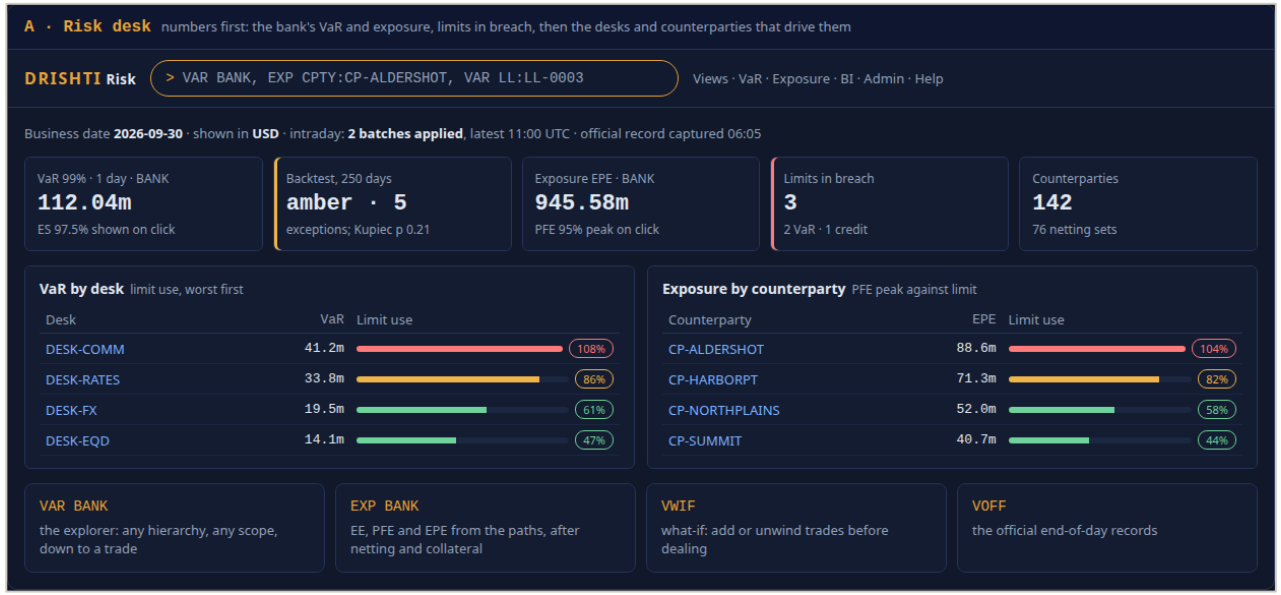
07:00–09:00	official run, VaR by desk, limit use, breaches
Intraday	VWIF and XWIF pre-deal, scopes, baskets
Close	official captures
Month-end	backtest; LIMIT_USE in BI; stressed VaR
Year	SWIN: the stress window review

What was painful before

- the morning figures came from an overnight batch, one fixed cut
- VaR and exposure lived in different systems with different trees
- a pre-deal check meant a request to another team

Key insight One lens, opened from a link: the risk manager's whole day without leaving it.

The Risk desk landing: numbers first, then who drives them



The Risk desk landing (the approved design; figures illustrative)

The screens and commands, in order

VAR BANK · VAR DESK RATES
the explorer: any hierarchy, any scope, down to a trade

EXP BANK · EXP CPTY:CP-ALDERSHOT
EE, PFE and EPE from the paths, after netting and collateral

VWIF · XWIF
add or unwind trades before dealing: VaR and exposure

VRES · VOFF · HIER · SMAT
results, official records, hierarchies and limit letters, the scenario matrix

PLAY VAR · VBT · SWIN
the VaR menu's check and report items

Computed and explained

every tile is the engines' own answer for the date; a tile the engines do not serve is left out, never faked.

The result

- VaR 99% BANK 112.04m; limit letter LL-0003 16.07m; EPE 945.58m (the guides' figures on 2026-09-30)
- landings: Risk desk, Command first, Risk map

The first line: my books, my limits, and does this trade fit?

Who they are

Trader
the P&L of their books, inside delegated limits; a fast pre-deal check

Desk head
the desk's P&L, limits and breaches, and each trader's limit use

Business manager
the desk's view across books and counterparties

What they need

- my books on one page, scoped to me
- VaR and limit use per book, near-limit warnings
- one answer for a new trade: fits, near or breaches
- today's blotter: new, amended, cancelled

Their day

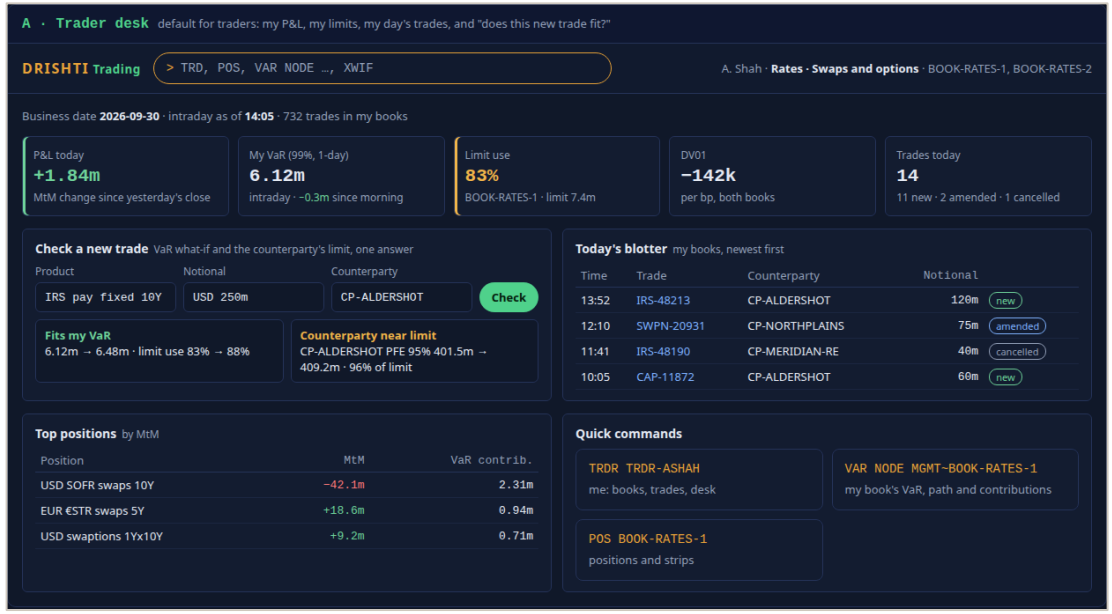
06:30–08:00	before the open: books, limits, overnight changes
Intraday	check a new trade; the blotter
16:00–19:00	the close: P&L against VaR
Month-end	limit use over the month
Year	the desk's limits renewed

What was painful before

- a pre-deal check meant asking risk, and waiting
- VaR and the counterparty limit came from two systems
- my books meant filtering a bank-wide report

Key insight **The trader asks one question, does it fit, and gets the VaR and the counterparty limit in one answer.**

The Trader desk, and a hedge tested before dealing



The Trader desk landing (the approved design; its figures illustrative)

The screens and commands, in order

`TRD · CTRD · PTRD · LTRD`
the blotter: trades by counterparty, portfolio, legal entity

`BOOK · POS · VAR NODE MGMT~BOOK-RATES-1`
my book's positions, VaR, path and contributions

`VWIF DESK:RATES`
the VaR what-if of a hedge, before dealing

`XWIF`
the exposure pre-deal on the counterparty's paths

Computed and explained

Check a new trade runs one VaR what-if and one exposure pre-deal, and answers fits, near (0.8 of a limit) or breaches.

The result

- the hedge on DESK:RATES: VaR 74,633,680 → 47,599,220 (−27,034,460)
- ES 75,377,415 → 56,247,260; stressed VaR 85,318,588 → 59,556,766

The market-data team: are today's curves, surfaces and fixings right?

Who they are

Market data analyst
curves, surfaces and fixings in on time and clean

Valuation control (IPV)
independent checks of the prices behind valuations

Market risk methodology
the history behind VaR's 509 scenario days

What they need

- today's moves across curves, vol and FX at a glance
- stale data, jumps and gaps flagged, not found by accident
- one search for any curve, surface or fixing
- the impact of a bad point on VaR (designed)

Their day

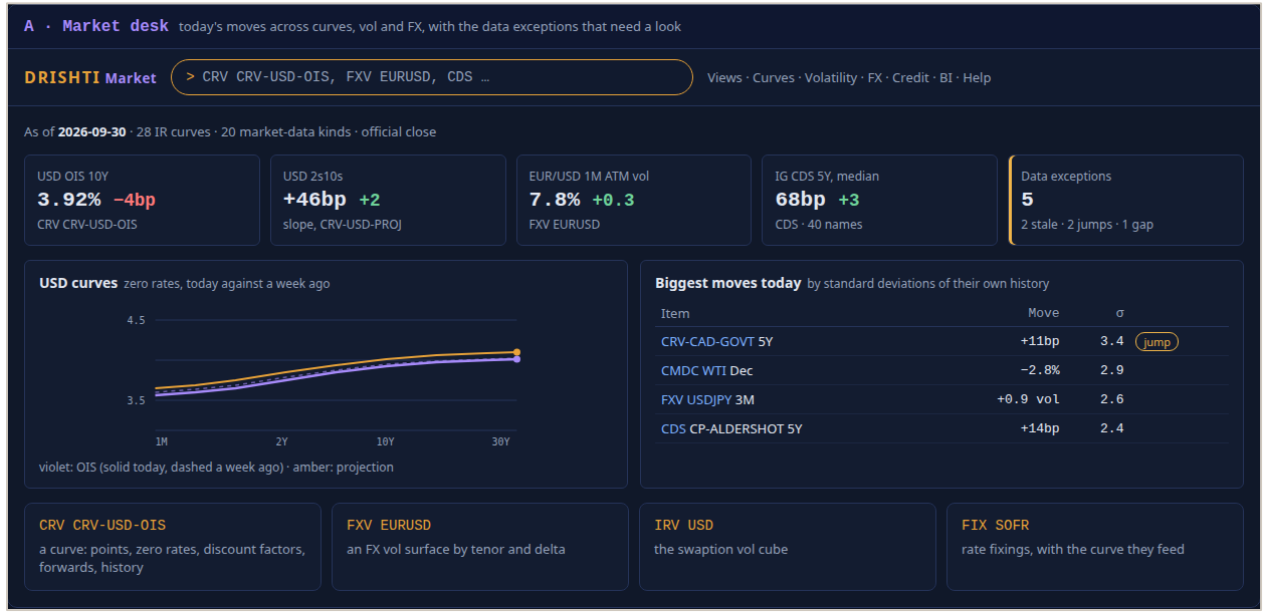
17:00–20:00	the close and the batch window
Morning	exceptions before the desks open
Intraday	a curve someone questions
Month-end	IPV
Year	the history behind the scenarios

What was painful before

- checks lived in scripts and inboxes, not on the screen
- a jump was found after VaR moved, not before
- each data kind had its own viewer

Key insight **The data that feeds every valuation and every VaR gets a desk of its own, with its checks on it.**

The Market desk: moves, curves and the exceptions to look at



The Market desk landing (the approved design; its figures illustrative)

The screens and commands, in order

`CRV CRV-USD-OIS`
a curve: points, zero rates, discount factors, forwards, history

`FXV EURUSD · IRV USD`
the FX vol surface; the swaption vol cube

`FIX SOFR · CDS · INFC · CMDC`
fixings, credit, inflation and commodity curves

`CORR · DATAMAP`
correlations computed from prices; the map of all data

`Quality landing`
stale, gaps and jumps by kind

Computed and explained

the checks are computed from the data each time the landing opens; VaR at stake and two-source checks are designed.

The result

- stale after 2 business days; biggest moves over 21 days in σ; a jump at 3σ or more
- on the committed demo data every check is clean

Context from public data: economies, markets, companies and funds

Who they are

Economist, macro strategist
growth, inflation, debt and forecasts, country by country

Country and credit analysts
sovereigns, companies, legal-entity groups, coverage

KYC and reference data
who owns whom: legal entities and their parents

What they need

- an economy, a company or a fund one command away
- linked: a company to its filings, its index, its legal entity
- groups and sectors computed from members
- licences respected: internal-only data marked

Their day

Morning	the world desk: indicators, news, filings
Intraday	a country or a company in question
Month-end	sectors and groups reviewed
April, October	the IMF's new vintages
Year	ten fiscal years of filings

What was painful before

- public data downloaded by hand into spreadsheets, then stale
- company, legal entity and fund data never linked
- group totals kept as copies that drifted

Key insight **Public data as packs: linked, offline, refreshed by script, and computed where it can be.**

The World desk: one command line across economies, companies and funds



The World desk landing (the approved design; its figures illustrative)

The screens and commands, in order

ECON CAN · MACRO · FCST
an economy: growth, inflation, debt, labour, forecasts

USEQ AAPL · FACT · FILING
a company: financials, filings, index membership

LEI · LEIG · LEIC
a legal entity, its ownership group, a country's entities

FUND SPY · HELD · TRACK
a fund's holdings and the index it tracks

NEWS · QUOTE
headlines and the offline market snapshot

Computed and explained

ownership groups, countries and sectors computed from their members when opened (LEIG, LEIC, SIC, ECG), never stored.

The result

- the 200 largest US funds and ETFs; ten fiscal years of XBRL
- World Bank data (CC BY 4.0); IMF and Yahoo data internal only

If this counterparty defaulted, how much would we lose?

Who they are

Counterparty credit risk officer
exposure against credit limits; pre-deal approval; collateral and netting

Credit committee
the largest counterparties and their netting sets

Their day

Morning	the bank's exposure board; limits
A request	a counterparty and its netting sets
Pre-deal	XWIF: the trade on the paths
Close	the official record

What they need

- exposure on an average path and on a bad one: EE, PFE, EPE, EEPE
- from the bank to one netting set and one trade
- the effect of collateral, path by path
- a pre-deal answer that is exact, not an add-on

What was painful before

- exposure stored per counterparty by an overnight run
- collateral applied as an add-on, not on each path
- pre-deal checks by a rough add-on table

Key insight Exposure is computed from the simulator's paths for any scope, with collateral on each path.

From the bank to a netting set, collateral path by path



A netting set's 200 paths and its collateral

The screens and commands, in order

`EXP BANK`
peak PFE 95% 2,897,620,439 at 2W; EPE 947,202,991; 74 netting sets, 23 counterparties

`EXP CPTY:CP-ALDERSHOT`
peak PFE 401,456,810; limit use 5.0%; four netting sets

`EXP NSET:NS-ALDERSHOT-LDN`
peak PFE 85,063,192 at 2M with collateral, 478,546,912 at 3Y without

`XWIF`
an unwind on the paths, exact: peak PFE −3,099,344, EPE −615,288

Computed and explained

the netting set's collateral applied on each of 200 paths; the trade's share of the set's EE shown; the official record kept.

The result

- 200 paths, a 24-date grid, 8,022 live trades
- CSA: 10-day margin period, 10m thresholds, 500k minimum transfer

A business analyst builds a report and a board, without a data team

Who they are

Business analyst
builds reports for their desk or function; knows Power BI

Report reader
opens the report or the board and sees only what they may

Their day

Start	pick a model, or a pack dataset
Build	cards, a map, a bar, a slicer
Publish	own workspace, shared, the catalogue
Run	subscriptions and alerts; a board in kiosk mode

What they need

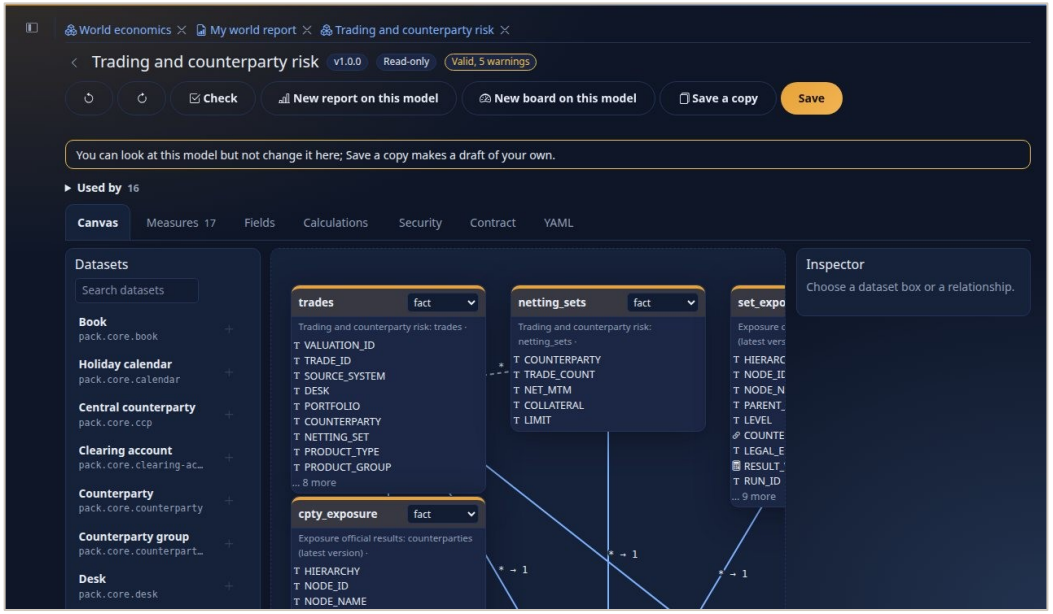
- a governed model to start from, not a raw table
- drag-and-drop visuals, slicers, drill-through
- two facts side by side, filtered by one slicer
- a live board on a wall; a subscription by email

What was painful before

- an extract requested from a data team, then a copy in a BI tool
- masks and row rules rebuilt in the BI tool, or forgotten
- the report and the terminal disagreed on the same figure

Key insight **The BI author starts from a model the packs already describe, under the reader's own masks.**

A model, a report, two facts in one model, published



Two fact tables in one model, joined through the counterparty

The screens and commands, in order

`BI <GO> · Models · World economics`
seven measures the model owns; GDP per capita a ratio of sums

`New report`
cards, a map drilled world › country › region, a bar with top 10, a slicer

`Two facts in one model`
trades and netting sets, each added up on its own, one slicer filters both

`Wizard`
a dataset narrowed from a pack, a derived column, a model in one click, no YAML typed

`Publish · Subscribe`
a workspace, the catalogue; PDF or Excel by email, each reader's masks

Computed and explained

every number by the server; the browser only draws; the reader's row rules and masks applied before any sum.

The result

- one slicer filters trades and netting sets together
- a join that would repeat rows is refused, not summed

A new feed lands every night: load it, announce it, prove it

Who they are

Data engineer
owns the nightly feeds from the pricing and trade systems into the lake

Operations
knows by 07:00 whether every expected load arrived

What they need

- one data standard for every feed: JSON Lines
- a loader into Delta, Iceberg or S3 Tables, one commit a date
- a signal that data has landed, and an alert when it has not
- a way to see and download what is there

Their day

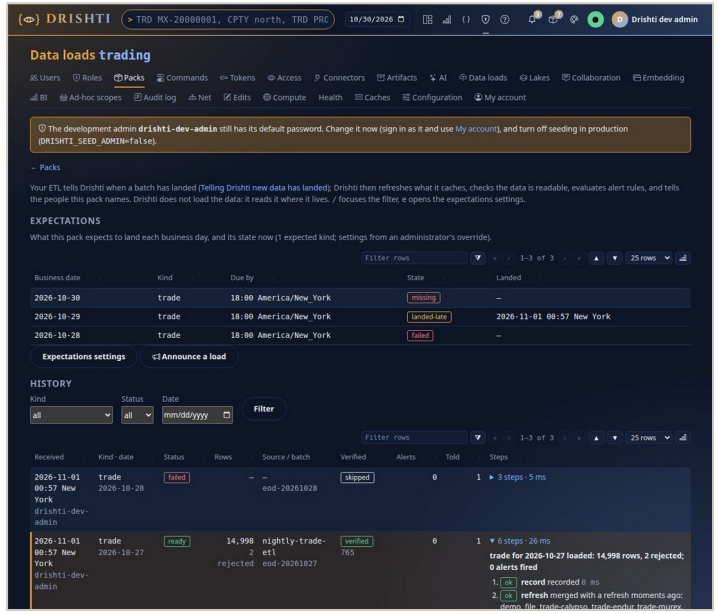
Night	the pricing systems write JSON Lines
Load	lake build; one commit per date
Signal	data landed: the pack acts on it
07:00	the loads page: expected, arrived, late
Any time	DATAMAP and the Data view

What was painful before

- each feed in its own format, with its own loader script
- a missing day found by the first user of the morning
- no map of what data exists and how it joins

Key insight One format in, one signal when it lands, and the platform shows what it holds.

From a folder of JSON Lines to a dataset anyone can find



The Data loads page: every announced load, its batch and its state

The screens and commands, in order

`drishti.py lake build`
JSON Lines into Delta or Iceberg; --engine java for the parallel loader

`drishti.py data landed`
announce a load: ready or failed; the pack acts on it

`Admin → Data loads`
expected, arrived, late; an alert for a missing day

`DATAMAP · DATA <GO>`
what the packs hold and how it joins; any dataset downloaded

Computed and explained

every join on the data map is checked against the data; every download is streamed, masked and audited.

The result

- 10,000 positions × 510 scenarios loaded in 0.19 to 0.29 s
- the demo lakes: 60 business days, 85 tables, about 15 minutes

The second line: are we within appetite, and who approved what?

Who they are

Market risk oversight

the second line, reporting to the CRO: appetite, limits, independent challenge

The CRO's committee

escalations, temporary limit increases, the board pack

What they need

- why VaR moved overnight, by cause, adding up exactly
- a breach as a record: status, commentary, approvals
- temporary limit increases with four eyes
- escalation from the desk to the business to the CRO

Their day

DESIGNED

07:30–09:00	the morning meeting: why did VaR move?
Breach	explained, approved or remediated, closed
Escalation	desk, business, CRO
Committee	the board pack

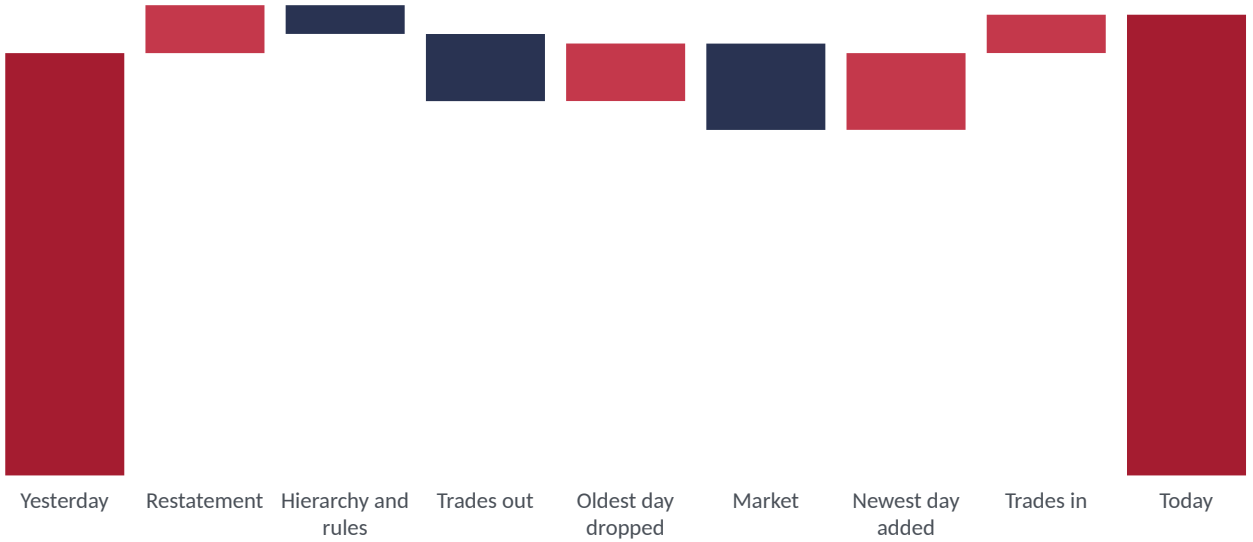
What was painful before

- the VaR change explained by hand, steps that never added up
- breaches tracked in email and spreadsheets
- approvals with no record of who and when

Key insight Why did VaR move is the most-asked question of the morning meeting; the design answers it exactly.

Why VaR moved, step by step; a breach from open to closed

RO1 · the VaR change, from yesterday's record to today's



Illustrative bar heights. Each step changes one thing at a time, so the steps add up exactly; a Shapley view of the order is optional (at most 720 orders).

RO2 · a breach as a record

DESIGNED

Open BR-<year>-<seq>, raised by the limit check

↓

Explained the desk's commentary

↓

Approved or remediated a temporary increase with four eyes, or the position cut

↓

Closed with its history

- escalated at any point: desk, business, CRO
- views: queue, board, timeline

Key insight **Designed on what is built: official records, captures, hierarchies and limits; all 131 nodes in well under a second.**

Data in

10



One JSON object a line: every generator writes it, every export offers it

Generators
every tool under tools/ writes JSONL; .jsonl.gz where the repository gzips

→

JSON Lines
one object a line, UTF-8; generating and loading are separate steps

→

Load
load_rows.py, lake build: the lake byte for byte the same as before

→

Delta or Iceberg
the lake every connector reads

Readers

- JSON Lines, JSON arrays and per-entity .json still read
- CSV deprecated: one warning a file, named in Admin → Health
- drishti.py convert FILE: CSV, a JSON array, gzip or a folder into JSONL

Exports

- Server: search, SQL, BI runs, /export/jsonl, statements, subscriptions, scheduled reports, VaR and exposure
- Console: every download and menu; CLI: --jsonl, --format jsonl
- CSV, Excel, PDF and PNG all stay

Records

- Official VaR and exposure results as JSONL beside their Delta tables
- Sutra test cases: one cases.jsonl per Sutra, 197 folders converted
- Sample catalogues, staged lake rows, seals: JSONL

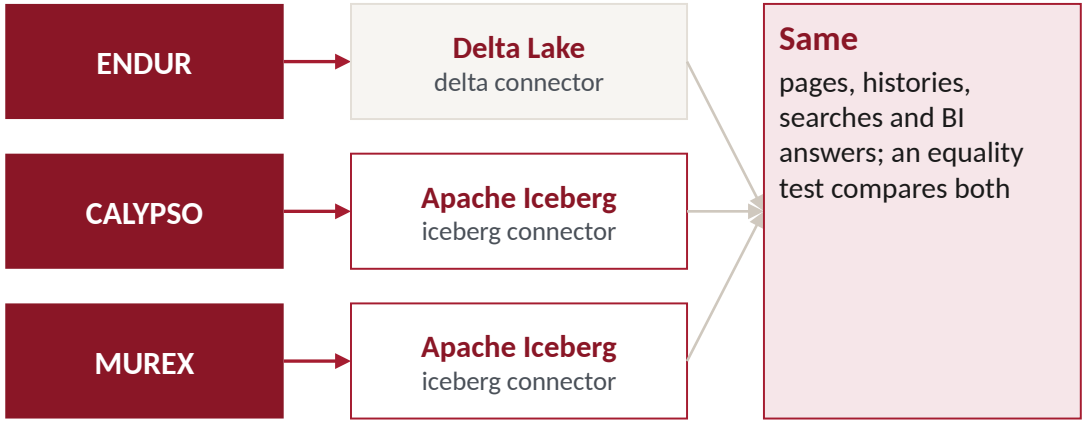
Guarded in the drill: `test_generators_write_jsonl.py` runs the generators and reads every data file; `test_exports_offer_jsonl.py` checks every export entry point.

Key insight **A pipeline streams any Drishti input or output a line at a time; CSV, Excel, PDF and PNG stay for people.**

Each system's trades in Delta or Iceberg — the same answers

One switch per source system

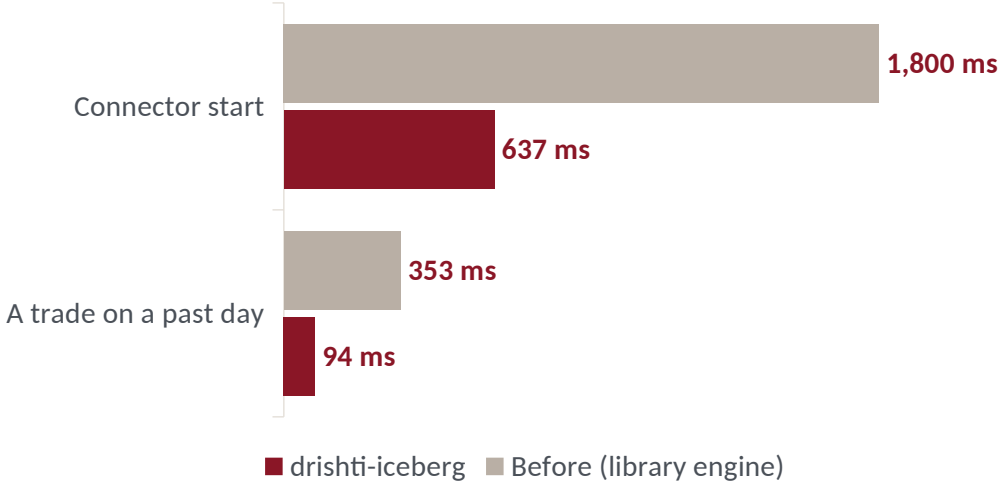
```
build_demo_lakes.py --store endur=delta \
    --store calypso=iceberg --store murex=iceberg
```



The iceberg-external connector also reads tables Drishti did not write: Spark, Flink, Trino, Snowflake, Glue.

drishti-iceberg: Iceberg without Hadoop

Drishti's own engine, the default — so Iceberg runs on Windows



2.8×[1]
faster connector start

3.8×[2]
faster read of a past day

Key insight **The store is an admin's decision per system, not a rewrite: the pages cannot tell which format answered.**

[1] 1,800 ms ÷ 637 ms = 2.8: connector start, library engine against drishti-iceberg, both measured (RELEASE_NOTES, DRISHTI_ICEBERG.md) [2] 353 ms ÷ 94 ms = 3.8: a trade on a past day, both measured

AWS S3 Tables and Delta Lake on S3: one connector file and a script each

AWS S3 Tables (Iceberg REST, SigV4)

```
plugin: iceberg
kinds: [pnl-vector, scenario-pv, …]
settings:
  catalog: s3tables
  table-bucket-arn: ${DRISHTI_S3TABLES_ARN:}
  domain: risk
```

- **One setting:** the table bucket ARN gives the endpoint, region, signing name and prefix
- **AWS keeps them in shape:** S3 Tables compacts and expires; Drishti runs no maintenance there
- **Writing:** the loaders write the same tables; each date one commit

Delta Lake on S3 (native engine, no Hadoop)

```
plugin: delta
kinds: [pnl-vector, scenario-pv, …]
settings:
  root: s3://${DRISHTI_DELTA_S3_BUCKET:drishti-lake}/delta
  domain: risk
  s3.region: ${AWS_REGION:us-east-1}
```

- **Exactly as on local disk:** each table at s3://bucket/prefix/domain/table; VaR reads the same tables
- **No key in the file:** the AWS default chain (the ECS task role, a profile); least-privilege IAM listed
- **Commits:** S3's conditional writes; a check script proves a bucket honours them

Tool scripts `s3-tables.sh · delta-s3.sh  create | check | remove` Python with boto3, .sh and .ps1, no AWS CLI, --dry-run, no secret printed

Key insight Tested against recorded AWS signatures and stand-ins; the live run waits for an AWS sandbox account.

Arrow where it pays; our own formats where they are faster

Boundary	Format	Why
The lake	Parquet (Delta or Iceberg)	Arrow's partner on disk: compressed columns, statistics to skip files
VaR hot files	our own float32 rows, memory-mapped	summing runs near memory bandwidth, about 70 GB/s[1]; Arrow would not be faster
Shard partials	raw vectors of doubles	already minimal; Arrow would add framing
Server ↔ Python workers	Arrow IPC, memory-mapped	no copy, no parse
VaR contributions to the browser	**Arrow** on request; Arrow JS	fewer bytes, faster fetch: measured, right
BI queries	Arrow on request; JSON in the console	mostly text: hardly smaller after gzip, parsed slower
P&L paths, entity pages	JSON	small answers; Arrow is no smaller
Search results	HTML drawn on the server	no table reaches the browser

Where Arrow paid

MEASURED

VaR contributions	JSON	Arrow
by trade, 1,281 rows	631,873 B	244,304 B
by position, 1,173 rows	692,324 B	243,392 B
fetch, through the console	81–91 ms	59–61 ms[3]

61–65% fewer bytes[2], read with Arrow JS (vendored) in the VaR explorer; painted in the same time.

Kept as JSON, measured

BI grids: 8–14%[4] smaller after gzip, parsed slower, painted no sooner; drawing the rows is the cost.
P&L paths: 509 numbers, no smaller.[5]

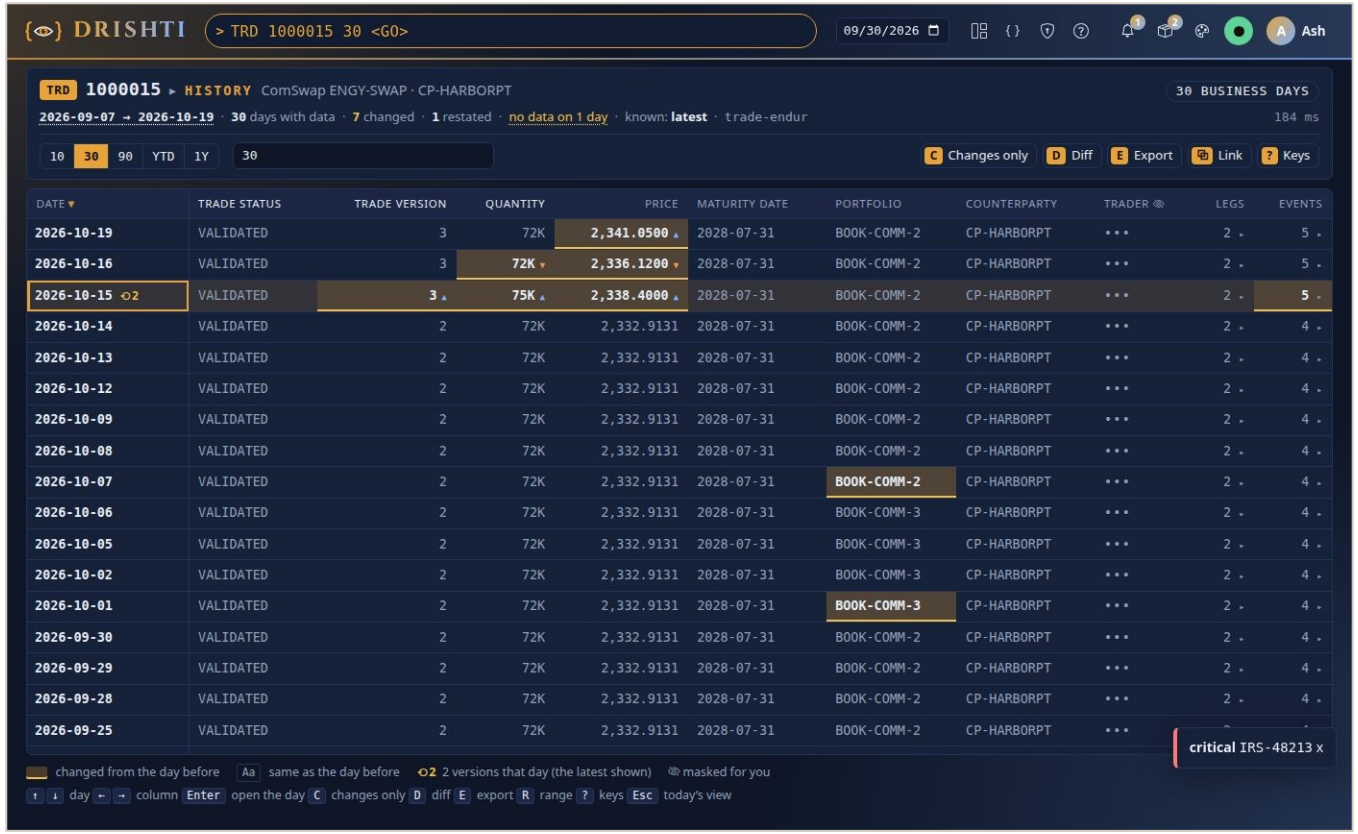
PLANNED

Next: a client-side renderer; search drawn in the browser with it, where Arrow can carry the rows.

Key insight Measured, Arrow paid in one place: the VaR contribution tables. Elsewhere JSON, Parquet or our own formats are as fast or faster.

[1] 1,000,000 trades × 509 × 8 B = 4.07 GB summed in 58 ms (VarScaleBenchmark, measured, VAR_GUIDE.md 27.2) = 70 GB/s [2] 1 − 244,304 ÷ 631,873 B = 61% (by trade) and 1 − 243,392 ÷ 692,324 B = 65% (by position): bytes as the console sends them; measured, DATA_FORMATS.md section 6 [3] fetch through the console, against 81–91 ms as JSON (Chromium, one machine); measured, DATA_FORMATS.md section 6 [4] gzip 122,953 → 112,656 B (5,000 trades) and 1,216,777 → 1,048,434 B (50,000 trades) = 8% and 14% smaller; measured, DATA_FORMATS.md section 6 [5] the P&L path: 21,607 B as JSON, 22,032 B as Arrow; measured, DATA_FORMATS.md section 6

Type a range after an id and see the entity over time



TRD 1000015 30: one row per business day, changed cells marked ▲/▼, restated days flagged (tutorial 21).

One grammar, any range

```
TRD 1000015 30
TRD 1000015 3M DIFF
TRD 1000015 FROM 2026-09-01 TO 2026-09-30
trade 1000015 last month
```

13 ms
30 days of one trade among 10,000, cold

< 1 ms
the same, warm

0.3 ms
from the history cache

Generic: any kind with dated data, no code per kind; fast paths on Delta, Iceberg, Redis, PostgreSQL, DuckDB, MongoDB, Aerospike and Cassandra; masks hold on every day's row.

Key insight Across a cluster, a node reads an entity's days from the peer that holds them in one call, masked on both.

US companies, legal entities and funds — linked, offline, refreshed by script

us-equities

586

US companies from SEC EDGAR, Nasdaq and GLEIF: financials, filings, index members

`USEQ · IDX · SIC · EDGAR`

LEI

gleif

9,512

legal entities in 540 ownership groups; groups computed from DIRECT_PARENT

`LEI · LEIG · LEIC`

LEI

funds

200

largest US mutual funds and ETFs from N-PORT, N-CEN and prospectuses; 5.6 MB gzipped

`FUND · HELD · TRACK · FUNDS`

Refresh script
`tools/publicdata/<pack>.py .sh and .ps1, status, dry-run`

→

Committed snapshot
`data/testdata/<domain>/<kind>/*.jsonl.gz`

→

Demo lakes
`build_demo_lakes → Delta`

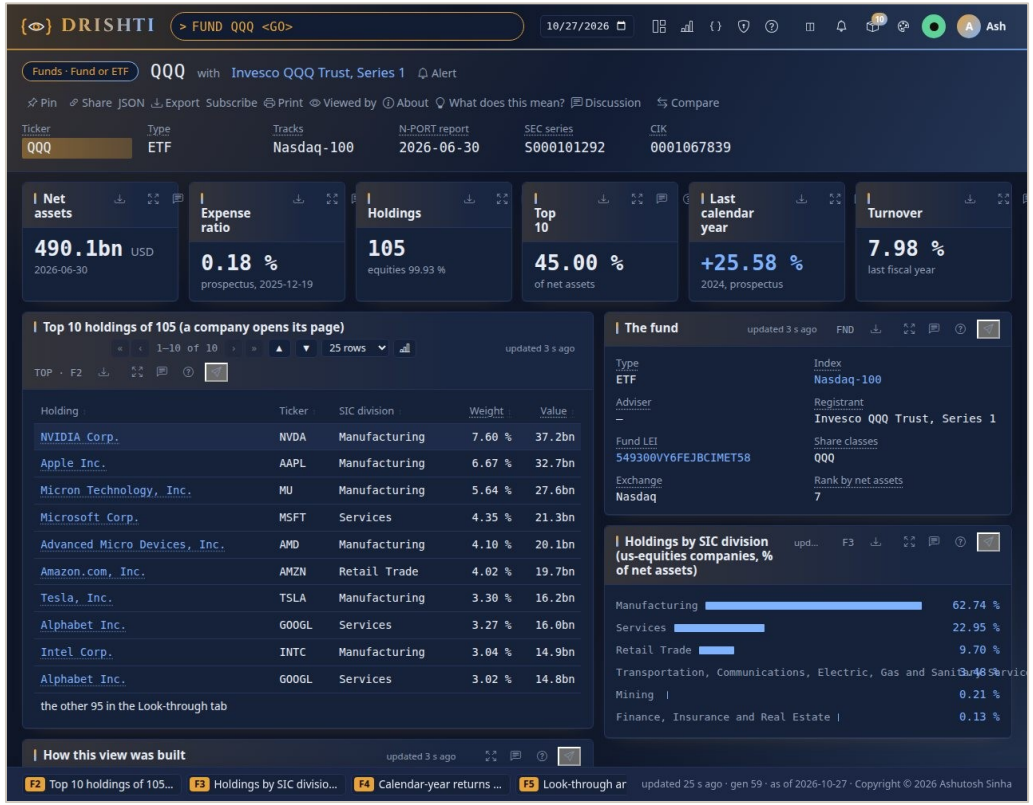
→

A connector per pack
`served offline, like any source`

External data, one page: every source with its URL, licence and use; IMF, Yahoo and Google News marked internal only and refused in the repository; sectors are SIC divisions because GICS is proprietary — the pages say so.

Key insight **Public data arrives the way a bank's data does: a contract per kind, a lake, a connector — and no network at demo time.**

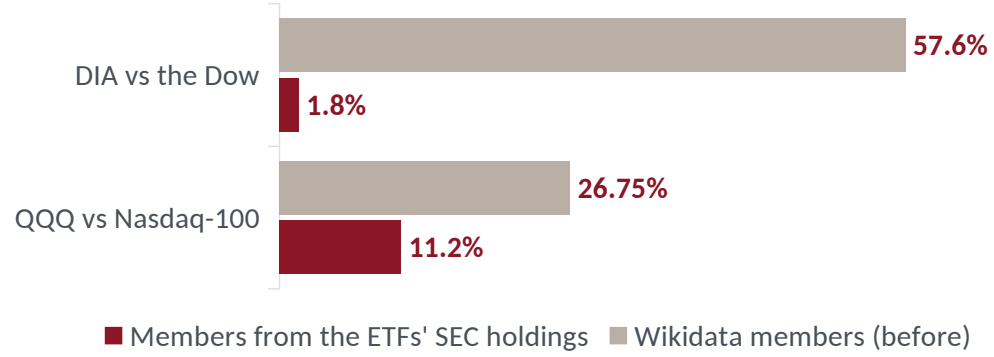
Index members from what the tracking ETFs report to the SEC



FUND QQQ: fact sheet, top holdings, look-through by SIC division (tutorial 28).

Earlier the S&P 500, Nasdaq-100 and Dow members came from Wikidata's lists, which lacked Exxon Mobil, BlackRock, NVIDIA in the Nasdaq-100 and about 100 more.

Active share against the index, before and after



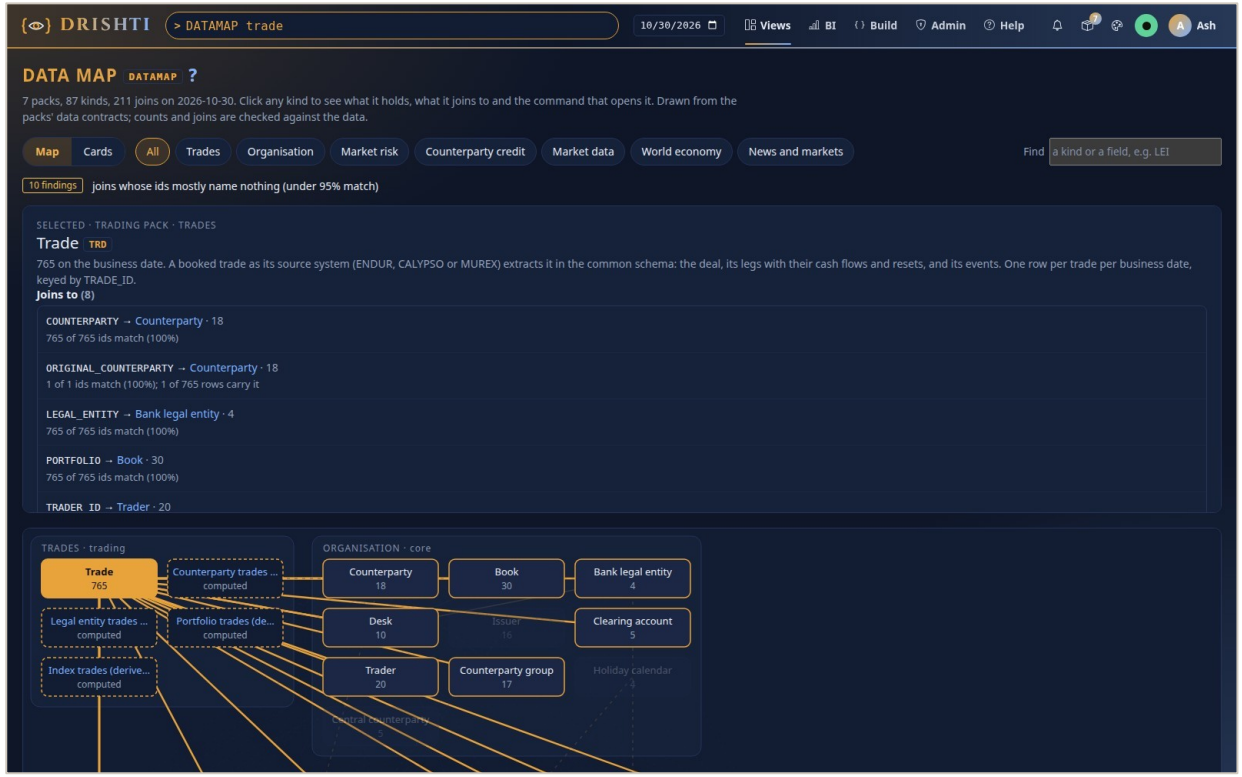
503 → 586
companies in the pack

98 of 99
listed Nasdaq-100 members held by QQQ

162
WHAT terms, 23 new for funds

Key insight The Dow is weighed by price and the Nasdaq-100 capped, as their methods do — the active shares fell to what an investor expects.

A map of all your data that draws itself — and one true surprise a day



DATAMAP trade: the selection across the top (Trade, its joins and match rates), the map under it.

- **Nothing hand-drawn**
  Kinds from the enabled packs, joins from the data contracts, counts from the lake; a new pack appears with no setup.
- **Every join checked**
  How many ids match; joins whose ids mostly name nothing are listed as findings (e.g. a jurisdiction code where an id is expected).

HOME · THE FACT OF THE DAY

One surprising, true sentence from your data, computed as you, on your business date.

Seven ship: a book over its VaR limit while its desk is within · a desk that lowers VaR · posted collateral raising exposure · VaR against peak PFE · the most-held company · two managers holding one portfolio · a fund's weights above its net assets

Key insight Each pack declares its facts; only those that hold today are shown, with 'Why does that happen?' and the command.

Security, drawn as flows

11



Authentication, authorization and audit are kept apart

Authentication
Who you are

- Local passwords: PBKDF2-SHA256, 240,000 iterations
- Or SSO over OIDC: Keycloak, Entra, Okta, Ping
- Lockout after 5 failures in 15 minutes
- Disable a user and their sessions end everywhere

Authorization
What you may see and do

- Roles name the kinds you may open
- Powers: raw, author, approve, calc, layout; BI: bi, bi-author, bi-steward
- Masked fields shown as ••• without raw
- Packs assigned per user

Audit
What happened

- Sign-ins, changes, token use, purges
- Access log: who viewed what, for which date
- Hash-chained collaboration, sealed
- Legal holds and an eDiscovery export

Key insight A mask is applied where a value leaves — in views, search, emails, chat bridges and snapshots alike.

Authentication — a person signs in to the console

Browser → Console → Server

```
1. Browser → console: any page but landing, help
   → 303 to /login (or to the identity provider)
2. Password: server checks PBKDF2-SHA256 hash
   or OIDC: server verifies the ID token itself
3. Server → signed session; console sets cookie
   HttpOnly, SameSite=Lax, Secure by default
4. Every write: same-origin check (else 403)
   and JSON only (else 415)
5. Every 10 s: is this user still enabled?
   disabled, deleted or reset → session ends
```

- **Lockout**
  Five failed sign-ins lock the account for fifteen minutes; an admin's reset unlocks it.
- **The last admin is protected**
  You cannot remove the last administrator, on any node of a cluster.
- **Forced change**
  The development admin is flagged in every page until its password changes.

Key insight A cross-site form cannot post: writes must come from the console and must be JSON.

Authorization — a script acts as its owner, never as more

Scope	Opens	Still required
(none)	Read: views, search, explain, read-only commands	The owner's roles and packs
design:write · bi:write	Create and save designs and BI items	The author or bi-author power
design:approve · bi:approve	Approve someone else's design or BI item	The approve power; never your own
packs:admin	Deploy, roll back and switch packs	The admin role
loads:write	Say a batch of data landed	The pack in the owner's packs

- **Hashes only**
  Secrets are stored as SHA-256; shown once.
- **Write scopes expire**
  At most 90 days.
- **Never open**
  Users, roles, tokens, sign-in, caches and collaboration state.
- **Audited**
  Every write and every refusal.

Key insight A scope only opens the endpoint — the owner's roles are still checked on every call.

Token exchange — the host proves itself, and the user

Host app → Server → Console → Browser

```
1. Host server → POST /api/v1/embed/token  (RFC 8693)
   client: secret (hash stored) or private_key_jwt
   subject: the IdP's ID token, or a host-signed
            assertion (≤ 60 s, used once)
2. Server → ES256 token, 300 s (max 900)
   audience = app, origin-bound, scopes embed:view
3. Browser → console /embed/v1/... with the token
   console checks signature (JWKS), audience, origin
4. Server checks again: app, user, origin, kind,
   rate limit — and always masks
5. Live stream: event: token → renewed in place
```

- **Off by default**
  drishti.embed.enabled; each host app registered by an admin.
- **Always masked**
  An embedded view never shows raw values, whatever the user's roles.
- **CORS, exactly**
  Only registered origins; never *, never credentials.

Key insight The token proves who and where; the server still decides what — on every call.

Evidence, not just conversation

Chained

Every revision of a thread or share hashes the one before

Sealed

Seals cover comments, thread heads, holds and audit; verify on demand

Held

Legal holds by entity, user, thread or all (423 while held)

Exported

eDiscovery zip with a manifest; one-shot download

- **Retract and hide, never delete**

Retention keeps for ever by default; a hidden comment is still in the record.

- **Measured**

An export of 8,300 comments builds in 2.0 s and downloads (19.8 MB) in 3.1 s.

- **Only for compliance**

Legal holds and the export need the compliance power; an export downloads once.

Key insight **Tamper evidence is the honest claim: Drishti cannot stop a database administrator, but it shows when one acted.**

Evidence, by the numbers

12

Over 3,400 Java test methods, three browsers, one gate

3,400+
Java test methods (46 parameterised); the run expands them

~2,250
Console test functions, 640 of them in a browser

~640
CLI, generator and tool test functions

3
Browser engines for embedded views

- **Gates in every build**
  Error Prone and Spotless; ArchUnit for module direction; licence headers; no file over 1,500 lines; branding literals; JSONL from every generator and on every export; READMEs for every pack.
- **Pages at three widths**
  Every page at 390, 1,600 and 2,560 px; at 390 with touch: real content, 44 px targets, nothing clipped.
- **Numbers checked against an oracle**
  VaR and ES recomputed from the input files and compared with the server; intraday equals one run, to the cent; Delta equals Iceberg; a sharded cluster equals one server.

Key insight **Any failing test in any of the three suites stops the push — on Java 21 and on Java 25.**

VaR and exposure from Iceberg equal Delta, to the cent

Test	What it proves	Latest run
`VarFormatEqualityTest`	the same lines in Delta and Iceberg: VaR, ES, component and incremental VaR, worst scenarios, limits and limit letters, USD and EUR; intraday batches; official records	green
`ExposureFormatEqualityTest`	EE, PFE, EPE, profiles, contributions and official records, both formats	green
`LakeAppendsTest`	a Kafka window lands on an Iceberg risk lake as one append	green
`OfficialSinkTest` · `OfficialS3Test`	official tables in Delta and Iceberg, a folder or S3; six writers at once; a Delta commit on S3 is an exclusive create	green
`OfficialReadersTest`	BI's readers follow the official format by themselves	green

NOT RUN The Glue and S3 Tables runs on AWS (GlueLiveTest, check-glue.sh): written, never run; they need an AWS account.

On the demo data

Figure	Delta	Iceberg
VAR BANK, 2026-09-30 (USD)	112,041,646	112,041,646
Risk tables moved	—	14 tables, 955,124 rows, 579 s

A server on the Iceberg lake answered VAR BANK, the limit letter LL:LL-0003 and XPO BANK and published its official records.

Demo figures from Drishti's synthetic data, one workstation: an observation, not a sizing rule. Delta stays the default; Iceberg is chosen by configuration.

Key insight One connector switched from plugin: delta to plugin: iceberg, the same rows, the same numbers.

Trees rebuilt from columns; upgrades that keep every edit

`HierarchyEquivalenceTest`

MGMT and LEGAL rebuilt from columns

- The same node ids, parents, levels and members as the risk-hierarchy rows
- Aggregates equal to the bit; VaR, ES, components and worst scenarios to the cent, USD and EUR
- Limits equal; every limit letter the sum of its rows

Latest run: 3 tests green

`HierarchyUpgradeWithoutLettersTest`

Data written before the letters

- P&L vectors and exposure paths without LIMIT_LETTER
- Only LL is skipped, with a DRS-5211 warning and its fix
- VaR, exposure, MGMT, LEGAL and the official captures answer as before

Latest run: 1 test green

`ConnectorUpgradeTest`

A server on the previous release's files

- Every unedited file equals the current template
- The edits stay (an edited risk-store and var-whatif); missing keys added
- var-result, var-whatif with ccy and exposure-result answer

Latest run: 3 tests green

ColumnHierarchyBenchmark: 10,000 trades, 700 limit letters: the trees build in 65–99 ms against 53–197 ms without them; the VaR of all 1,412 nodes takes about 30 ms.

Key insight A new tree is configuration; an upgrade never loses an edit or a date.

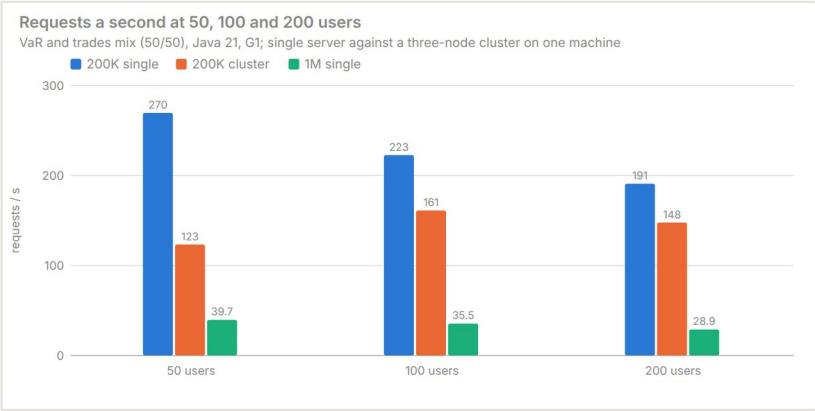
Measured: 200,000 and 1,000,000 positions side by side

One workstation: one server at each size, and a cluster of three at 200,000[1]

Measure	200K, one server	1M, one server	200K, cluster of 3
Requests a second, 50 / 100 / 200 users	270 / 223 / 191	39.7 / 35.5 / 28.9	123 / 161 / 148
p99 a VaR node, 50 / 200 users	0.72 / 5.39 s	1.41 / 26.0 s	0.46 / 5.08 s
p99 the bank minus a desk, 50 / 200 users	0.92 / 5.44 s	9.48 / 30.7 s	1.11 / 5.15 s
A date's first open	4.5 s	26.3 s	10.8 s
Load: five kinds, two dates, Java loader	13.9 s	27.6 s	the same lake
Heap after GC, two dates open	0.88 GB	7.5 GB	1.03 / 0.82 / 0.53 GB
Resident memory (RSS) at the end	5.0 GB	21.2 GB	4.9 / 5.6 / 5.4 GB
Disk a date: lake / hot file	0.29 / 0.56 GB	1.43 / 3.14 GB	the same lake

- **A cluster of three at 1,000,000 positions:** not measured on one machine[2]
- Mix: half VaR (nodes, the bank minus a desk, 50,000-position baskets, contributions, P&L paths), half trade pages; 60 s a step after a 45 s warm-up
- Single server -Xmx8g at 200K, -Xmx16g at 1M; the cluster -Xmx4g a member, VaR sharded, 2 replicas a shard; Java 21, G1

Requests a second, by users

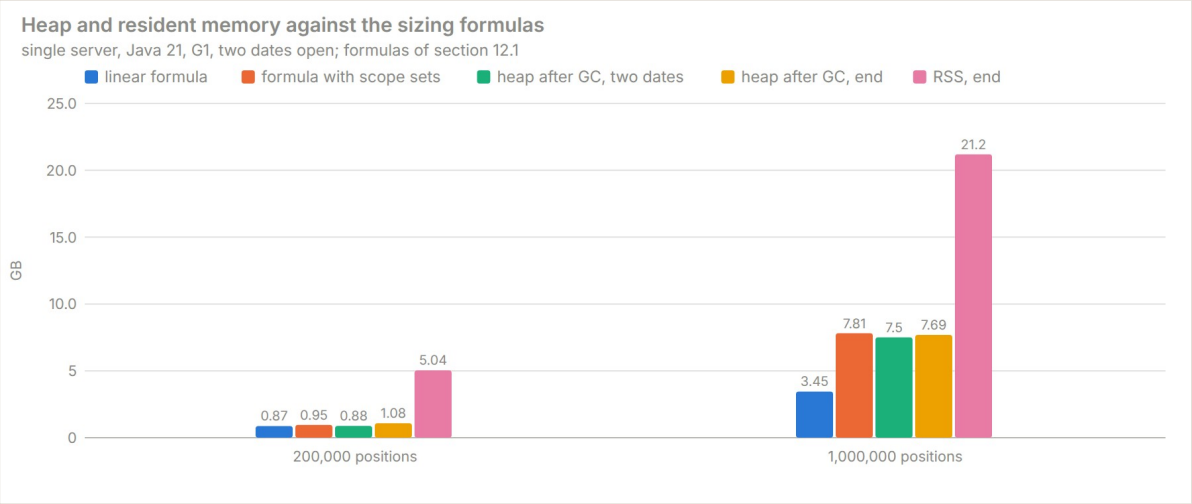


- 200K, one server: 270 a second at 50 users, every p99 under 1 s; at 200 users the wait is queueing on 24 threads
- 1M: a node, a basket and contributions stay under 1.5 s at p99 with 50 users; the bank minus a desk sets the pace (next slides)

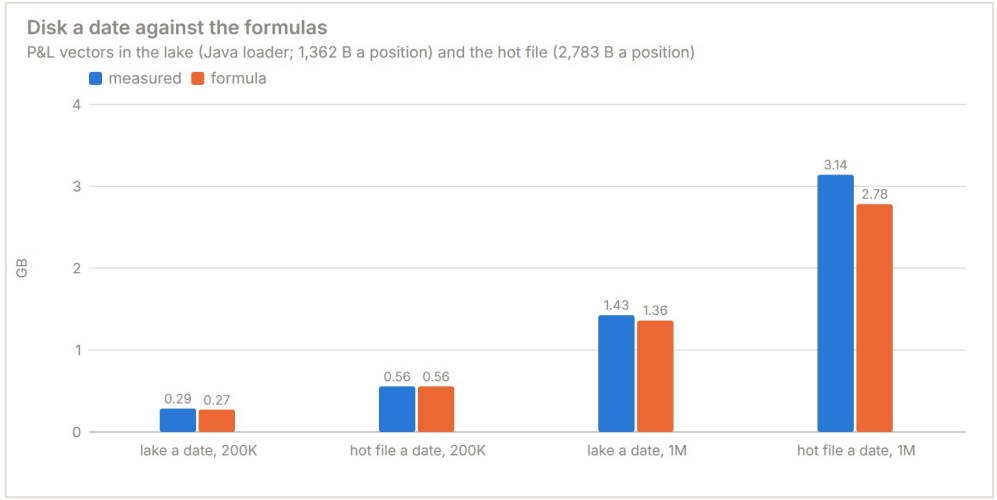
[1] Measured on one workstation (12 cores, 24 threads, 61 GB, NVMe): BENCHMARK.md section 4 [2] Three 16 GB servers exceed one 61 GB machine

Heap and disk against the formula, at both sizes

Heap and resident memory, two dates open



Disk a date: the lake and the hot file



The heap formula[1]

H = 128 MB + 980 B × P + 2 × (1,023 B × P + 23.6 KB × N + P × N ÷ 16)

P positions, N nodes, two dates open: 982 MB against 881 MB measured at 200,000; 7.8 GB against 7.5 GB at 1,000,000. The P × N ÷ 16 term is the nodes' ordinal sets; without it the formula meets 200,000 only. -Xmx = (H + 0.13 GB) ÷ 0.61: 13 GB at 1,000,000, run at 16 GB.

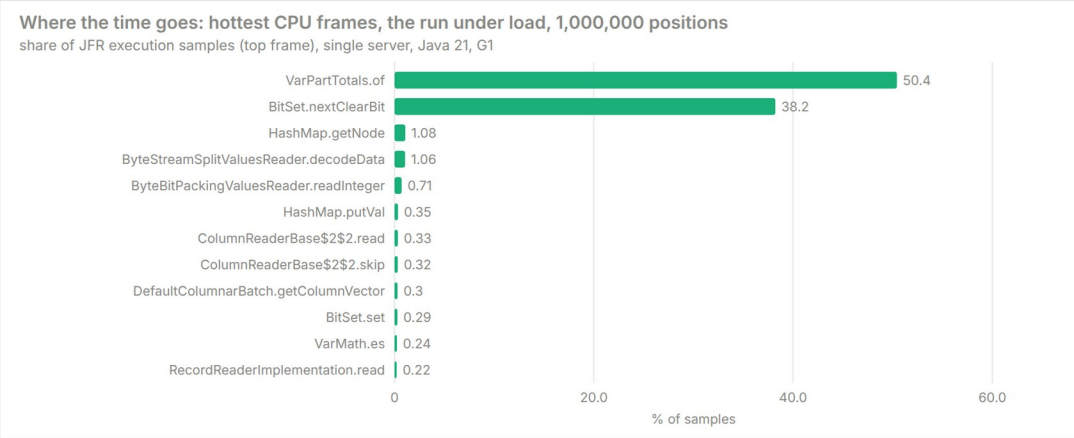
Disk a date[1]

The lake: 1,362 B a position, 5% above it at both sizes. The hot file: 2,783 B a position, met at 200,000, 13% above at 1,000,000 where 27,024 nodes add 639 MB of aggregates; 2,500 B × P + 23.6 KB × N fits both. Resident memory is about three times the heap: the mapped hot files count in it.

[1] Fitted on both measured sizes, one server, Java 21, G1: BENCHMARK.md sections 4.8 and 7.1

Where the time goes and what we do next

Hottest CPU frames under load, 1,000,000 positions[1]



- Summing a scope's parts position by position, with its bit-set scan, takes 89% of the CPU under load at 1M; a third at 200K
- One user alone: a VaR node 3 ms at 200K, 4 ms at 1M; the bank minus a desk 68 ms and 1.46 s
- Heap: 3.2 GB a date at 1M, mainly each node's ordinal set, as long as the date
- A date's first open: decoding the lake's Parquet is two thirds of its CPU at both sizes

What we do next, in order

1 Sum a scope's parts by runs

a position is one run of ordinals: the totals walk runs, not positions, so the bank minus a desk no longer grows with the bank

2 A few hundred MB a date[2]

the 3.2 GB a date comes mainly from the nodes' bit sets: compressed run sets and off-heap arrays take it to a few hundred MB

3 Ten dates in memory

a configured number, ten by default, the least recently used out first; month-, quarter- and year-ends pinned; a byte budget as the safety net

4 Then Java 25, and BLAS later

Java 25's accelerated paths (hot files as memory segments, the Vector API behind a setting); later BLAS through OpenBLAS, for many scopes at once

[1] Measured: JFR execution samples, one server, Java 21, G1 (BENCHMARK.md 4.9) [2] Target, not yet measured (HIGH_PERFORMANCE_NATIVE.md section 3.3)

Years on disk, ten dates in memory, and the dates people return to pinned

Open in memory
the mapped hot file and the date's structures; ten dates by default, least recently used out; a byte budget as the safety net

A hot file on local disk
the built file of a date not open; opens again in seconds, without reading the lake

The lake (Delta or Iceberg)
every date, for the years kept; opening builds the hot file once

The official record
each date's figures of every node; history, backtests, statistics and reports read it

At 1,000,000 positions a date

	Measured	Target
Heap a date open	3.2 GB	about 0.4 GB[1]
Hot file a date	3.14 GB	unchanged
Lake a date	1.4 GB	unchanged
Dates in memory	2 newest + 1	10, LRU[1]

Pinned, from the calendar

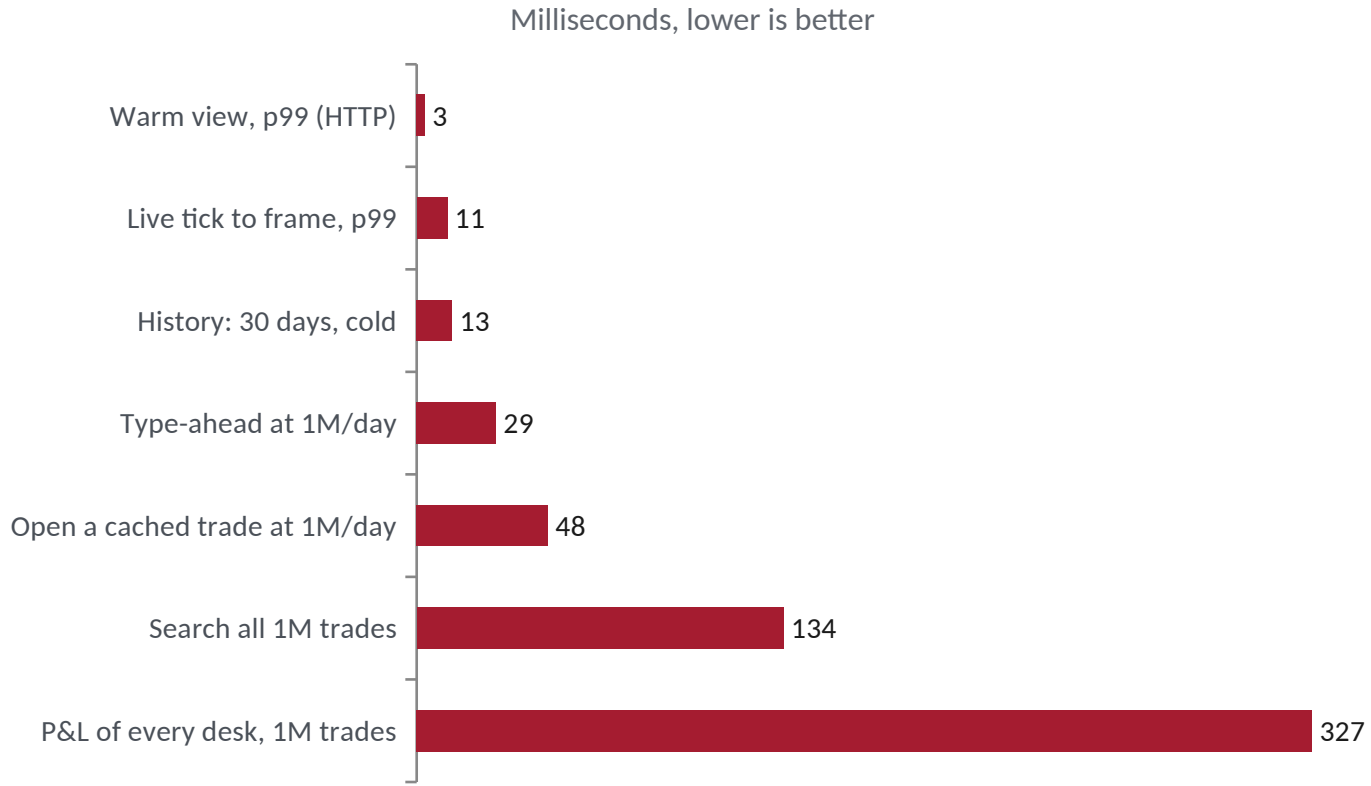
- month-, quarter-, year- and fiscal-year-ends (USNY+CATO)
- the stress window's first and last days
- any date the bank names: a crisis day, an audit date
- kept on disk by default, or held in memory

Key insight **Memory grows with the dates open at once, a setting, never with the years kept.**

[1] Target, to be measured: the 1M re-benchmark after the memory work (HIGH_PERFORMANCE_NATIVE.md section 3.3; the plan's theme memory, scale and mobile)

PERFORMANCE · MEASURED

Fast where people wait



Gate	Target	Measured
Warm view p99	< 50 ms	3.0 ms
Live p99	< 40 ms	11.2 ms
History, 30 days	—	13 ms cold, < 1 ms warm

- **One workstation**
  24 cores; 1,000,000 trades a day for three days, 5.2 GB.
- **Cold vs cached**
  Opening a trade cold is 137–157 ms; from the cache 48 ms.

Key insight **The gates are written down first and measured after — and every one has headroom.**

SCALE · 35.7 MILLION TRADES

Tens of millions of trades, fifty users, no failures

35.7M
Trades: ENDUR, CALYPSO, MUREX, 30 days, loaded while serving

179/s
Requests a second, 50 users against the server, 0 failures

0.21 s
Form search p50, the form kept per data version

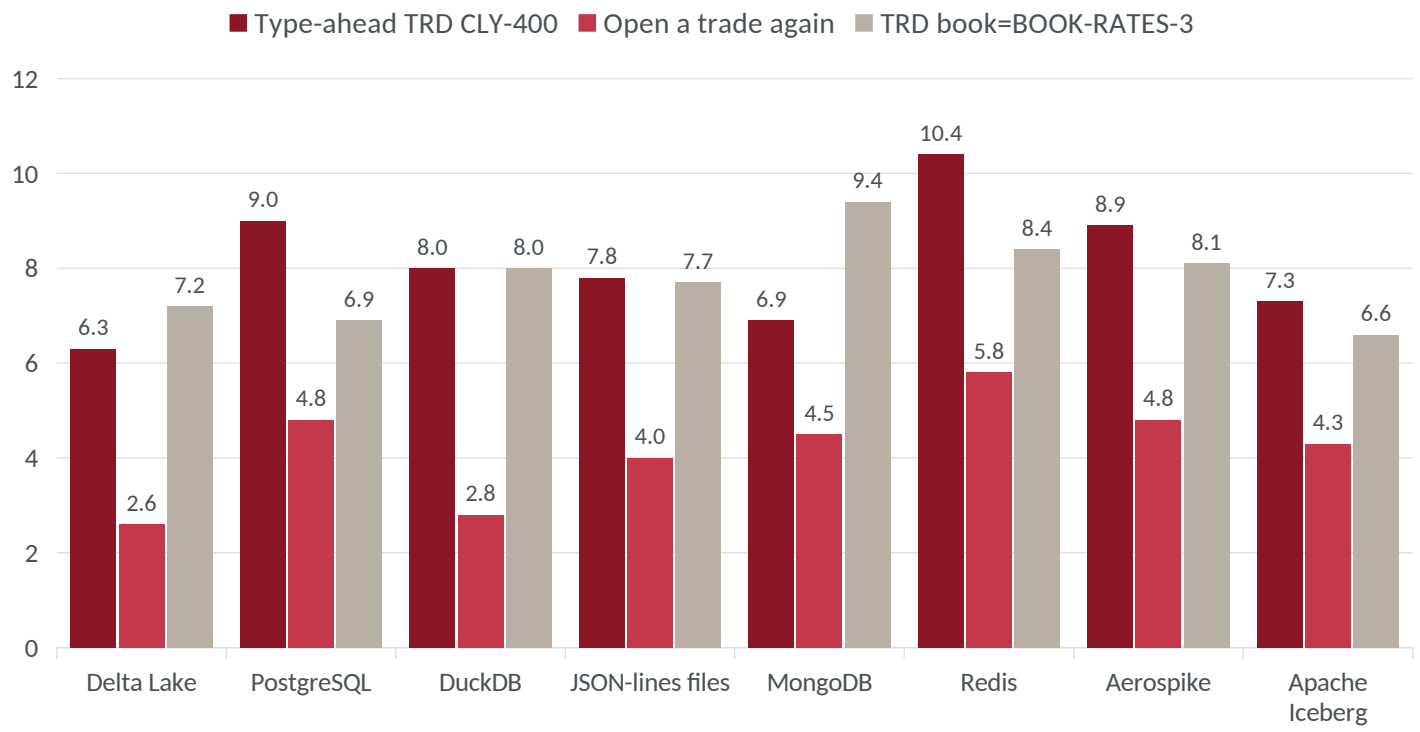
182/s
Console pages a second, 50 users, 0 failures

Data	Measure	Users	Result
35.7M trades	Single trade / grouped page, p50 / p95	50	0.16 / 0.73 s, 0.17 / 0.73 s
35.7M trades	Server heap while serving	50	3.7 GB of 8 GB
400,000 restated	A day reloaded, 10 % amended, while serving	30	2 min 54 s, 202 requests/s, 0 failures
8.2M trades	Single trade / grouped page from disk, p50	50	18 ms / 70 ms

Key insight One workstation, an 8 GB heap: the figures and their method are in PERFORMANCE.md and SCALE_STRATEGY.md.

The same book in eight stores, the same questions over HTTP

Median milliseconds at 50,000 trades a day (lower is better)



Store	Size	Load
Delta Lake	270 MB	10.2 s
PostgreSQL	739 MB	22.3 s
DuckDB	122 MB	22.3 s
JSON-lines	1,322 MB	15.4 s
MongoDB	202 MB	17.1 s
Redis	183 MB	20.2 s
Aerospike	1,176 MB	18.8 s
Iceberg	133 MB	26.4 s

Measured on one workstation, 2026-10-01. Four stores were also measured at a million trades a day.

Key insight Every store answers an interactive question in single-digit milliseconds at this size; the choice is about operations.

VaR and exposure answers in milliseconds, at 100,000 trades in memory

VaR (VarScaleBenchmark, VarTierBenchmark)

Measured	Time	On
A node's VaR and ES	0.48 ms	tiered store, 100k trades
The bank from float32 rows	3.4 ms	tiered store, 100k trades
The bank from float64 rows	58 ms	heap tier, 1M trades
A basket of 50,000 positions	2.6 ms + ≈ 45 ms ids	tiered store, 100k
First load of a date	0.8–1.2 s	tiered store, 100k
Loading the demo date	572 ms	9,572 trades, the server

Exposure (ExposureTierBenchmark)

Measured, 100k trades, 9,901 sets	Time
Load a date: rows into value matrices	785 ms
Collateral, E, N, hot file written	1,250 ms
Reopen the date from its hot file	157 ms
Scope BANK (every set)	61 ms
Scope CPTY: (20 sets) · NSET: (1 set)	0.22 · 0.06 ms
Pre-deal exact, Java · proxy	0.11 · 0.007 ms

Why it is fast

Vectors add: a node's vector is summed once and kept; a hot file per date is memory-mapped (float32 rows, float64 sums); sums run on every core in a fixed order — the same bits every time.

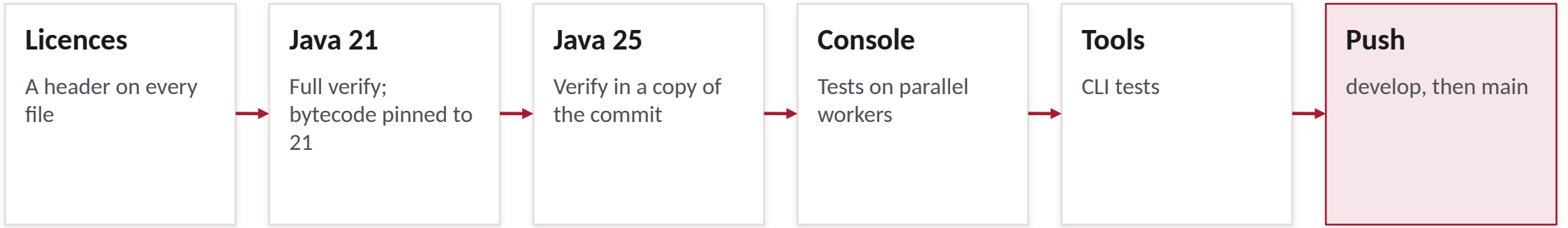
What is extrapolated, and said so

1M trades × 50 dates (VaR) and a 1M-trade, 100,000-set bank (exposure) are extrapolated from these runs[1]; a workstation run stays at 100,000 trades in memory.

Key insight Measured on one workstation (24 threads, 61 GB, JDK 25, NVMe); the full million-trade runs are for a real server.

[1] linear in trades (× 10 from 100,000 to 1M) and in dates from the measured runs above (VAR_GUIDE.md 27.3, EXPOSURE_GUIDE.md 27.4); not measured, to be confirmed on a server

Nothing reaches main without the full gate



~1 hour

A full drill: Java 21 verify, then Java 25, console and tools side by side

45

Tagged releases, 30 September to 29 October 2026

2,471

Commits, every push through the drill

Key insight **A failed test stops the push; a flaky one is retried once and written down — never ignored.**

Operating and deploying

13



From a clone to a live view in about fifteen minutes

Seven commands

```
./mvnw -q package -DskipTests -pl drishti-server -am
uv venv --python 3.13 drishti-console/.venv
uv pip install --python drishti-console/.venv/bin/python \
    -r drishti-console/requirements.txt
python tools/drishti.py lake demo
java -Djava.io.tmpdir=data/tmp/server -Xmx4g \
    -jar drishti-server/target/drishti-server-<v>-exec.jar
python drishti-console/run_drishti_web.py
# open :17480/t → TRD MX-20000001 <GO>
#                 VAR BANK <GO>   EXP BANK <GO>
```

GETTING_STARTED

About 15 minutes: build, the demo lakes, a view

IDE_GUIDE

Server in IntelliJ, console in PyCharm

CLI_GUIDE

pack make, history, var, exposure, convert

USER_GUIDE

Every page, every key; 29 tutorials beside it

Key insight Start with your own JSON Lines: one pack make, one deploy, and your records have a terminal.

From a folder of JSON Lines to a deployed pack

One command

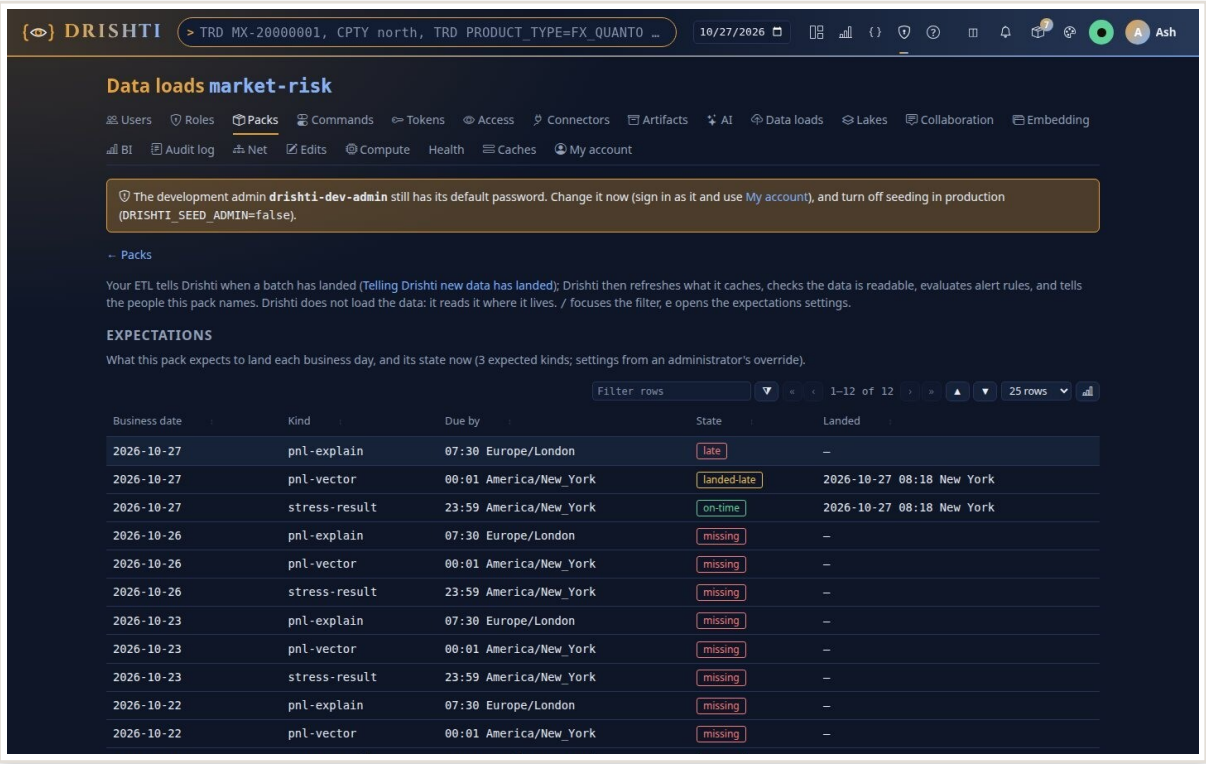
```
drishti.py pack make ./trades --pack desk \
    --kind trade --key TRADE_ID \
    --match PRODUCT_TYPE --date TRADE_DATE
# → desk/ pack, sutras per product type,
#   Delta lake by date, desk.tar.gz + sha256,
#   README.txt, run scripts
```



- **Checked before it lands**
  Checksums, paths, server version, Sutra tests and an optional Ed25519 signature.
- **Never stuck**
  The previous version is kept; a failed deploy rolls back; history has Roll back.

Key insight **Archives carry the pack, never the data — where data lives stays an admin's decision.**

Tell Drishti new data has landed — and it acts on it




```
drishti.py data landed --pack trading \
    --kind trade --date 2026-10-05 --rows 10000
# or POST /api/v1/packs/trading/loads
```

- **Refresh, verify, alert, notify**
  Caches refreshed, the date checked, alert rules run on it, bell, inbox and email.
- **Late is a fact, said once**
  Per-kind expectations with time zone and calendar; Health shows 'data late'.
- **Safe to repeat**
  Idempotent per pack, kind, date and batch; token scope loads:write.

Every build step from the command line

Group	What it does	Examples
sutra · pack	Lint, test, preview Sutras; make, sign, deploy packs	sutra test, pack make, deploy, rollback
data · lake · convert	Ingest JSONL, build lakes, turn CSV into JSONL	landed, lake demo --names, convert
server · connector · net	Packs and commands on or off, connectors, members	packs off, commands off, net members
bi · design · view	Models, SQL, runs, statements; designs; a view	bi sql --format jsonl, bi runs
history · contract	An entity over days; every command's contract	history TRD 1000015 30, contract check
var · exposure · compute	The risk engines as you see them; reactive kinds	var show … --ccy EUR, exposure profile
change · edits · doctor	Write-back changes, Admin → Edits, a setup check	change submit, edits status, doctor

Key insight Anything the console can build, a script can build — and a pipeline can check it.

Ready means ready; drain before stop; upgrade one node at a time

Each node, in turn: the rolling upgrade runner does exactly this



Order: consoles → BI and data-serving → coordinators → shard owners, one zone at a time. Any failure stops the run and names the node; resume with --from.

Readiness, for routing; liveness, for restarts

Readiness	HTTP	When
`UP`	200	joined, newest date open, owned shards ready
`DOWN`	503	joining, a date opening, or a shard not ready
`OUT_OF_SERVICE`	503	drained before a stop

- Liveness (the JVM answers) drives container restarts: a node waiting for a date is never restarted for waiting
- GET /api/v1/admin/node says what readiness waits for, and whether the node is drained and safe to stop
- Each gate can be switched off: drishti.node.readiness.joining, .engines, .shards

The runner and the stop hook

- tools/rolling-upgrade.sh --inventory my-cluster.yaml --version <v> --dry-run prints every step; without --dry-run it runs them (Windows: rolling-upgrade.ps1)
- The inventory (deploy/rolling.example.yaml): platform vm or ecs, the waits (drain 600 s, ready 1,800 s), zones, the nodes' real addresses
- The stop hook drains a node on every stop (systemd, Docker, ECS); Admin → Net has Drain for a stop
- The drain needs an administrator's token with the node:drain scope

Every server measures itself and raises its own alerts

Platform metrics, every 15 seconds

What	Prometheus names (examples)
Caches, the memory budget	`drishti_cache_bytes`, `drishti_memory_budget_*`
Disk folders and free space	`drishti_disk_folder_bytes`, `drishti_disk_free_bytes`
Sources and feeds	`drishti_source_up`, `drishti_feeds_attention`
Loads	`drishti_loads_last_ready_age_seconds`
Pages and alerts	`drishti_view_seconds`, `drishti_alerts_open`

- Exported on /actuator/prometheus with the JVM's and the HTTP server's own
- The CIO lens reads the same names
- Admin → Health shows them, with the alerts open now

With or without Prometheus: each server evaluates the rules itself; deploy/prometheus/drishti-alerts.yaml is the same rules in PromQL for a site's own Alertmanager.

The alert rules as shipped (config/alerts/platform-alerts.yaml)

Rule	Opens when	Severity
`cache-near-limit`	above 95% of its limit, an hour	warn
`heap-after-gc-high`	heap after GC above 85%, 10 min	warn
`direct-memory-high`	direct memory above 80%	critical
`disk-free-under-guard`	free disk under the guard	critical
`source-down`	a source down for 5 min	critical
`feeds-need-attention`	a load late, missing or failed	warn
`views-slow`	pages above 150 ms at p99, 5 min	warn
`member-suspect`	a member silent for 2 min	warn

- When one opens: logged, counted, and every administrator told in the inbox; listed on Health until it closes
- Rules are configuration: a metric, a threshold, for how long, a severity and a summary line; read again when changed

Tidy by design: scopes, one settings document, archives, API references

Admin → Ad-hoc scopes

every person's saved VaR baskets in one list

- Owners delete their own; an administrator may delete anyone's, with an optional reason
- Every delete audited (var-basket-delete: who, whose, the reason); the owner told in the inbox
- Also from the command line and the API (drishti.py var basket-delete)

One configuration document

docs/admin/CONFIGURATION.md, and Help → Configuration

- Every setting of every kind: key, type, default, environment variable, meaning, restart or live
- Generated from the files themselves and checked on every build: never out of date
- Administrators see the value in effect beside each default, and where it came from; secrets never shown

Loaders read archives

zip, tar.gz, tgz and tar, read in place, local or s3://

- JSON Lines members, plain or compressed, read in name order; other files skipped with a note
- No folders: one dataset named after the file; one level of folders: a dataset per folder
- One rule in the Python and the Java loaders, tested on the same archives

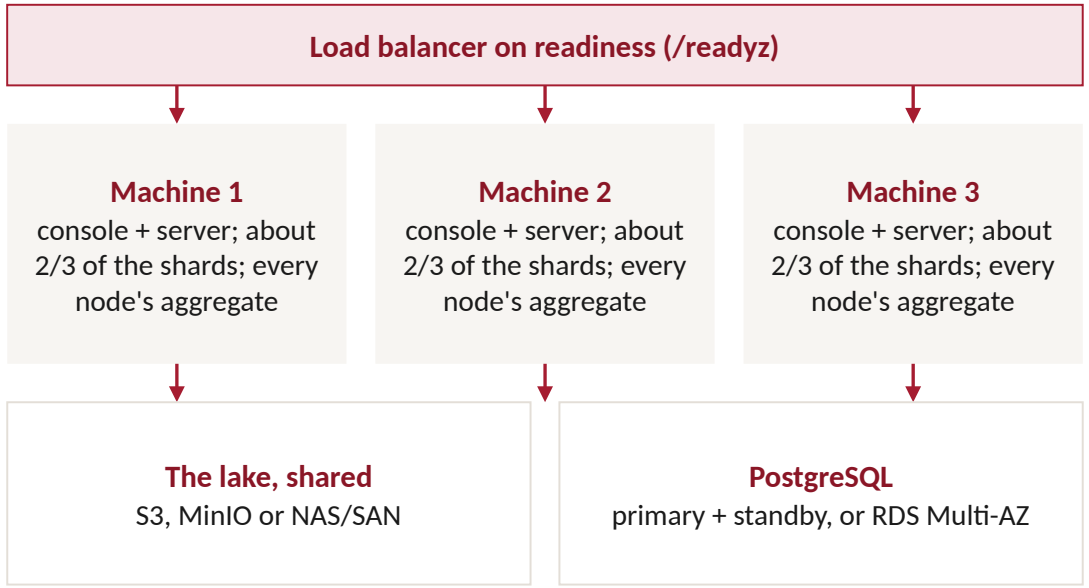
API references in the console

Help → Developer

- The REST API (OpenAPI), the renderer's TypeScript API
- Java and Python references generated from the code: plugin and connector SPI, shard machinery, compute SDK, drishti.py and its client
- The build fails on an undocumented public member

Three machines, on premises or on AWS

The smallest highly available Drishti: three machines, the rest shared



- One fails: every shard still has a replica; the others rebuild the second copy from the lake
- Upgrades one at a time: drain, stop, upgrade, ready; two copies throughout
- No consensus protocol: a split leaves read answers identical; writes go to the shared database
- Two machines: pilots and UAT only (no spare)

A machine (extrapolated, to confirm at 1M)[1]

	On premises	AWS
Machines	3 × 24–32 cores, 128 GB, 1–2 TB NVMe	3 × m7i.8xlarge, 32 vCPU, 128 GB; gp3 1 TB
Lake	S3-compatible (MinIO) or NAS/SAN	S3
Database	PostgreSQL + standby	RDS PostgreSQL Multi-AZ
Balancer	a pair + VIP, or F5	ALB across three zones

- 128 GB carries 1-day VaR to 2M (heap 5.2 GB, -Xmx 12g) and every horizon to about 1M; 192 GB for 2M with every horizon
- The lake is small: about 1–3 TB for several years at 1–2M positions (1-day)

AWS, a month (Canada (Central), list prices)[2]

On demand: USD 5,400–6,100[3]

1-year Savings Plan: USD 4,400–5,100[4]

3-year Savings Plan: USD 3,500–4,200

Check every price in the AWS Pricing Calculator before quoting.

[1–4] How these figures were derived: A-1, in the appendix (Three machines)

Where it is going

14



Ten themes; each metric ships with its playground

1 Memory, scale and mobile

- a date's heap shrunk; ten dates in memory
- phone tiers; lens menus on a phone
- S3 Tables and Delta on S3

2 Oversight

- why VaR moved, by cause (RO1)
- breaches and escalation (RO2)
- the board pack; market data checks

3 Java 25 and the accelerated paths

- hot files as memory segments
- Vector API kernels, bit-exact
- Java 21 and 25 builds side by side

4 Regulatory reporting

- BCAR 90.010, Pillar 3 MR1 and MR2
- FFIEC 102, §217.212, Volcker
- the CRO decision queue

5 Counterparty engines

- SA-CCR and its playground
- the CCR cube core
- the Exposure playground; FVA

6 Sensitivities and capital

- position sensitivities
- FRTB-SA and the MR cube
- SIMM

7 Native performance

- BLAS through OpenBLAS, many scopes at once
- DuckDB on Arrow, direct lake scans
- within the parity rule

8 XVA and wrong-way risk

- CVA, BA-CVA, SA-CVA
- MVA and KVA
- wrong-way risk

9 Advanced market risk

- FRTB-IMA
- QLRisk pre-deal pricing

10 Executive lenses

- the CRO and CIO lenses
- liquidity, capital and climate

Key insight At most two large items a theme, each proved by its own drills before the next begins.

Appendix: reference

A

248 error codes, every one documented

```
$ curl -s server/api/v1/views/trade/NOPE-404
Content-Type: application/problem+json
{ "status": 404,
  "code":   "DRS-1001",
  "title":  "entity not found",
  "detail": "no source holds the entity" }
```

Family	Area	Codes
DRS-1xxx	Entities and sources	14
DRS-2/3xxx	Grammar, Sutras, inference	11
DRS-4xxx	Commands and search	19
DRS-5xxx	Requests, Drishti*net, VaR, exposure	51
DRS-6xxx	Users, roles, tokens	11
DRS-7xxx	Collaboration	16
DRS-8xxx	Embedding	6
DRS-9xxx	Business intelligence, runs	120

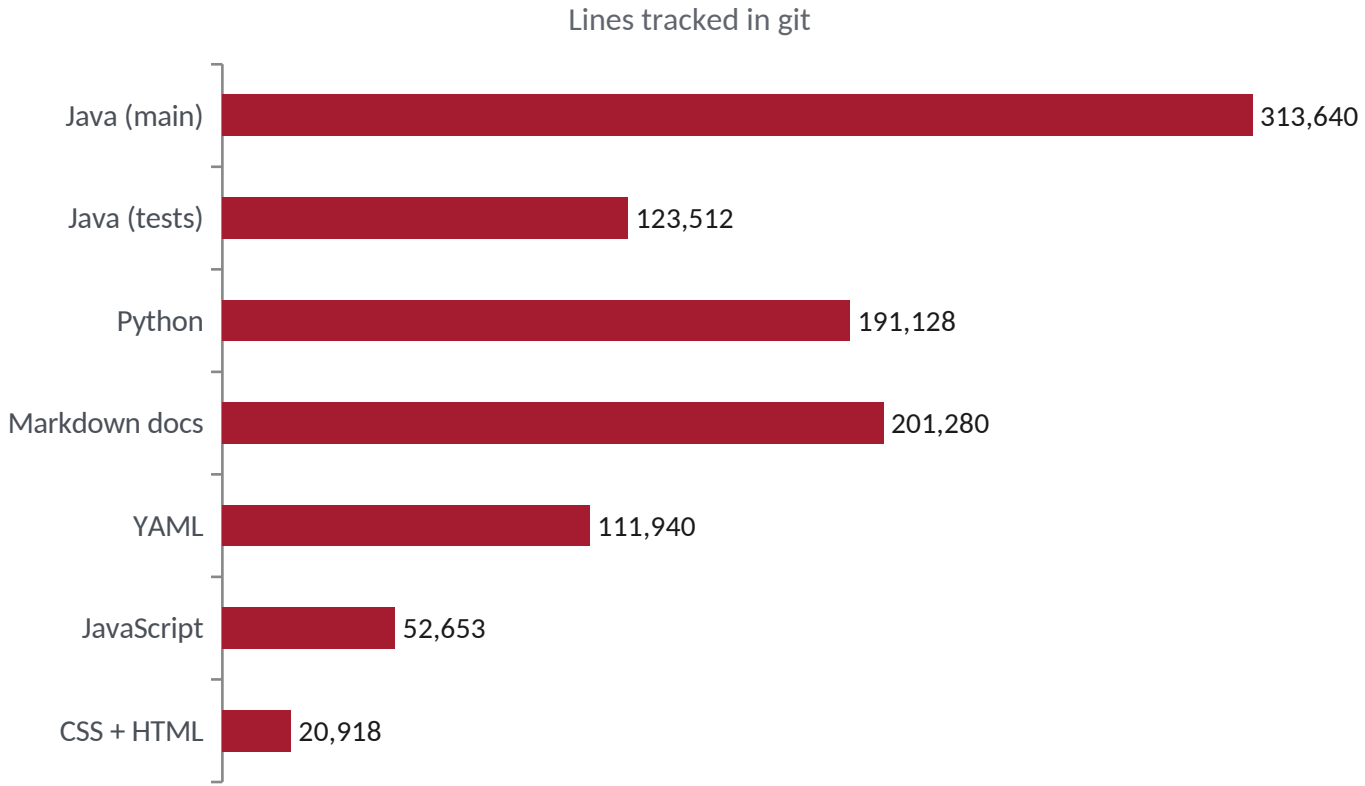
248
Codes in nine families, one table

47
Sutra problem codes in the grammar reference

RFC 7807
problem+json, the same shape everywhere

Key insight **Errors are a contract: a stable code, an HTTP status and a row that says what to do.**

Where the work is



- **193 guides and references**
  User, admin, CLI, BI, packs, connectors, Knowledge, 29 tutorials and 22 ADRs; 113 generated contract pages; two documents per measure.
- **Every screen documented**
  Over 800 pictures under docs/, taken by scripts on the current UI over the demo lakes.
- **21 modules, 19 plugins**
  Plus a server, integration tests and benchmarks; the risk engines in drishti-compute.

How the numbers were derived

How it scales: the algorithms

[1] A scope's set: one bit a trade, N ÷ 8 bytes: 200,000 ÷ 8 = 25,000 B ≈ 25 KB; 2.3M ÷ 8 = 287,500 B ≈ 290 KB (VAR_GUIDE §27a.1). The standalone cache: 8 B a position and lens: 200,000 × 8 = 1.6 MB; 2.3M × 8 = 18.4 MB ≈ 18 MB (§27a.3).

[2] Is a position whole: a scan of its own bits, (to − from) ÷ 64 machine words (a position of up to 64 trades: one word); a scope: one pass over its positions. A standalone VaR keeps the k worst of S values, O(S × k): k = ⌈509 × 0.01⌉ = 6 at 99% (§27a.1, §27a.3).

[3] Measured, one user, p50 (BENCHMARK.md §4.5): a node 3 ms at 200,000 and 4 ms at 1,000,000 positions, its aggregate whatever its size; the bank minus a desk 68 ms and 1.46 s, linear in the scope's positions while its parts are summed position by position: × 2.3 = 3.4 s at 2.3M (EXTRAPOLATED, §12.1), the walk that summing by runs removes.

Three machines

[1] Per member of three, replicas 2 (each holds 2/3 of the shards), two dates open; EXTRAPOLATED from the 200,000-position benchmark, to be confirmed at 1M: heap H = 0.323 + 2 × (1,023 B × P + 19.5 KB × 20,000 nodes; 15,000 portfolios + 5,000 hierarchy nodes, ASSUMPTION) = 5.2 GB at 2M; -Xmx need (H + 0.13) ÷ 0.61 = 8.7 → 12g; RAM = heap need + two open dates of hot files (2,783 B × P × 2/3) + a load + 7 GB = 29 GB ÷ 0.61 → 48 GB; every horizon (the capacity model × P ÷ 2.3M): 55 GB at 1M, 97 GB at 2M → 192 GB.

[2] AWS list prices, Canada (Central), on demand, 730 h (ASSUMPTION: verify before quoting): m7i.8xlarge = m7i.16xlarge $3.5952 ÷ 2 × 3 × 730 = $3,937; gp3 3 × 1 TB, 6,000 IOPS, 500 MiB/s = $363; S3 2 TB $60; ALB, zones, loads, NAT and endpoints, CloudWatch, secrets as the cost lines = $476.

[3] RDS PostgreSQL Multi-AZ $600–1,300 (ASSUMPTION, not in the price constants: db.m7g.xlarge to 2xlarge, us-east-1 list + ~10%): total $5,436–6,136, about 5,400–6,100.

[4] Compute Savings Plans, no upfront, on EC2 and Fargate only: 1 year −26.9% / −20.0% = −$1,065; 3 years −49.5% / −45.0% = −$1,964.

Related documents

Document	What it holds
README.md · docs/guides/GETTING_STARTED.md	the name and the tagline; from a clone to a live view in about 15 minutes
docs/guides/USER_GUIDE.md · CLI_GUIDE.md	every page and key; the command line
docs/architecture/IMPLEMENTATION_PLAN.md	what is built, planned and parked; known gaps
docs/architecture/EDITIONS.md · docs/admin/EDITIONS_GUIDE.md	Drishti Risk, World and Market; editions and lenses
docs/guides/DATA_FORMATS.md · docs/guides/COMMAND_CATALOG.md	JSON Lines as the data standard; a contract per command
docs/guides/VAR_GUIDE.md · EXPOSURE_GUIDE.md	risk computed on demand: VaR and exposure
docs/architecture/COMPUTED_ATTRIBUTES.md	the attributes on the pages, derived from trade-grain inputs
docs/admin/PERFORMANCE.md · docs/architecture/SCALE_STRATEGY.md	the measured numbers and how they were taken
RELEASE_NOTES.md · CHANGELOG.md	the release history (this deck is version-free)
docs/knowledge	the ideas behind the data
docs/publications/Drishti-VaR-Engine.pptx	the VaR engine in depth
docs/publications/Drishti-on-AWS.pptx	Drishti on AWS: the lake, Delta or Iceberg



Drishti दृष्टि

Thank you

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Ashutosh Sinha • November 2026